Rheinland-Pfälzische
Technische Universität
Kaiserslautern
Landau

Multi-Person Human Mesh Recovery with the Anny Body Model

Master's Thesis

by

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November 8, 2026

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Use of AI tools

AI assistants (Claude, Anthropic, and Codex, OpenAI) were used under the author's direction for code development, debugging, analysis scripts and assistance with drafting and revising the report. Numerical provenance is recorded in the repository and in Appendix A. **[TODO: Author review required: confirm the exact tools and uses, verify the report and numerical claims personally, and approve the disclosure wording against the applicable university rules before submission.]**

Abstract

The problem in this thesis is to turn one photo of several people into a 3D body for each person, at a lower computational cost. I trained a small ViT-B network, the student, that learns from a larger ViT-L network, the teacher. Both use Multi-HMR, which finds all people and their bodies in one pass, and the Anny body model, with named shape parameters such as age and height.

AnnyOne images are synthetic and have exact Anny labels. 3DPW photos are real, but labelled in another body model, SMPL. I fitted an Anny body to each 3DPW label, using points that correspond geometrically in both models and virtual torso joints, for 16,694 images with 22,749 labelled people.

The teacher was trained with 30 % real images per batch and two corrections to its targets. The student gets copies of the teacher's prediction heads, and a learned projector widens the student's features to fit them. The student learns from the labels and, by knowledge distillation, from the teacher's features and outputs.

The student has 101.38 million parameters and needs 603.3 GFLOPs for one person, the teacher 318.38 million and 1927.4 GFLOPs. On one L40S GPU the student's forward pass took 28.5 ms and the teacher's 59.7 ms.

But the student is also less accurate. Its per-vertex error (PVE, the mean distance to the true surface) was 119.6 mm on held-out AnnyOne images and 110.1 mm on 3DPW, the teacher's 73.7 and 89.0 mm. On 3DPW the student had a lower PVE than the released Multi-HMR-Anny baseline (126.9 mm), which was not trained on my converted 3DPW labels. The baseline's AnnyOne result most likely comes from its training images. All 3DPW errors are against converted labels and not directly comparable with published SMPL results.

In a training-length experiment I trained the student for 60 instead of 30 epochs. Its holdout PVE fell from 119.6 to 107.7 mm, but its 3DPW PVE went from 110.1 to 110.9 mm. So longer training helps on synthetic images but not on real photos, and the student still did not reach the teacher's accuracy. I kept the 30-epoch student. A sweep of the distillation temperature, which softens the features before they are compared, changed the errors only within the run-to-run noise.

For video I added temporal filtering and landmark alignment, without retraining. An exponential moving average blends each tracked person's pose, shape and position with earlier frames, and then the body's position and depth are fitted to detected 2D landmarks. On a walking clip the student kept a tracked body in 356 of 356 frames and the released baseline in 354; these counts show tracking only, not accuracy. By eye the student's output is steadier and better placed, but the clip has no ground truth. The holdout also has no children, the shape on real photos is uncertain and depth errors remain.

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1. Introduction

1.1. Motivation

The problem in this thesis is to turn one photo of several people into a 3D body for each person, using less computation. A body model builds each body as a mesh, a surface of small triangles, from a list of numbers for pose and shape. A 3D body per person is useful, for example, to show how people move or to understand a scene through the people in it.

The first of three difficulties is to recover the whole body: its pose, its shape and its distance from the camera. People can overlap or be partly hidden, and the model still has to find each of them.

Multi-HMR [Bar+24] does this for all people in one forward pass. A Vision Transformer (ViT) encodes the image, a detection head picks one keypoint per person, and a Human Perception Head predicts each body and its distance. NAVER has released a version of Multi-HMR that predicts bodies in Anny, a newer body model that I describe below [Bré+26; NAV26b]. So predicting Anny bodies for several people is not new in my work. I use that released model as the baseline.

The second difficulty is the data. AnnyOne [Bré+26] has synthetic images, made by a computer, so the exact body of every person is known. But these images look different from real photos. 3DPW [Mar+18] has real video frames, but its labels use another body model, SMPL [Lop+15], and my network predicts Anny. Think of SMPL and Anny as two differently built digital mannequins. Before I can train on 3DPW, I have to convert its SMPL labels into Anny's format.

SMPL describes shape with statistical coefficients learned from body scans of adults, and these numbers have no everyday meaning on their own. Most methods use the first 10 of them. Anny instead uses 11 named values, called phenotypes, such as age and height, and it has 163 bones for the whole body (Section 2.1, Table 2.1). So Anny can represent a child directly, but that does not mean a network can recognise a child in a photo.

The third difficulty is cost. A large image encoder gives good features, but it has many parameters and takes longer to run. For this reason I train a smaller student network with help from a larger trained teacher. This is called knowledge distillation: the larger model guides the smaller one during training. The student still handles all people in one forward pass, and I measure how much accuracy it gives up.

1.2. Problem statement and scope

The input is an RGB image and the camera intrinsics (focal length and image centre). Camera information tells the model which direction each image location points into 3D space. For each detected person the model predicts an Anny body: its bone rotations, its phenotype values and the position of one head joint relative to the camera. I call this joint the anchor.

The model I deploy is the ViT-B *Student*. Its teacher is a larger ViT-L model, which my experiment records call *Final*. The teacher is also my accuracy reference, so I compare the student's error with the teacher's. NAVER's released *Multi-HMR-Anny* is the external baseline.

Three questions define the scope:

1. **Real-image supervision.** How can the SMPL labels of 3DPW be converted into Anny labels, and which parts of pose and shape do the converted labels really pin down?

2. **Compact reconstruction.** How much smaller, cheaper and faster is a ViT-B student that starts from the teacher's prediction heads and learns from the teacher's features and outputs? And how much more error does it make than the teacher and the released baseline?

3. **Video.** Can the image-based student run on a clear single-person video? And does smoothing its predictions over frames (temporal filtering) and moving the body onto detected 2D landmarks (landmark alignment) make its output steadier and better placed, without retraining?

I do not train a network that looks at several frames at once. The student always sees one frame at a time. I evaluate reconstruction on AnnyOne and on converted 3DPW. For age and gender I use Relative Human, a set of real photos that come with age and gender tags. I do not collect new shape labels for real images, and I do not try to estimate a person's age in years from a photo.

1.3. Contributions

My work makes four practical contributions:

1. **Anny labels for real images.** A joint-based fitting pipeline converts the standard 3DPW splits into Anny labels. It matches body points that mean the same thing in both models and adds virtual torso joints. I checked the converted labels independently, with mesh overlays on the images and a separate joint comparison.

2. **A compact reconstruction system.** The ViT-B student gets a copy of the teacher's prediction heads at the start of training. A small learned projector turns the student's features into the width those heads expect.

During training the student also imitates the teacher's features and outputs. It has 101.38 M parameters, against 318.38 M in its teacher and 380.94 M in the baseline.

3. **A controlled evaluation.** I compare the three models with the same labels and the same evaluation settings, and I assess pose, placement and shape separately. Two smaller teacher experiments (ablations) test the effect of a consistent head-anchor label and of a fixed bone correction. Two more experiments study the student's training: 30 against 60 epochs, and a sweep of the distillation temperature (Section 5.4).

4. **A video pipeline.** The student and the baseline process the same frames. For the student, an exponential moving average blends each frame's pose, shape and position with the same person's earlier frames. A second step then moves the body and adjusts its depth so that its joints line up with 2D landmarks detected in the image (Section 4.9.1). I also compare the two models on a still photo of several people on a stage.

The student has 3.14 times fewer parameters than its teacher. It also ran 2.10 times faster when I timed both on the same L40S GPU. PVE is the mean distance between matching surface points of the predicted body and the labelled body, after both are aligned at the pelvis. The student's PVE was 119.6 mm on held-out synthetic AnnyOne images and 110.1 mm on converted 3DPW. The teacher stayed more accurate on both, with 73.7 mm and 89.0 mm. On 3DPW the student had a lower PVE than the baseline (126.9 mm). But only my models were trained on 3DPW frames with converted Anny labels, so this does not show a better network design.

Multi-HMR reconstructs several complete bodies from one image. My work builds on this framework with Anny, converted 3DPW labels and a smaller distilled student, and adds temporal filtering and landmark alignment to improve the video outputs.

1.4. Thesis structure

Chapter 2 introduces the two body models and the related work on mesh recovery, knowledge distillation and video. Chapter 3 describes the datasets, how I convert the 3DPW labels and what the converted labels can and cannot test. Both come before the method in Chapter 4, because the method depends on them. Chapter 5 gives the reconstruction and efficiency results, the training length (Section 5.3), the temperature sweep (Section 5.4) and the video results (Section 5.5). Chapter 6 interprets them and Chapter 7 concludes. The appendices give reproduction details, the joint correspondence, settings and diagnostics.

2. Background and related work

My network predicts bodies in one body model, Anny, but the real photos it trains on, 3DPW, are labelled in another, SMPL. I have to convert those labels, so I describe both models before the data. The new parts of my work are listed in Section 1.3.

2.1. The body models SMPL and Anny

A body model turns a few numbers into a 3D mesh of a person, a surface of points (vertices) joined into triangles. The pose $\boldsymbol{\theta}$ says how the body bends and the shape $\boldsymbol{\beta}$ how it is built. The function $M(\boldsymbol{\theta}, \boldsymbol{\beta}) \to \mathbf{V} \in \mathbb{R}^{N\times 3}$ is differentiable, so a network can be trained through it. The shape deforms a template mesh in its rest pose. Then the pose turns a skeleton of bones, and the skin follows by *linear blend skinning* (LBS):

$$\mathbf{v}'_i \;=\; \sum_k w_{ik}\, \mathbf{G}_k(\boldsymbol{\theta}, \boldsymbol{\beta})\, \mathbf{v}_i(\boldsymbol{\beta}), \tag{2.1}$$

where $\mathbf{v}_i(\boldsymbol{\beta})$ is the shaped rest vertex, w_{ik} the skinning weight of bone k and $\mathbf{G}_k$ the bone's world transform, found by chaining the rotations of all bones from the root down to k. In words, every vertex moves with the bones it is attached to, mixed by its weights. Because the transforms chain, a wrong rotation high up in the chain of bones moves everything below it. For example, in Appendix F two neck bones are each turned by half a turn. The head still points the right way, but the skin between them is twisted.

The SMPL family. SMPL [Lop+15], the body model of most earlier work and of the 3DPW labels, has 6,890 vertices and 24 joints. Its shape space was learned from registered scans of adults as principal components, and most methods use the first 10 coefficients, called betas. These are statistical directions, not named properties: no coefficient means "older" or "more muscular". Because the scans are of adults, SMPL represents child proportions poorly. SMIL [Hes+18] is a separate infant model, and AGORA [Pat+21] and BEV [Sun+22] blend the adult template towards an infant one with one extra coefficient. SMPL-H [RTB17] adds MANO hands with 15 joints each, and SMPL-X [Pav+19] also adds a face, for 55 joints on 10,475 vertices. SMPL is released under a research licence. The assets, datasets and trained weights each have their own terms, which have to be checked separately. Multi-HMR, the network my work starts from, was published for SMPL-X [Bar+24]. NAVER's later Anny release replaced its output layer, and my teacher starts from that release's backbone weights and uses its own heads (Appendix A).



Figure 2.1.: *Anny at the mean AnnyOne shape in rest pose, with only the age value changed from 0 to 1. The skeletal height (head joint to mean ankle) grows from 0.53 m to 1.54 m. Body model only, no network; one camera for all panels, so the relative size is visible. The approximate years printed in the panel titles come from one of three age scales that disagree (Section 2.1) and are not used in this thesis.*

Anny. Anny [Bré+26] is a body model by NAVER LABS Europe. It is built from the open MakeHuman project [Mak], an artist-made human model, and not from scans. Its code is under the Apache 2.0 licence and its assets under CC0 [NAV26a]. I use the full rig of 163 bones, because the AnnyOne labels are written for it. The Anny package loads a pruned rig of 104 bones by default, and that rig does not fit the checkpoint or the labels. So the package, its rig and its mesh have to be the ones the checkpoint and the labels were made with. Pose is one rigid transform per bone, applied by Equation (2.1). Of the bones, 88 are helper bones (toes, breasts, small face bones) whose rotation an image cannot show. I keep them at rest, which leaves 75 predicted bones. The neck has three stacked bones, `neck01` to `neck03`, followed by `head`.

Shape is set by 11 *phenotype* values between 0 and 1, such as `gender`, `age`, `weight` and three blend weights `african`, `asian` and `caucasian` (all in Table 2.1). Each value drives blend shapes, artist-made offsets of the rest-mesh vertices. Because the values have names, a predicted body can be read as "older" or "heavier". Gender 0 is the male end and 1 the female end, with rest statures of 1.753 m and 1.632 m. Age blends five anchor bodies spaced evenly from $-\frac{1}{3}$ to 1. Age 0 is the baby, 0.333 the child, 0.667 the young adult and 1 the old adult. The newborn would need an age below 0, which a network output squashed into $(0, 1)$ never gives.

Figure 2.1 shows what age does. At the mean AnnyOne shape, moving age from 0 to 1 raises the skeletal height from 0.53 m to 1.54 m (1.37 m at 0.5), so most of the change happens below about 0.5. A short body alone does not make a child. At age 0.6 the smallest reachable body is 1.22 m tall and still has adult proportions. What marks a child is narrow shoulders relative to the head. The shoulder-to-head width ratio is 2.24 at age 0.1 against 3.00 at 0.6, and only a low age produces it. I never convert ages to years, because three scales in the code disagree. I call a person *child-shaped* if the age value is below about 0.45.

SMPL and Anny side by side. Table 2.1 and Figure 2.2 put the two models side by side. They differ most in the skeleton (24 joints against 163 bones) and in the shape (10 unnamed coefficients against 11 named phenotypes). SMPL stores the pose as one axis-angle rotation per joint, a vector of three numbers whose direction is the axis and whose length is the angle. My network gives each Anny rotation as 6 numbers (the 6D form), which are turned into a valid

	SMPL	Anny
Built from	registered scans of adults	the open MakeHuman project (artist-made)
Mesh	6,890 vertices	one artist-made mesh; vertex count not stated[a]
Skeleton	24 joints (23 + root): 6 on the centre line, 9 on each side	163 bones: face 59, hands 38, toes 28, arms 14, legs 12, spine 5, neck and head 4, breasts 2, root 1
Shape	10 betas: principal components of the scans; no single value has a meaning of its own	11 named phenotypes in $[0, 1]$: gender, age, muscle, weight, height, proportions, cup size, firmness, African, Asian, Caucasian
Pose	one axis-angle rotation per joint, $24 \times 3 = 72$ numbers	one rigid transform per bone
Children	needs a separate infant model or a blend towards one	covered by the age phenotype (baby at 0)
Licence	research licence terms	code Apache 2.0, assets CC0
In this thesis	the 3DPW labels; refitted with Anny on 24 matched points (root and 18 bones move, 144 stay at rest)	what the model predicts: 163 rotations in 6D form ($163 \times 6 = 978$ numbers; 88 helper bones set to rest, 75 used), 11 phenotypes and a 3D position (distance and image offset, 3 numbers)

[a] The vertex count of the Anny mesh depends on the release and on whether unattached vertices are kept; it is not stated here.

Table 2.1.: *The two body models of this thesis. SMPL is the body model of the 3DPW labels and of most earlier work; Anny is the body model the network predicts. Bone counts per region are those of the full rest rig used by the AnnyOne labels and by my model (Figure 2.2).*

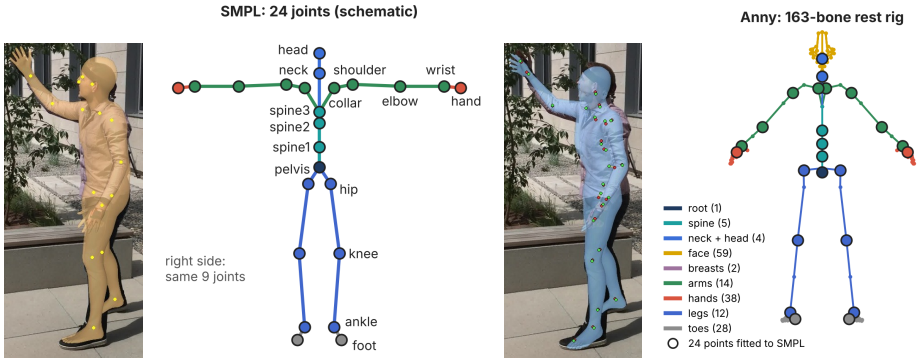
rotation matrix.

The only dataset with Anny labels, AnnyOne, is synthetic, and the real photos I have, 3DPW (Section 2.5), come with SMPL labels. This is why Chapter 3 converts 3DPW.

AnnyOne. AnnyOne is NAVER's synthetic dataset for Anny: computer-rendered scenes of several people, each with exact Anny labels (all bone transforms, all phenotypes and the camera). Its authors describe more than 780,000 images [Bra+26]; the release copy I train on holds 579,960. It is my only source of complete Anny labels, shape included (Section 3.1).

2.2. Image features and attention

A Vision Transformer (ViT) [Dos+21] cuts an image into square patches and turns each into a vector, a token. Attention mixes information between tokens



(a) *SMPL* **(b)** *SMPL, schematic* **(c)** *Anny* **(d)** *Anny rest rig*

Figure 2.2.: *The two body models. (a) The 3DPW SMPL label of one person (courtyard_basketball_01, image 570); the yellow dots are OpenPose 2D points of the image, not SMPL joints. (b) SMPL's 24 joints as a schematic (positions not taken from the model file). (c) My converted Anny label of the same person and frame; red dots are the 3DPW SMPL joints, green dots the Anny joints. (d) The bone heads of Anny's 163-bone rest rig, coloured by region with the number of bones per region; the large dots are the 24 points fitted to the SMPL joints during the conversion. (a) and (c) are labels, not model predictions. Side by side in Table 2.1.*

through queries $\mathbf{Q}$, keys $\mathbf{K}_a$ and values $\mathbf{V}_a$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}_a, \mathbf{V}_a) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_a^\top}{\sqrt{d_k}}\right)\mathbf{V}_a, \tag{2.2}$$

where d_k is the length of a key. In words, each query takes an average of the values, weighted by how well it matches each key. The subscript a keeps these keys apart from the camera intrinsics $\mathbf{K}$ (Chapter 4). Multi-HMR uses self-attention between image patches and cross-attention, where a person query reads from all image tokens. Both backbones are DINOv2 [ODM+23] ViTs as NAVER released them with Multi-HMR, a ViT-L for the teacher and a ViT-B for the student. The ViT-B produces shorter token vectors than the ViT-L, so the student needs a projector (Section 2.6).

2.3. Single-person and multi-person mesh recovery

Human mesh recovery (HMR) estimates the mesh of a body model from an image.

Regression and fitting. HMR [Kan+18] regressed body-model parameters for one cropped person with a convolutional encoder, an iterative regressor and an adversarial pose prior (a second network that tells real poses from impossible ones). HMR 2.0 [Goe+23] replaced the encoder by a ViT and showed that a large ViT trained on a broad mix of datasets works well without

further tricks. Multi-HMR uses the same recipe, and so does my model. The network outputs the parameters directly, with no optimisation at test time. SMPLify [Bog+16] goes the other way and fits SMPL to 2D keypoints by optimisation, and SPIN [Kol+19] puts such a fit inside the training loop. My 3DPW converter (Chapter 3) is an optimisation of the SMPLify kind, but in 3D, between two body models, and run once offline.

Single-shot multi-person methods. Two-stage methods run a person detector and then a single-person network on each crop, so the cost grows with the number of people. ROMP [Sun+21] does both in one pass (single-shot): it predicts a heatmap of body centres and reads the body parameters at the detected centres. Multi-HMR [Bar+24], the architecture of my work, keeps this idea. Its ViT turns the image into one token per 14×14-pixel patch, which gives a 48×48 grid of tokens for the 672×672-pixel input. Each token scores whether a person's primary keypoint, the head joint in the released code [NAV26b], lies in it. High-scoring tokens become the queries of the *Human Perception Head* (HPH), which attends to all image tokens and predicts the body parameters and the distance of the person's head joint from the camera. I call this head joint the anchor. It is at the centre of the depth defect in Appendix F. Section 4.2.2 gives the details.

The released SMPL-X checkpoints reach a PVE (mean vertex distance, Section 4.8) of 94.1 mm on the 3DPW test set with ViT-L in 74 ms, and 94.0 mm with ViT-B in 43 ms on a V100 GPU [NAV26b]. The ViT-B is as accurate and faster, which is why my smaller model is a ViT-B student. These are SMPL-X errors and cannot be compared with the Anny-space errors of this thesis.

BEV and Relative Human. BEV [Sun+22] extends ROMP in two ways: it also predicts how far each person stands from the camera, and its body model has an age parameter. Its authors collected Relative Human, real photos in which every person has an age group (adult, teenager, kid, baby), a depth layer and keypoints, but no 3D body. It is the only real data with age labels I could use, and I use it for evaluation only (Section 3.7).

Multi-HMR-Anny and Anny-Fit. NAVER has released a Multi-HMR checkpoint that predicts Anny, trained on BEDLAM, AnnyOne and further real images [NAV26b]. So a multi-person network for Anny already exists, and I do not claim to be the first. This checkpoint, Multi-HMR-Anny, gives my teacher's backbone weights and is the baseline I compare against. On the same backbone its decoder has 8 cross-attention blocks and 76.57 M parameters, against 2 blocks and 14.01 M in mine. It was trained on AnnyOne, so it has most likely seen my AnnyOne test images during training (Section 5.2). Anny-Fit [Bra+26], by the same group, refines such predictions per image by optimisation and adapts them to children at test time. My model stays a single forward pass. I did not use Anny-Fit. For future work on children it would be the first method to compare against.

2.4. Synthetic training data

AGORA [Pat+21] renders scanned people with SMPL-X fits into realistic scenes, and BEDLAM [Bla+23] showed that synthetic images alone can train competitive networks. AnnyOne does the same for Anny. Synthetic labels are exact and complete, but the images look different from photos, so a network trained on them can work less well on real photos. In my experiments the shape predictions followed the labels on the synthetic holdout, the AnnyOne images I keep out of training, but collapsed towards the training average on real photos (Section 5.7.1). This is why Final, my teacher, mixes 30 % real images into every batch, and why those images need Anny labels.

2.5. Real-data labels for a new body model

3DPW [Mar+18] is the standard real-world test set for mesh recovery: 60 video sequences with SMPL labels for one or two actors each. SMPL and Anny have different skeletons and different shape spaces, so I have to refit the labels in Anny. The SMPL-X code transfers fits between SMPL-family models through a vertex correspondence [Pav+19], and the Anny package has an inverter that fits Anny to a target mesh [NAV26a]. Neither turns a 3DPW label into an Anny label directly, so I wrote my own converter (Section 3.3). My fit uses only joint positions, so it can only recover what the joint positions fix. Section 3.6 lists what the converted labels can and cannot test. SHAPY [Cho+22] instead supervises shape on real images with body measurements and word attributes. Anny's named phenotypes would suit such labels, but I collected none.

2.6. Knowledge distillation

A large network is accurate but slow. Knowledge distillation [HVD15] trains a small *student* to imitate a large, trained *teacher*, originally by matching its class probabilities softened by a temperature. FitNets [Rom+15] let the student also match an intermediate representation of the teacher through a small learned regressor. DINOv2 [ODM+23] distilled its smaller ViTs from its largest one, so a DINOv2 ViT-B is a good starting point for a student. My Student uses both forms and matches the teacher's image features and outputs (Section 4.5). The Student has 101.38 M parameters and Final has 318.38 M. What I want to know is whether the Student keeps at least 95 % of Final's accuracy on both test sets.

The projector. The student starts with a copy of the teacher's prediction heads. Its features have 768 channels, but the teacher's features and the copied heads use 1024. A projector, one linear layer, converts one width into the other, so the student's features can be compared with the teacher's and fed into the copied heads. Because the heads read its output, the projector stays in the deployed student.

The temperature. After the projector, each patch token is a list of 1024 channel values. Teacher and student tokens are compared in three steps. First, the values are divided by the temperature T. Then a softmax turns them into positive weights that add up to 1. Finally, the Kullback–Leibler (KL) divergence measures how differently teacher and student spread these weights. It is zero when they are equal.

A larger T makes the weights softer. For example, two channels with weights [0.88, 0.12] at $T = 1$ get about [0.62, 0.38] at $T = 4$ (an illustration, not a measurement). Softer weights let the student also learn from the smaller channel responses. The weights describe feature patterns, not probabilities of body parts. Dividing by T shrinks the gradients by about T^2, so the loss is multiplied by T^2 to keep the gradient size comparable across temperatures [HVD15].

The output distillation is a weighted L1 loss. It takes the absolute differences between the student's and the teacher's rotations, phenotypes and distances and weights each term as in the supervised loss. It has no temperature. The reported Student uses $T = 4$. Section 5.4 tries other values.

2.7. Temporal models for video

Most video methods refine the per-frame features of an image network: HMMR [Kan+19] with a temporal convolution, VIBE [KAB20] with a recurrent network and a motion discriminator tied to SMPL, and TCMR [Cho+21] with branches that predict the current frame from its neighbours to reduce jitter (small frame-to-frame shaking). MPS-Net [Wei+22] and GLoT [She+23] use attention over frames. SmoothNet [Zen+22] is a learned network that smooths the output poses of any estimator over a window of frames. PHALP [Raj+22] and 4DHumans [Goe+23] track people through video, the first step of any multi-person video method, and WHAM [Shi+24] places the motion in world coordinates.

My student sees one frame at a time, so the body it predicts can shake or change size between frames. I keep it frame-wise and add two steps at inference time, without retraining (Sections 4.9 and 4.9.1). First, an exponential moving average blends each tracked person's new bone rotations, phenotypes and position with the earlier ones. Like SmoothNet, it smooths outputs without image features. But it is a fixed formula, not a learned network, it is causal (only the current and earlier frames count), and it works on body-model parameters. Second, the pipeline moves the body's position and depth so that they match 2D landmarks from a separate detector. The learned methods above remain future work.

2.8. How my work relates to prior work

My method keeps the single-shot design of Multi-HMR. It is smaller because it uses a smaller image encoder, with a projector and copied heads to connect it to the teacher. I did not change the attention itself. Table 2.2 relates the prior work to mine.

Approach	Relevant idea	Relation to this thesis
Multi-HMR and its released Anny model [Bar+24; NAV26b]	single-shot multi-person detection and body regression	framework; teacher's backbone weights; baseline
FitNets and knowledge distillation [Rom+15; HVD15]	feature and output transfer from a teacher	the student's projected feature matching and output distillation
BEV [Sun+22]	age-aware reconstruction and real-photo age-group labels	reason to evaluate child proportions; not used in the student
VIBE, TCMR and SmoothNet [KAB20; Cho+21; Zen+22]	temporal features or output refinement	SmoothNet is the closest relative of my inference-time filter; the learned temporal models are future work

Table 2.2.: *Prior work and how it relates to this thesis.*

3. Datasets and Anny label generation

Every number in this thesis comes from one of the datasets below, and I describe them before the method. Table 3.1 lists each split, its size and its use. Most of the chapter is about converting 3DPW to Anny, because every real-image number in this thesis is measured against those converted labels. I follow two rules in every experiment. No test image is ever trained on, and I never put a number on the old 3DPW labels next to a number on the final v23 labels.

split	size (images / people)	used for	chapters
AnnyOne training	579,860 images	training of every model; 70% of each mixed batch	4, 5
AnnyOne holdout	100 images	headline synthetic evaluation (V1); never trained on	F, 5
diagnostic manifest	40 + 15 entries	per-person diagnostics; excluded from every fine-tune	F
3DPW-Anny train, v23	5,882 / 7,584	30% of each mixed batch	4, 5
3DPW-Anny val., v23	2,622 / 3,311	300 images scored after each epoch; monitoring only	5
3DPW-Anny test, v23	8,190 / 11,854	reported 3DPW numbers; never trained on	5
3DPW-Anny old labels	16,631 / 20,028	historical experiments only; not used by Student	E
Relative Human val.	635 / 2,130	children and gender on real photos; evaluation only	F, 5
demo images	12 images	by-eye comparison; no metric	F, 5
walking video	356 frames	`man_walking.mp4`; video comparison, no labels	5

Table 3.1.: *Every split of the thesis, its size and its use. 3DPW-Anny sizes are images / labelled people after the drop rules (v23 labels, Section G.2). The reported teacher and 30-epoch Student use the last checkpoints of their schedules; validation only monitors training.*

3.1. AnnyOne

AnnyOne [Bré+26] is the synthetic dataset released with Anny. It holds 579,960 rendered images of 1280×1280 pixels, with 2.73 people per image on average. Because the images are rendered, every visible person has an exact label: the Anny pose (163 bone transforms), the 11 phenotypes (Anny's shape values, Section 2.1) and the camera intrinsics (focal length and image centre).

The network sees each image resized to 672 pixels. The field of view changes from image to image, from 51° to 110° on a small sample of five images. For this reason training and evaluation use each image's own intrinsics.

Split. AnnyOne has no official split, so I made one. Indices 0 to 579,859 are the training set. The last 100 images, indices 579,860 to 579,959, are my holdout, which no model trained in this thesis has seen. NAVER's model was trained on the AnnyOne release, which includes the 100 images I hold out. Its 60.3 mm holdout PVE (the mean vertex error, Section 4.8) is therefore most likely an in-sample number, in other words an error on images it had already seen during training. My models have never seen these images.

The 3DPW test script also scores the last 100 AnnyOne images, but in setting V0. In V0 the error counts Anny's helper bones (toes, breasts, face and subdivision bones); in my headline setting V1 they are masked out (Section 4.8). The script's "AnnyOne test-100" value is therefore not the holdout number, which I always give in setting V1.

Few children. In Anny the child shape appears below an age value of about 0.4 to 0.5 (Section 2.1). Only 2.53% of AnnyOne's people are younger than 0.3 (1.06% below 0.2, 4.45% below 0.4). The median age is 0.601, and 41% of all people lie between 0.5 and 0.6. In a training batch, 2.1% to 2.3% of the supervised people are younger than 0.3. The holdout has nobody younger than 0.513, which means it tests adults only. This is why I added Relative Human (Section 3.7).

3.2. 3DPW

3DPW [Mar+18] is a real video dataset of 60 sequences filmed with a hand-held phone camera at 1080×1920 pixels (portrait) and 30 frames per second. Inertial sensors on the actors give SMPL ground truth. The standard release has 24 training, 12 validation and 24 test sequences, with one or two labelled actors each: 87 person-sequences (62 men, 25 women). Bystanders have no labels. I keep the splits as 3DPW defines them. One validation sequence, `outdoors_crosscountry_00`, gives no labels at all (Section G.2), and 11 validation sequences remain.

Why 3DPW must be converted. My network predicts Anny bodies. AnnyOne is the only dataset with Anny labels, and it is synthetic. 3DPW has real photos, but its labels are SMPL bodies, which have a different skeleton and different shape controls (Table 2.1). For this reason I fit an Anny body to every labelled 3DPW person and write new labels in AnnyOne's format.

3.3. Converting SMPL to Anny

My converter (`smpl_to_anny.py`, version 2.3) works in four steps:

1. Read the source labels: joint positions, shape, camera and frame number of each person.
2. Fit an Anny body: move, turn and bend it until its joints match the 3DPW joints, first the root, then the bone rotations, finally the free shape values.
3. Keep one body shape per person: the median shape over the sequence, so the proportions stay the same through the video.
4. Build the image labels: refit every frame's pose with that shape, move the body into camera coordinates and save one label per image.

Objective. There is no formula that turns joint positions back into Anny's pose and shape (Anny's skinning has no analytic inverse). Instead I fit them by gradient descent: the fitter starts from a guess, measures the joint mismatch and changes the values to reduce it. For each person and frame I search Anny's 163 rotations $\mathbf{r}$ (6D form, two columns of each rotation matrix), the phenotype logits $\boldsymbol{\ell}$ (values that the sigmoid σ maps into $(0,1)$) and a translation $\mathbf{t}$ so that Anny's joints land on the 24 SMPL joints $\hat{\mathbf{J}}^S$ of the frame:

$$\begin{aligned}\min_{\mathbf{r},\boldsymbol{\ell},\mathbf{t}} \frac{1}{24}\sum_{k=1}^{24}\left\|\mathbf{p}_{\pi(k)}(\mathbf{r},\sigma(\boldsymbol{\ell}))+\mathbf{t}-\hat{\mathbf{J}}^S_k\right\|_2^2 \\ +\lambda_r\sum_{b\neq\text{root}}\|\mathbf{R}_b-\mathbf{I}\|_F^2+\lambda_\phi\sum_{j\in\text{free}}\left(\sigma(\ell_j)-\tfrac{1}{2}\right)^2,\end{aligned} \tag{3.1}$$

where π is the joint correspondence, $\mathbf{p}_b$ the point attached to bone b (its head, or a virtual joint, Section 3.4) and $\mathbf{R}_b$ the bone's rotation matrix. In words, the joints of the Anny body must land on 3DPW's joints. The two extra terms penalise twists that move no joint and extreme shape values.

The first regulariser ($\lambda_r = 10^{-4}$) pulls every non-root rotation towards identity, because a twist about a bone's own axis moves no joint but corkscrews the mesh. The second ($\lambda_\phi = 10^{-3}$) pulls the free phenotypes towards 0.5, so that one remaining joint gap cannot drag a phenotype to an extreme. Adam (learning rate 0.05) fits in the order of step 2. Shape comes last because it is the weakest signal. Each frame starts from the previous frame's solution (150 / 400 / 250 iterations per stage on a person's first frame, 20 / 80 / 60 afterwards). The pose refit of step 4 takes 100 iterations per frame. The targets are 3DPW's own per-frame `jointPositions`. I do not recompute them with the neutral SMPL model, because 3DPW's shape coefficients belong to the gendered models, and the neutral model would move the joints by 37 mm on average.

Bones and shape. Only the root and 18 of the 23 mapped bones may rotate; 144 of 163 bones stay at rest, among them `neck02`, `neck03`, `head` and every finger bone. The joint positions say nothing about most of the phenotypes. Weight, muscle, cup size, firmness and the three ethnicity values move no joint,

and adult age barely does. The converter therefore reads gender from 3DPW, fixes age at 0.55 (an adult value) and the seven others at 0.5, and fits only height and proportions. Since version 2.3 it writes men as 0 and women as 1, Anny's convention (`-gender_convention anny`).

Frames, camera and label files. I fit every third valid frame (10 frames per second). Since version 2.3 one grid holds for every person of a sequence (frame numbers divisible by 3, `-frame_grid shared`), so both actors are fitted on the same images. This matters for multi-person detection, because all labelled actors must appear in the same image label. The label builder writes one file per image in the AnnyOne format and checks that the image exists. It drops a person when the fit error exceeds 80 mm, less than 30% of the mesh is inside the image, or the box is narrower or shorter than 40 px. A dataset class pads the portrait images to a square and shifts the principal point (the image centre in the intrinsics). The training and evaluation code then run on these labels unchanged.

3.4. Geometric correspondence and virtual joints

SMPL and Anny define some joints at different anatomical points. For example, both have a hip joint, but not at exactly the same place, and matching the two directly would make the fitter distort the body to close that gap. This means I cannot pair joints by name alone. Instead my final procedure puts both skeletons in their rest pose and lines them up at the hip midpoint and along the body axes. It then chooses each Anny hip bone by distance and each spine bone by its normalised height along the torso. Appendix B gives the full correspondence.

For a torso target k attached to Anny bone $b = \pi(k)$, the fitter uses a virtual joint

$$\mathbf{p}_b(\mathbf{R}, \boldsymbol{\phi}) = \mathbf{h}_b(\mathbf{R}, \boldsymbol{\phi}) + \mathbf{Q}_b(\mathbf{R}, \boldsymbol{\phi})\boldsymbol{\delta}_k, \tag{3.2}$$

where $\mathbf{h}_b$ is the posed bone head, $\mathbf{Q}_b$ its orientation, and $\boldsymbol{\delta}_k$ a constant offset in the bone's rest frame, measured from the aligned rest skeletons. In words, the point used for fitting is moved by a fixed offset that turns with the bone. The two skeletons are then compared at the same body location, and Anny's rig stays intact. Limb targets use the bone head itself. The offsets only shrink the difference in where the two models put a joint; SMPL and Anny are still two anatomically different bodies.

3.5. Label construction and validation

I call the final labels v23. They use the gender convention and the shared frame grid of version 2.3, and every label has its image. Each actor has one shape for the whole sequence, and the pose changes per frame. Table 3.2 compares the v23 labels with the two earlier sets.

	old labels	rebuilt	v23 labels
converter fits	v2.2	v2.2	v2.3
image files of 8 sequences	missing	re-extracted	re-extracted
train, images / people	4,451 / 5,183	6,437 / 7,589	5,882 / 7,584
validation	2,458 / 2,995	2,777 / 3,314	2,622 / 3,311
test	9,722 / 11,850	9,722 / 11,850	8,190 / 11,854
total	16,631 / 20,028	18,936 / 22,753	16,694 / 22,749
gender reversed	yes	yes	no
actor missing in two-person frames	yes	yes	no
labels missing (no image file)	yes	no	no
used for training	history models only	never	Final, Control, Student

Table 3.2.: *The three 3DPW-Anny label sets. All are built at every third frame with the same drop rules. The v23 labels have fewer images than the rebuilt set but the same people, because both actors of a two-person sequence now share each frame.* `outdoors_crosscountry_00` *yields no labels in any set (Section G.2). Numbers measured on the old labels are retired and never appear next to numbers on the v23 labels.*

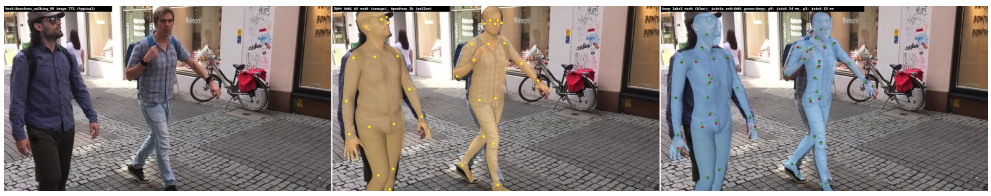


Figure 3.1.: *A label-conversion example, not a network prediction: 3DPW test frame (*`downtown_walking_00`*, image 771), v23 labels. Left: image. Middle: 3DPW's SMPL label (orange; the yellow dots are OpenPose 2D keypoints). Right: converted Anny label (blue; red dots 3DPW's SMPL joints, green dots Anny joints). The check's median body-joint discrepancy over all checked labels is 14.1 mm. By eye, one frame.*

The optimiser's loss only says how well the fit met its own objective. It cannot tell me whether the finished labels are right. For this reason I run a separate check that reads the finished labels and compares four things with 3DPW's source data: the body-joint positions, the projected keypoints, whether every actor is present, and the gender convention. The check samples images and cannot prove that every label is free of error. On the v23 labels the median body-joint discrepancy was 14.1 mm, and the check's overall grade was WARN (Table 3.3 lists the grades). The warnings that remain come from limb proportions and from joints that SMPL and Anny define differently, the hip among them. They are not the defects of the earlier label sets, which the re-fit corrected (Appendix G).

I also looked at the labels. In Figure 3.1 the converted Anny mesh covers the same two people as 3DPW's SMPL mesh, and the SMPL joints (red) and Anny joints (green) lie close together. This is one frame judged by eye.

check	old labels	v23 labels
overall grade	FAIL	WARN
body joints vs 3DPW, median / p95 (mm)	18.7 / 24.5	14.1 / 21.1
surface, Anny→SMPL / SMPL→Anny (mm)	18.9 / 18.3	18.9 / 16.1
2D error over raw 3DPW (px)	+1.1	+0.4
gender labels	reversed	men 0, women 1
refit with gender flipped	better for 72 of 73 men	worse: men 13.9 → 20.0 mm
two-person images, an actor unlabelled	407 of 960	0 of 1,080
fitted frames; median frame error (mm)	22,915; 18.2	22,912; 14.0
head turn vs chest, labels / SMPL (°)	9.5 / 20.2	19.6 / 19.3
post-check of the re-fit, all fitted frames, 87 people		
mean joint error, men / women (mm)	19.0 / 15.0	14.7 / 15.2
two-person frames, a valid actor unfitted	4,457 of 8,274	0 of 6,055
still WARN on the v23 labels		
limb ratio Anny / SMPL, forearm / thigh		0.948 / 1.064
people with a joint more than 100 mm off		3.2%

Table 3.3.: *Independent conversion check of the 3DPW-Anny labels before (old labels, v2.2 fits) and after the re-fit (v23 labels, v2.3 fits). The check samples images from every sequence and compares the labels with 3DPW's own joints, surface and 2D keypoints. The middle block is the re-fit's own post-check over all fitted frames; old and new fits fitted different frames, so it is not a paired comparison.*

The main experiments use only the v23 labels. The earlier converter versions and the checks that found the label defects are in Appendix G.

3.6. What the 3DPW labels can and cannot test

The v23 labels are converted labels, not ground truth. The way the converter is built limits what they can test, in four ways.

Eight constant phenotypes. The labels test three phenotypes: gender (from metadata), height and proportions (fitted; medians 0.47 and 0.51). The other 8 are the same for every person: age 0.55 and seven values at 0.5. A model trained on 3DPW is pulled towards these constants for every real person, by the shape loss and the mesh terms. By my estimate, 3DPW people are about 16% of the supervised people in a 70/30 batch. The 3DPW error is also measured against these neutral-body meshes. I expect that it rewards a neutral prediction, but I did not test this. The shape mask of Section 4.4 removes the shape loss on the constants (Final-masked). At age 0.55 every label is a full adult (Figure 2.1). 3DPW has no children, and these labels can neither teach nor test them.

Head and hands at rest. `neck02`, `neck03` and `head` stay at identity in every label. The head turns only through `neck01`, and the labels cannot test the

half-turned neck bones of Appendix F. 3DPW's SMPL ground truth has no finger joints, so every label has rest-pose hands. The training masks their rotation terms for 3DPW people (Section 4.4).

Residual label error. The checked labels differ from 3DPW's source body joints by 14.1 mm at the median and 21.1 mm at the 95th percentile. This number describes label quality. A network is compared with the fitted Anny targets and not with the source joints. The residual is therefore no universal error floor, and it must not be subtracted from a network's PVE (Section 4.8).

Not comparable with published numbers. I never evaluated against the original SMPL meshes, my targets carry the residual error above, and my MPJPE (the mean joint error) averages all 163 Anny joints (Section 4.8). For these three reasons my 3DPW numbers cannot be compared with published 3DPW results, and I write "lowest error on our 3DPW-Anny labels". Bystanders have no labels, and a model that detects one of them correctly still counts as a false detection. For this reason the precision I measure on 3DPW (the share of detections that match a labelled person) is only a lower bound.

3.7. Relative Human

Neither the holdout nor 3DPW contains children, and I needed a third dataset for them. Relative Human, introduced with BEV [Sun+22], is such a dataset. It holds 8,741 real photos, many crowded and with children. Each person has a box, an age group (adult, teen, kid, baby) and a gender coded 0 for male, as in Anny; there are no 3D labels. The copy I use came without a licence file. Its redistribution terms still need checking. I use only the validation split, for evaluation: 635 images with 2,130 labelled people (1,680 adults, 101 teens, 209 kids, 117 babies, 23 of unknown age).

The model runs once on each whole photo with a field of view of 55°, and detections are paired with labelled boxes as in Section 4.8. A predicted age below 0.45 counts as child-shaped, a predicted gender below 0.5 as male. For each age group I report recall (the share of labelled people found) and the mean predicted age. I also report the gender accuracy and the child-against-adult AUC of the predicted age. This AUC is the chance that a kid or baby gets a lower predicted age than an adult; a value of 0.5 means chance level. Every number covers detected people only, and recall always stands next to it.

3.8. The diagnostic manifest and the demo images

For the checks of Appendix F I picked 55 fixed (image, person) entries from the AnnyOne labels alone, with a script and a fixed random seed, so the choice can be repeated: 40 from the training range and 15 from the holdout (groups in Table F.1). Base, the AnnyOne-only model that every later run starts from, saw the training entries during its own training. Every later run gives their

images a sampling weight of zero, and no model sees them again. This keeps them the same already-seen reference for all models. The manifest children therefore show whether a model can fit a child it has seen, not whether it generalises.

For by-eye comparisons I use 12 real images: 4 photos and 8 3DPW test frames, one middle frame from each of 8 test sequences. No model trained on them and no metric is computed on them. The photo comparison with Multi-HMR-Anny (Section 5.5.1) and the walking video (Section 5.5) have no labels either. I can only show their meshes, and no reconstruction error can be computed for them.

4. Methodology

The model I deploy is the ViT-B Student. It learns from a larger ViT-L teacher, which my experiment records call Final. The teacher gives the student two things: a copy of its prediction heads at the start of training, and targets to imitate during training.

4.1. System overview

The real 3DPW training images come with SMPL labels, and the converter of Chapter 3 turns these into Anny labels. With these labels I train the teacher on synthetic and real images together. The student then learns from the labels and from the teacher, which stays frozen, meaning training does not update its weights. Only the student runs at deployment (Figure 4.1).

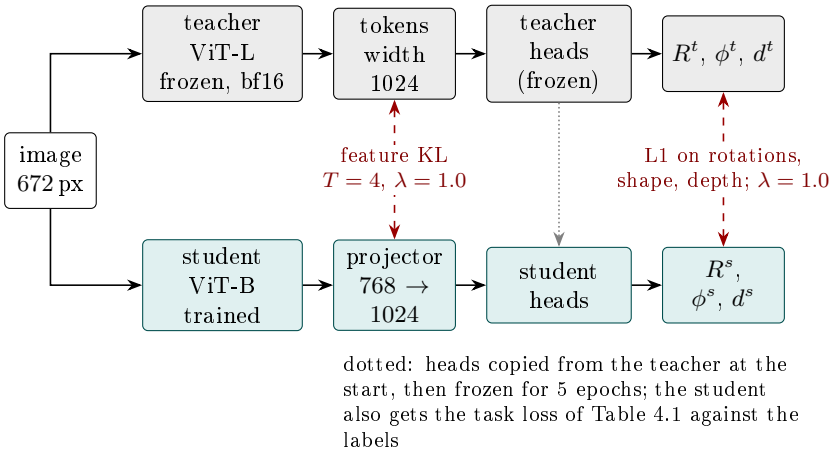


Figure 4.1.: *Teacher–student training (schematic). Both branches see the same image. The student's ViT-B features pass through a 768-to-1024 projector into heads copied from the frozen teacher, and the student is trained with the supervised and two distillation losses. At inference only the student runs; its projector stays, because the heads read its output.*

The student's ViT-B encoder f_B turns an image I into N_p tokens of $C_s = 768$ channels, $\mathbf{Z}^s = f_B(I)$. A trainable linear projector maps them to the teacher's width:

$$\widetilde{\mathbf{Z}}^s = \mathbf{Z}^s\mathbf{W}^\top + \mathbf{b}, \qquad \mathbf{W} \in \mathbb{R}^{C_t \times C_s}, \quad C_t = 1024. \tag{4.1}$$

So each student token gets the 1024 channels that the teacher's heads read, and this is what makes it possible to copy the heads. Most of the saving in

parameters and running time comes from the smaller encoder. Using a ViT-B encoder, or distillation, is not new in itself.

4.2. Multi-HMR for Anny

4.2.1. The task

The input is one RGB image with camera intrinsics $\mathbf{K}$, the matrix of focal length and image centre. It shows an unknown number of people. For each person i the network predicts the rotations $\mathbf{R}_i \in SO(3)^{163}$ of Anny's bones, the phenotypes $\boldsymbol{\phi}_i \in [0,1]^{11}$, and the image position (u_i, v_i) and depth z_i of one anchor joint a, the head. The Anny body model M [Bré+26] turns $(\mathbf{R}_i, \boldsymbol{\phi}_i)$ into vertices M_V and joints, and the mesh is moved so that its anchor joint $M_{J,a}$ sits at the back-projected anchor pixel:

$$\mathbf{V}_i = M_V(\mathbf{R}_i, \boldsymbol{\phi}_i) - M_{J,a}(\mathbf{R}_i, \boldsymbol{\phi}_i) + \mathbf{t}_i, \qquad \mathbf{t}_i = z_i\, \mathbf{K}^{-1}\,(u_i, v_i, 1)^\top. \tag{4.2}$$

So the head is first moved to the origin, and then the whole body is shifted so that the head lands at depth z_i on the camera ray through its pixel.

4.2.2. Architecture

The teacher follows Multi-HMR [Bar+24] with outputs for Anny (Figure 4.2). A DINOv2 ViT-L [ODM+23] with weights from NAVER's Anny checkpoint turns the image into tokens. One MLP scores each token for the anchor joint (the head, `--person_center head`), and a second predicts the anchor's 2D offset inside the token. Each detected token becomes a query of the Human Perception Head (HPH, Section 2.3), which lets the people attend to each other and to all image tokens.

The Anny head is the part that differs from the SMPL-X version. Each rotation is predicted as 6 numbers, which Gram–Schmidt orthogonalisation turns into a valid rotation matrix [Zho+19]. A sigmoid keeps the phenotypes between 0 and 1, and the depth is Multi-HMR's logarithm of the distance, normalised by the focal length. Right after Gram–Schmidt, four bones are multiplied by one constant rotation each (Section 4.7.2). Then a fixed mask sets 88 helper bones (toes, breasts, face and subdivision bones) to the identity, both in the prediction and in the label, and keeps the other 75 bones.

The Anny layer runs inside the network. It uses the same full rig as the code that builds the training targets, and it places the body with Equation (4.2). This means the 3D and 2D losses are computed on the mesh, and their gradients flow back into the network through this layer. The student keeps everything after the encoder, replaces the encoder by ViT-B and adds the projector (sizes and cost in Table 4.2).

4.2.3. What the model predicts at inference

At inference there are no labels, so the model estimates everything from the image and $\mathbf{K}$. Camera information tells the model which direction each image

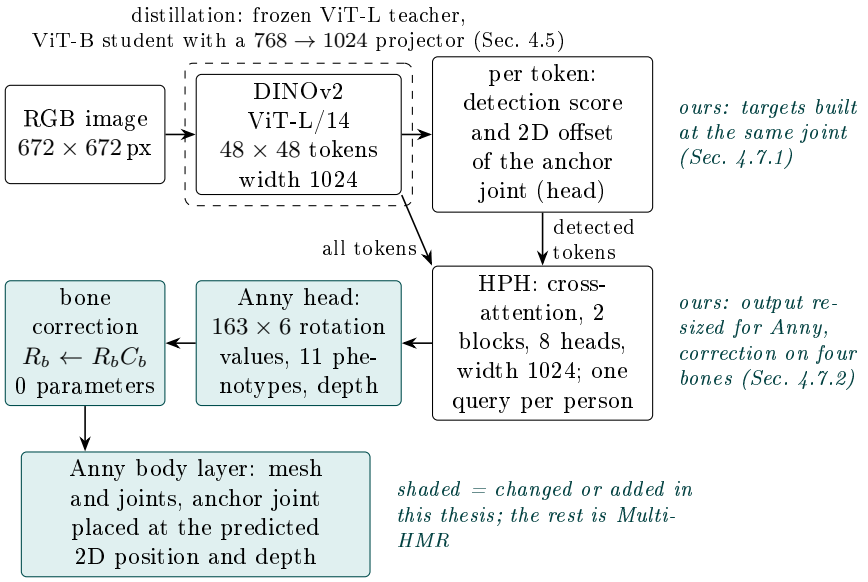


Figure 4.2.: *Teacher architecture (schematic): 672-pixel input, 48 × 48 ViT-L tokens with a head-anchor score and offset, then per person 163 × 6 rotation values, 11 phenotypes and a depth, the constant correction of four bones and the Anny layer. Shaded boxes are new in this thesis; the dashed box is what distillation replaces.*

location points into 3D space. When the calibration is unknown, as for the demo photos and the video, I build **K** from an assumed field of view, and a wrong guess changes the recovered depth. Per detected person the network gives a detection score, a 2D offset, 163 bone rotations (how the body bends), 11 phenotypes (its proportions) and the head-anchor depth. Anny turns these into a mesh and joints, placed by Equation (4.2).

4.3. Supervised training objective

A loss is a numerical penalty for a mismatch during training. The target vertices and joints come from running Anny on each person's true parameters, and the anchor targets come from one joint, the head (Section 4.7.1). The total loss is

$$\begin{aligned}\mathcal{L} = {} & \alpha_{\text{det}}\mathcal{L}_{\text{det}} + \alpha_{\text{off}}\mathcal{L}_{\text{off}} + \alpha_{\text{rot}}\mathcal{L}_{\text{rot}} + \alpha_{\text{shape}}\mathcal{L}_{\text{shape}} \\ & + \alpha_{\text{dist}}\mathcal{L}_{\text{dist}} + \alpha_{\text{t}}\mathcal{L}_{\text{t}} + \alpha_{\text{j3d}}\mathcal{L}_{\text{j3d}} + \alpha_{\text{v3d}}\mathcal{L}_{\text{v3d}} \\ & + \mathbb{1}[e \geq e_{\text{2D}}]\,(\alpha_{\text{j2d}}\mathcal{L}_{\text{j2d}} + \alpha_{\text{v2d}}\mathcal{L}_{\text{v2d}}) + \lambda_{\text{kd}}\mathcal{L}_{\text{kd}} + \lambda_{\text{kd-out}}\mathcal{L}_{\text{kd-out}},\end{aligned} \tag{4.3}$$

with e the epoch, e_{2D} the first epoch of the 2D terms, and weights α and λ; the two distillation terms exist only for a student. So every wrong part of the prediction costs something, and the weights decide how much. Table 4.1 gives what each term compares and its weights.

Term (log name)	Supervised quantity	Default	Final	Control	Student
Detection (`bce`)	focal loss on the per-token score "an anchor lies here"	10	10	10	10
Offset (`offset`)	L1 on the anchor's position inside its token	1	1	1	1
Rotations (`rotmat`)	L1 over the 9 entries of each predicted/target bone rotation	0.1	0.1	0.1	0.1
neck and hand boost	renormalised: $\times 2.41$ neck, head, hands; $\times 0.60$ rest	1	4	4	4
Shape (`shape`)	L1 on the 11 phenotypes after the sigmoid	1	20	20	20
Distance (`dist`)	L1 on the anchor's focal-normalised log-depth	1	5	1	5
Translation (`transl`)	L1 on the anchor's 3D position in the camera frame	1	5	1	5
3D joints (`j3d`)	L1 on the 163 joints, each at its own pelvis	100	100	100	100
3D vertices (`v3d`)	the same for the mesh vertices	100	100	100	100
2D joints (`j2d`)	L1 in pixels on the joints projected from anchor and depth	1	0.1	0.1	0.1
2D vertices (`v2d`)	the same for the projected vertices	1	0.1	0.1	0.1
first 2D epoch	`--start_2d_epoch`	10	0	0	0
Global rot. (`global_rot`)	L1 on camera-frame head, neck03, wrist orientation	0	0	0	0
Feature KD (`kd`)	KL of softened student and teacher tokens, $T = 4$	–	–	–	1.0
Output KD (`kd_out`)	L1 to the teacher's rotations, phenotypes, depth	–	–	–	1.0
Target joint	`--gt_center_joint` (four anchor targets)	pelvis	head	pelvis	head
Bone correction	neck02, head, metacarpal1.L, finger4-1.R	none	on	none	on

Table 4.1.: *Loss terms and their weights. "Default" is the value in `loss.py`; Final, Control and Student are the values of those runs (model names in Table 4.3). The total loss is the weighted sum of all rows above the rule. In plain words, detection and offset find each person, rotations and shape get the posture and the proportions right, distance and translation put the body at the right distance and place in the image, the 3D terms match the body geometry, the 2D terms make the mesh line up with the photo, and the two KD terms make the student imitate the teacher. For people from 3DPW the rotation and global-rotation terms skip the 38 finger, thumb and metacarpal bones in all three runs; the mesh terms are not masked. The shape mask for 3DPW people is off in all three runs and on only in Final-masked (Section 4.4). The Student keeps the teacher's bone correction during training and is evaluated with its own, re-estimated after training (Section 4.6).*

4.3.1. Detection and anchor localisation

Let s_q be the detection probability at token q, $y_q \in \{0, 1\}$ its target and $N_+ = \sum_q y_q$. The detection term is the focal loss of CornerNet [LD18]:

$$\mathcal{L}_{\text{det}} = -\frac{1}{\max(1, N_+)} \sum_q \begin{cases} (1 - s_q)^2 \log(s_q + \epsilon), & y_q = 1, \\ s_q^2 \log(1 - s_q + \epsilon), & y_q = 0. \end{cases} \tag{4.4}$$

The sum runs over all tokens of the batch. A positive token is one that holds a head anchor. The squared factors make the easy background tokens count less. The loss is large when the network misses a person or fires on background. For N_h supervised people with sub-token offsets $\mathbf{o}_i$,

$$\mathcal{L}_{\text{off}} = \frac{1}{N_h} \sum_i \|\widehat{\mathbf{o}}_i - \mathbf{o}_i\|_1 \tag{4.5}$$

grows when the anchor sits at the wrong place inside its token.

4.3.2. Pose, shape and placement

For predicted and target rotation matrices $\widehat{\mathbf{R}}_{ib}$ and $\mathbf{R}_{ib}$ of bone b of person i, pose and shape use an L1 loss, the sum of absolute differences:

$$\mathcal{L}_{\text{rot}} = \frac{1}{N_h} \sum_i \sum_b m_{ib} w_b \|\widehat{\mathbf{R}}_{ib} - \mathbf{R}_{ib}\|_1, \qquad \mathcal{L}_{\text{shape}} = \frac{1}{N_h} \sum_i \sum_k |\widehat{\phi}_{ik} - \phi_{ik}|. \tag{4.6}$$

Here m_{ib} is 0 for unlabelled 3DPW hand bones (Section 4.4) and w_b is the normalised bone weight. The two terms penalise wrongly turned bones and wrong proportions. Helper rotations are the identity on both sides. Final supervises all phenotypes, including the ones that the converter leaves at a constant. The two shape-mask models (Section 4.4) test that choice. The code first sums over the bones (or phenotypes) of one person and only then averages over people, so the size of each term, and with it the meaning of its weight, depends on this order of summing. An L1 loss is smallest at the median of the plausible answers, so weak image cues pull the shape toward the training median (Appendix F).

For predicted and target log-depth d_i and camera-space anchor translation $\mathbf{t}_i$,

$$\mathcal{L}_{\text{dist}} = \frac{1}{N_h} \sum_i |\widehat{d}_i - d_i|, \qquad \mathcal{L}_{\text{t}} = \frac{1}{N_h} \sum_i \|\widehat{\mathbf{t}}_i - \mathbf{t}_i\|_1. \tag{4.7}$$

These penalise a wrong distance and a wrong position in the image. They are the only terms that see the absolute placement directly; the 2D terms see it indirectly.

4.3.3. Geometric and reprojection supervision

For joints or vertices $\mathbf{X}_i$, each body's own pelvis is subtracted first, $\overline{\mathbf{X}}_{i\ell} = \mathbf{X}_{i\ell} - \mathbf{J}_{i,\text{root}}$. With K_X points and weights w_ℓ ($w_\ell = 1$ for vertices, the configured

joint weights for joints),

$$\mathcal{L}_{X3D} = \frac{1}{N_h K_X} \sum_i \sum_{\ell=1}^{K_X} w_\ell \|\widehat{\overline{\mathbf{X}}}_{i\ell} - \overline{\mathbf{X}}_{i\ell}\|_1. \tag{4.8}$$

This compares posture and body geometry independently of where the person stands, which is also why it cannot replace Equation (4.7). The 2D terms project the uncentred camera-space points:

$$\mathcal{L}_{X2D} = \frac{1}{|\Omega_X|} \sum_{(i,\ell)\in\Omega_X} \|\pi_{\mathbf{K}}(\widehat{\mathbf{X}}_{i\ell}) - \pi_{\mathbf{K}}(\mathbf{X}_{i\ell})\|_1, \tag{4.9}$$

where $\pi_{\mathbf{K}}$ projects a point into the image and Ω_X holds the target points inside it. In plain words, the projected mesh must line up with the photo. If one of these terms comes out as NaN or infinite, that term is skipped.

These L1 losses are related to PVE (Section 4.8) but are not the same measure, because PVE averages Euclidean distances. Training runs in 32-bit floats (fp32) with the faster TF32 format for matrix products, because mixed precision gave NaN (not-a-number) values in the Anny layer, and steps with NaN or Inf gradients are skipped (16 of 10,000 in Final). I did not tune any of the loss weights in Table 4.1.

4.4. Training with real images

A weighted sampler draws on average 30% of each batch from 3DPW-train converted to Anny (v23 labels, 5,882 images) and 70% from AnnyOne. AnnyOne images hold more people, so by my estimate only about 16% of the supervised people are real. The test split is never used for training.

The hand mask. The converter fits only the root and the bones that carry a mapped SMPL joint, so the fingers, thumbs and metacarpals of every 3DPW label are at rest. With `--mask_3dpw_hands 1` the rotation terms skip these 38 bones for 3DPW people, and the fingers are learned from AnnyOne only. The mask removes the rotation terms only; the mesh terms still carry the rest-pose hands.

The shape mask. The converter leaves 8 phenotypes at constants: age, muscle, weight, cup size, firmness and the three ethnicity values (Section 3.6). With `--mask_3dpw_shape 1` the shape term ignores them for 3DPW people (Final-masked); the partial variant masks the 7 values other than age (Final-partial-mask). The mesh terms still carry the neutral body. I kept the unmasked model as Final and as the teacher, because on real photos both masked models lost the improvement in predicted gender that Final has over Base (Section 5.7.2).

4.5. Knowledge distillation

Knowledge distillation means using a larger trained model to guide a smaller one (Section 2.6). The frozen teacher sees the same image and guides the student through its features and its outputs.

Feature distillation. For image u and token n, the teacher token and the projected student token become distributions over the C_t channels, $p^t_{unc} = \mathrm{softmax}_c(\mathbf{z}^t_{un}/T)$ and $p^s_{unc} = \mathrm{softmax}_c(\widetilde{\mathbf{z}}^s_{un}/T)$. With batch size B, the loss is their Kullback–Leibler (KL) divergence, summed over tokens and averaged over images:

$$\mathcal{L}_{\mathrm{kd}} = \frac{T^2}{B} \sum_{u=1}^{B} \sum_{n=1}^{N_p} \sum_{c=1}^{C_t} p^t_{unc} \log \frac{p^t_{unc}}{p^s_{unc}}. \tag{4.10}$$

The loss works in three steps. The channel values are divided by the temperature T. A softmax then turns them into positive weights that add up to 1. Finally the KL divergence measures how differently the teacher and the student spread these weights. A larger T gives softer weights, so the student also learns from the smaller responses (example in Section 2.6); T^2 keeps the gradient size comparable [HVD15]. I set $T = 4$ without tuning it; Section 5.4 sweeps 5 values up to 16. Feature distillation follows the idea of FitNets [Rom+15]; here the student side is exactly the projected tokens the heads read, and the teacher side is detached, so no gradient reaches the teacher.

Head transfer and output distillation. The student starts with a copy of every teacher head tensor whose name and shape match. Both models run on the true person locations, so person i of one is person i of the other, and

$$\begin{aligned} \mathcal{L}_{\text{kd-out}} = \frac{1}{N_h} \sum_i \Big[& \alpha_{\mathrm{rot}} \sum_b \|\mathbf{R}^s_{ib} - \mathbf{R}^t_{ib}\|_1 \\ & + \alpha_{\mathrm{shape}} \sum_k |\phi^s_{ik} - \phi^t_{ik}| + \alpha_{\mathrm{dist}} |d^s_i - d^t_i| \Big] \end{aligned} \tag{4.11}$$

pulls the student's outputs toward the detached teacher's. It is a plain L1 difference with no temperature, and its rotation term has no per-bone boost and no hand mask, because the teacher predicts hands even where a converted label has none. The teacher can be wrong, so the supervised terms stay. Both weights are $\lambda_{\mathrm{kd}} = 1.0$ and $\lambda_{\mathrm{kd-out}} = 1.0$.

Warm-up freeze. The copied heads already know how to predict bodies, but at first the student's features differ from the teacher's. If the heads are trained on these features from the start, they give confident but wrong outputs, and the gradients from those outputs can damage the copied weights. For this reason I keep the heads frozen for the first epochs, so that the backbone and the projector can adapt first. After that, everything trains. Every head is built at the teacher's width (`--head_dim`), so all head tensors transfer (earlier designs: Section E.2).

4.6. The ViT-B student of Final

The Student uses this design with a ViT-B backbone (Table 4.2). The backbone starts from NAVER's Multi-HMR ViT-B, which is already trained for mesh recovery, and all of its weights are trained. The copied heads stay frozen for 5 epochs, and the targets are exactly the teacher's. There is no data augmentation. The job script names colour jitter, but no code reads that setting. The copied heads also carry the half-turn that the teacher's constant bone correction undoes (Section 4.7.2), so the student trains with the teacher's correction. Afterwards I estimate the student's own correction on 200 training images and use it in every evaluation. A second run of the same recipe for 60 epochs (its learning rate halves every 24 epochs, the same two-fifths rule) is only the training-length experiment (Section 5.3); the reported Student is the 30-epoch run.

I also set a retention criterion beforehand, the 95 % rule: for an error metric E, the student passes if $E_s \leq E_t/0.95$, with E_t the teacher's error. This rule only says in advance what would count as passing. It does not say that the student passes. I apply the same thresholds against Multi-HMR-Anny separately, because its AnnyOne error is most likely in-sample (Section 3.1) and it was not trained on my converted 3DPW images.

4.7. Training schedule and model names

The teacher's ViT-L backbone starts from NAVER's checkpoint, and its Anny heads were trained on AnnyOne; the result is Base, the start of every fine-tune (earlier stages in Appendix E).

Final and Control. Final trains from Base for 10 epochs of 1,000 iterations at batch 4 with a constant learning rate of 10^{-5}; only the last 4 backbone blocks and the heads train. Three changes come on together: the depth fix with distance and translation weight 5 (Section 4.7.1), the bone correction (Section 4.7.2), and 30% real images per batch with the hand mask. The 55 images of the diagnostic manifest (Chapter 3) are never sampled. Training took 4,292 s on one A100. Control is identical but keeps the pelvis targets with weight 1 and has no bone correction. So the difference between Control and Final comes from the two fixes alone, and the difference between Base and Control comes from the real images, together with whatever restarting the training from Base costs. I tested each fix alone only in short pilots (Appendix F).

Model names. Table 4.3 is the key to the names used from here on; Table A.1 in Appendix A gives each run's checkpoint, data, schedule and flags.

4.7.1. Consistent head-anchor supervision

The body is placed by its head joint (Equation (4.2)), so the detection, offset, depth and translation targets must all describe that joint. In Final the target

A. Recipe		
	Final (the teacher)	Student
Backbone	ViT-L/14 (width 1024) from Base; last 4 of 24 blocks trained; frozen in bf16 when teaching	ViT-B/14 (width 768) from NAVER's HMR-trained Multi-HMR ViT-B; fully trained
Heads	HPH at width 1024 (14.01 M)	copied from the teacher, frozen 5 epochs, fed by a projector 768→1024 (0.79 M)
Params / GFLOPs / ms	318.38 M / 1,927.4 / 59.7	101.38 M / 603.3 / 28.5 (3.14× / 3.19× / 2.10× less)
Schedule	10 epochs × 1,000 iterations × batch 4 = 40,000 images	30 epochs × 1,000 iterations × batch 8 = 240,000 images
Learning rate	10^{-5}, constant	3×10^{-5}, halved every 12 epochs
Targets	head anchor, depth weight 5, 30% 3DPW (v23), hand mask	identical
Distillation	–	feature KL $\lambda = 1.0$, $T = 4$; output L1 $\lambda = 1.0$
Bone correction	four bones, estimated on Base	teacher's in training; own one afterwards
B. The 95% bars: student error ≤ reference error / 0.95		
	against Final	against Multi-HMR-Anny
AnnyOne holdout PVE (V1)	73.7 → **77.6**	60.3 → 63.5 (likely in-sample)
AnnyOne holdout MPJPE (V1)	77.9 → **82.0**	63.3 → 66.6
3DPW test PVE (v23)	89.0 → **93.7**	126.9 → 133.6
3DPW test MPJPE (v23)	88.3 → 92.9	129.4 → 136.2

Table 4.2.: *The Student (ViT-B) next to its teacher Final. Part A: recipe and measured cost. Part B: the error bars of the 95% rule in mm; the job computes the bars from the teacher re-measured in the same job, so these values are for orientation. Cost at 672 px, one person; ms on one L40S, batch 1, both models in the same job. The rule also asks for fewer parameters, fewer FLOPs and a faster forward pass. Read as "error at most 1.05 times the teacher's", the bars against Final would be 77.4 and 93.4 mm. The Student's own errors are in Chapter 5.*

builder decodes the labelled body and takes the head's camera-space position and image projection; the visibility checks and the one-person-per-token assignment use it too. Base and Control build these targets at the pelvis, so targets and decoder disagree about where the body is, and the pelvis-centred losses cannot see it (Figure 4.3; measurements in Section F.3). Control and Final differ in both the anchor and the placement weight (5 against 1), so this comparison changes the two together.

Name	Meaning
Multi-HMR-Anny	NAVER's released ViT-L model for Anny; external reference and source of the teacher backbone
Base	end of the schedule, AnnyOne only; the start of every fine-tune
Control	Base, 10 epochs, 30% 3DPW (v23 labels), hand mask, neither fix
Final	larger ViT-L teacher: Control plus consistent head anchors and bone correction
Final-masked	Final plus the shape mask on all 8 converter constants
Final-partial-mask	Final plus the shape mask on the 7 constants other than age
Student	deployed ViT-B model distilled from Final, 30 epochs
Student, 60 epochs	the same recipe for 60 epochs; training-length experiment only

Table 4.3.: *Model names used in this thesis. Every name is one run; no run was repeated with another seed. Full definitions: Table A.1 (Appendix A).*

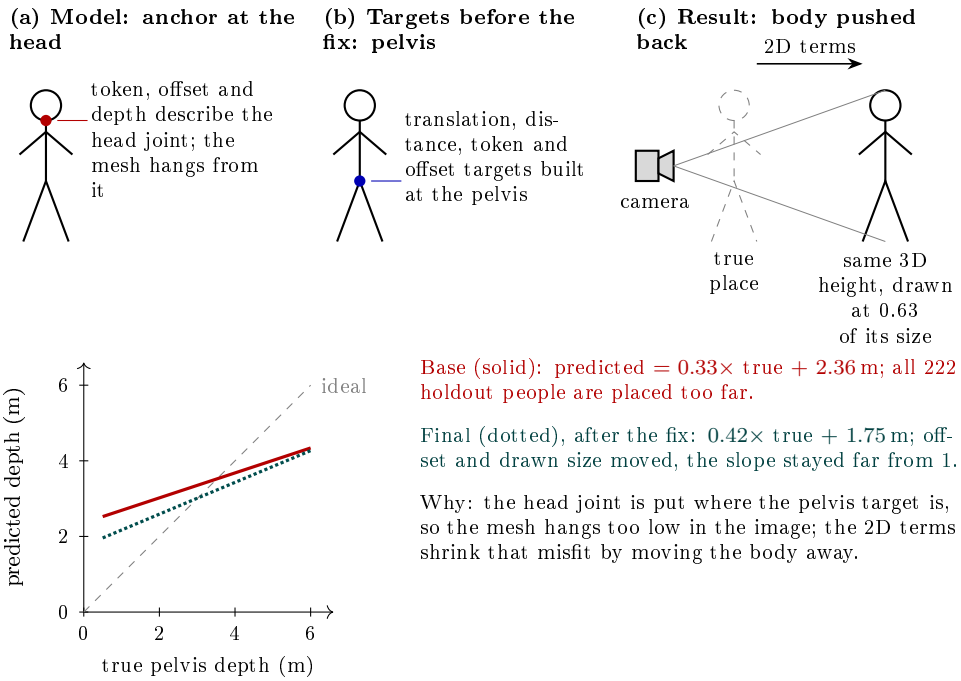


Figure 4.3.: *The body is placed by its head joint, so detection, offset and depth targets must refer to that point (schematic; pelvis targets against head-anchor targets). The measured depth scatter is in Figure 5.7.*

4.7.2. Fixed output rotation correction

In Base, neck02 and the head bone are each turned by about half a turn (Section F.4). I do not know where this flip comes from. I correct it with one constant rotation per affected bone, applied to the pose output before decoding:

$$\mathbf{R}_{ib}^{\text{corr}} = \widehat{\mathbf{R}}_{ib}\mathbf{C}_b, \qquad \mathbf{C}_b = \text{Proj}_{SO(3)}\left(\frac{1}{N_{\text{cal}}}\sum_i \widehat{\mathbf{R}}_{ib}^{\top}\mathbf{R}_{ib}^{\text{target}}\right), \tag{4.12}$$

where $\text{Proj}_{SO(3)}$ gives the nearest proper rotation to the mean residual over N_{cal} people. So $\mathbf{C}_b$ is the average turn from the prediction of bone b to its label. I estimate it on 200 random training images, never on evaluation images, and only for bones whose offset stays above 90°. For the teacher these are neck02, head, metacarpal1.L and finger4-1.R (Figure 4.4). The correction acts before helper masking, decoding and the loss, and adds no parameters. A constant cannot fix an error that differs per image, so I fine-tuned the teacher with it in place. The student trains with the teacher's correction, and I use its own correction, estimated after training, for its evaluation and the video (Section 4.6).

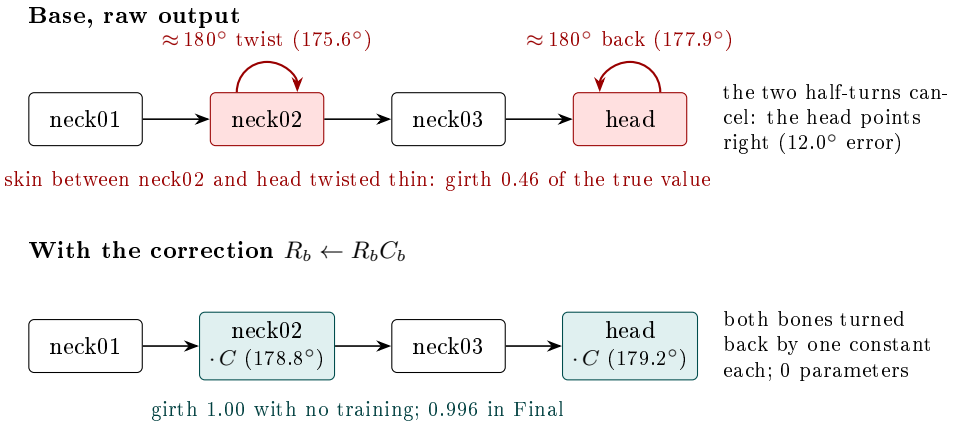


Figure 4.4.: *The neck chain neck01–neck02–neck03–head (schematic). Top: in Base, neck02 and the head bone are each turned about half a turn; the turns cancel, so the head points right, but the skin between them is twisted to 0.46 of its true girth. Bottom: one constant rotation per bone turns both back. Girth on the AnnyOne holdout, setting V1.*

4.8. Evaluation protocol

Errors are means over matched people: a greedy 2D matcher pairs each labelled person with at most one detection (Table 4.4).

For one person with predicted and target vertices $\hat{\mathbf{v}}_k, \mathbf{v}_k$ (N_V of them), joints $\hat{\mathbf{J}}_j, \mathbf{J}_j$ ($J = 163$) and pelvis joints $\hat{\mathbf{p}}, \mathbf{p}$,

$$\text{PVE} = \tfrac{1}{N_V} \textstyle\sum_k \left\| (\hat{\mathbf{v}}_k - \hat{\mathbf{p}}) - (\mathbf{v}_k - \mathbf{p}) \right\|_2, \tag{4.13}$$

$$\text{PA-PVE} = \tfrac{1}{N_V} \textstyle\sum_k \left\| s^* \mathbf{R}^* \hat{\mathbf{v}}_k + \mathbf{t}^* - \mathbf{v}_k \right\|_2, \tag{4.14}$$

$$\text{MPJPE} = \tfrac{1}{J} \textstyle\sum_j \left\| (\hat{\mathbf{J}}_j - \hat{\mathbf{p}}) - (\mathbf{J}_j - \mathbf{p}) \right\|_2, \tag{4.15}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}), \tag{4.16}$$

with $s^*, \mathbf{R}^*, \mathbf{t}^*$ the scale, rotation and translation of the least-squares fit of prediction to target. PVE is the mean distance between corresponding surface points once both pelvises sit at the same place. PA-PVE first turns, moves and resizes the prediction to fit the target (Procrustes alignment) and measures

Metric	What it measures	Blind to
PVE (mm)	vertex error after both pelvises are put at the same point	placement, depth, size
PA-PVE (mm)	PVE after the best rotation, scale and shift (Procrustes)	also global orientation
MPJPE (mm)	PVE on the 163 Anny joints	as PVE
Recall, precision (%)	labelled people matched; detections matched	false detections; bystanders
Root error (m)	3D pelvis distance, no alignment	pose and shape
Size ratio	drawn head-to-pelvis distance, predicted over true (1 is right)	3D body size
Depth slope / intercept	line fitted to predicted against true pelvis depth (1 / 0 m)	errors around the line
Bone angle error (deg)	angle of $\hat{R}_b^\top R_b$ for a parent-relative rotation; ≈180 = flipped	compensating bones
Head direction (deg)	camera-frame head orientation error	the chain below the head
Neck girth ratio	neck-skin radius, predicted over true (1 is right)	neck length
$\mathrm{PVE}_{\mathrm{neck}}$ (mm)	vertex error over the neck skin, centred at neck01	the rest of the body
Child AUC	chance that a child gets a lower predicted age than an adult (0.5 = chance)	undetected people
Gender accuracy (%)	predicted gender on the right side of 0.5 (50 = chance)	undetected people

Table 4.4.: *Metrics used in this thesis. All errors are means over matched people; the last two are on Relative Human; MPJPE is not comparable with published 3DPW numbers. On the AnnyOne holdout the headline setting is V1 (helper bones masked); V0 leaves them unmasked and reads 3.8 mm higher in PVE on the same people. 3DPW numbers are against our converted labels only (Section 3.6).*

what remains. MPJPE is PVE on the Anny joints. TP counts the labelled people found and FN those missed; finding 9 of 10 gives 90 % recall. For cost I report parameters, GFLOPs (one multiply-accumulate is two operations) and the median time of one forward pass on one GPU, at batch size 1 with one person.

AnnyOne holdout. The holdout is the last 100 AnnyOne images (indices 579,860 to 579,959). They were never trained on and contain no children (Chapter 3). The headline setting V1 masks the helper bones, as the demos do; V0 does not (Table 4.4). A trace script adds placement and per-bone metrics.

3DPW test. I evaluate on all 8,190 test images (11,854 people) with the v23 labels, with one script for every model and setting V0. The targets are converted labels and MPJPE averages all 163 Anny joints, so these numbers are not comparable with published 3DPW numbers (Section 3.6).

Relative Human and demo images. I use the Relative Human validation split (635 images, 2,130 people) for evaluation only, with boxes from the projected vertices and matching at IoU ≥ 0.3. I count a prediction as child-shaped when its predicted age is below 0.45. The 4 demo photos and 8 3DPW test frames place each body from its detection box, which hides depth errors, and I judge them by eye.

Training variation. Each model is one training run without a fixed seed; I did not repeat the Student or Final. Two identical short pilots differed by 7.6 mm PVE and 1.9° head direction (Section F.7), which gives no confidence interval but means I read small differences between single runs with care.

4.9. Applying the student to video

The network works on single images, so for a video I run it frame by frame. Each selected frame is letterboxed (resized and padded without distortion) to the checkpoint's input size, with an assumed field of view of 55° along the longer side for every model. The raw outputs keep the model's own camera-space placement; nothing is placed from a detection box. A geometric tracker follows one person by box overlap and normalised centre distance. A frame without a prediction is left empty, and when the track breaks, no frame-to-frame difference is computed across the break. The pipeline also records the pelvis-relative joint velocity and acceleration ($\mathbf{q}_{tj} = \mathbf{J}_{tj} - \mathbf{J}_{t,\text{root}}$, frame spacing Δt)

$$\mathbf{v}_{tj} = \frac{\mathbf{q}_{tj} - \mathbf{q}_{t-1,j}}{\Delta t}, \qquad \mathbf{a}_{tj} = \frac{\mathbf{q}_{tj} - 2\mathbf{q}_{t-1,j} + \mathbf{q}_{t-2,j}}{\Delta t^2} \tag{4.17}$$

and the phenotype spread and frame-to-frame change. These describe predicted motion only, not an error against ground truth, and natural or camera motion raises them too.

The input, the student and Multi-HMR-Anny are shown side by side on the same frames. Playback FPS is the source rate divided by the frame stride. Inference FPS uses batch size 1 and a clock synchronised with the GPU around the model call:

$$\text{FPS}_{\text{infer}} = \frac{N_{\text{timed}}}{\sum_{t \in \mathcal{T}} s_t}, \qquad s_t = \text{forward seconds of frame } t, \tag{4.18}$$

where $\mathcal{T}$ holds the N_{timed} frames after a warm-up of 10 frames, which stay in the videos and the measurements. The time s_t includes detection and Anny decoding and excludes preprocessing, transfers outside the model, rendering and video input and output. I time rendering separately. Wherever I give a speed, I also give the device, the precision, the resolution and the number of people it was measured with.

4.9.1. Temporal filtering and landmark alignment

Each frame is predicted on its own, so the body can change from frame to frame, and the raw student mesh sometimes extends beyond the person (Sec-

tion 5.5). I added two inference-time steps; the student is not retrained and has no learned temporal information. First, the student predicts a body per frame. Then an exponential moving average blends it with the track's earlier predictions. Then 2D landmarks from a separate detector correct its position and depth. Finally, that depth is smoothed.

Temporal filter. The filter is causal: it uses only the current and earlier frames. For each smoothed quantity it keeps a value $\bar{x}$ and updates it with each new prediction x_t:

$$\bar{x}_t = (1 - \alpha)\,\bar{x}_{t-1} + \alpha\,x_t, \qquad \alpha = 1 - \exp(-\Delta t/\tau_c),$$

with Δt the time between source frames and τ_c the time constant of quantity c. A longer time constant gives the new frame less weight, so the output is steadier but lags behind fast movement. The filter smooths the bone rotations ($\tau_c = 0.08\,\mathrm{s}$), the camera-space head anchor that places the mesh (0.12 s) and the phenotypes (1.0 s). Rotations are blended by shortest-arc quaternion interpolation, which keeps them valid; a plain average of turns of almost $+180°$ and $-180°$ would give no rotation at all. The filtered parameters are decoded again through Anny, so vertices are never averaged. The state resets after a missed detection, a new track or a gap above 0.25 s, and a missed frame is never filled with an old body. Raw and filtered outputs share one forward pass. The constants are the pipeline defaults (the five-panel run used exactly these, as its `run.json` records). They are initial settings that I inspected on the video and did not tune.

Landmark alignment. The filter cannot move a body that sits at the wrong place or is drawn too large. For this I run MediaPipe Pose, an existing 2D landmark detector, once offline on the CPU over the original video. Its shoulders, elbows, wrists, hips, knees and ankles match the heads of the Anny bones upperarm01, lowerarm01, wrist, upperleg01, lowerleg01 and foot. A least-squares fit that down-weights outliers finds one 3D translation of the whole body so that the projected Anny points land on the landmarks. A body moved farther away looks smaller, so this corrects an oversized or shifted mesh, but it cannot repair a wrongly bent arm or leg. A fit counts only with at least 4 landmarks of confidence 0.5 or more, covering a shoulder, a hip and a knee or ankle, and a depth within 0.5–2 times the current median joint depth; otherwise the frame stays unchanged and the depth history resets. Finally, the accepted depth is smoothed in log space (0.35 s, at most 0.5 log-units per second), so the body grows and shrinks less between frames; a reset can still produce a visible size change. The alignment acts on the filtered body when the filter is on.

What the two steps can and cannot show. The landmarks are detector estimates, not ground truth, so a smaller landmark error means closer agreement with that detector, not a better 3D body. The clip has no 3D labels and gives no PVE, and a steady body can still be wrong. A visible gain in the aligned output comes from this refinement, not from a new network. The panels also time

different things: the raw panels show the inference FPS of Equation (4.18), the filtered panel adds filtering and decoding, the aligned panel adds the fit but not the offline landmark extraction, and the input panel shows playback FPS. These are not one comparable number (Table 5.6, Figure 5.5).

5. Experiments and results

The Student is the model I deploy, so I start with it. I compare it with its teacher and with the released baseline. Then I look at what a longer training and another distillation temperature change, and at how the Student does on a photo and on a video. After that I study its teacher, Final. Older schedules and early student variants are in Appendix E.

5.1. Evaluation sets and settings

I use four sets, described in Chapter 3 and scored as in Section 4.8. The AnnyOne holdout is the last 100 AnnyOne images (222 people, no children), never trained on and scored in setting V1. A second script adds the placement numbers of Section 5.6.3, with each body at the depth its model predicts. The 3DPW test set is all 8,190 test images with the corrected labels, in setting V0. Its numbers hold only against my converted labels (Section 3.6). Relative Human gives real photos with an age group, a gender and a box per person. The 4 demo photos and 8 3DPW test frames have no labels, so I judge them by eye, with each body placed from its detection box.

Model names follow Table 4.3. Every model was trained once, without a fixed seed, and two identical pilots differed by 7.6 mm PVE (Section F.7). So a PVE gap of a few millimetres between two single runs can be noise.

5.2. The Student against its teacher and the released baseline

5.2.1. Reconstruction

The Student follows the recipe of Table 4.2 (Section 4.6). Table 5.1 lists the Student, its teacher and Multi-HMR-Anny on both test sets, with Control and Base for reference. The teacher has the lowest errors on our 3DPW-Anny labels, 89.0 mm PVE, against 110.1 mm for the Student and 126.9 mm for Multi-HMR-Anny. The Student's error is 1.62 times the teacher's on the holdout (119.6 against 73.7 mm) and 1.24 times on 3DPW. Table 5.3 applies my 95 % rule from that section. The Student misses the bar on every error metric on both sets. For example, on 3DPW it has 110.1 mm PVE against a bar of 93.7 mm. It passes only on recall and on cost, and it did not reach the teacher's accuracy.

Against Multi-HMR-Anny the 3DPW result is mixed. The Student has a lower PVE (16.8 mm better, inside the bar of 133.6 mm) and a lower MPJPE (112.0 against 129.4 mm). But its PA-PVE is slightly worse, 71.8 against

	AnnyOne holdout (V1)			3DPW test (v23)			
Model	PVE	PA-PVE	Recall	PVE	PA-PVE	Recall	3DPW training
Released baseline	60.3[a]	44.6[a]	89.0[a]	126.9	68.8	97	no
Teacher (Final)	73.7	50.3	87.0	**89.0**	**57.6**	99	yes
Student (ViT-B)	119.6	80.8	85.4	110.1	71.8	99	yes
Control	77.4	56.4	89.0	91.5	59.7	99	yes
Base	70.5	53.3	90.2	121.8	72.0	100	no

Table 5.1.: *Reconstruction comparison. Errors are in mm; recall is in percent. One checkpoint is evaluated per model; thesis-trained models were not repeated across seeds. 3DPW errors use converted Anny labels and are not comparable with published SMPL-space results. [a]The released baseline was trained on the AnnyOne release, which includes these hold-out images, so its synthetic result is potentially in-sample. The real-image training exposure differs between released baseline and student/teacher.*

Model	Params (M)	GFLOPs	Latency L40S (ms)	Latency A100 (ms)
Released baseline	380.94	≥1917.2[b]	n/m	n/m
Teacher (Final)	318.38	1927.4	59.7	77.9
Student (ViT-B)	101.38	603.3	28.5	n/m

Table 5.2.: *Model cost at 672 px input and one person. One multiply-add counts as two FLOPs. Latency is median batch-one forward time in fp32 with TF32; compare only within a GPU column. The student/teacher L40S pair was measured in the same job. n/m: not measured. [b]The baseline figure counts only its backbone, so total computation is higher. Base and Control share the teacher architecture; fixed rotation corrections add no trainable parameters.*

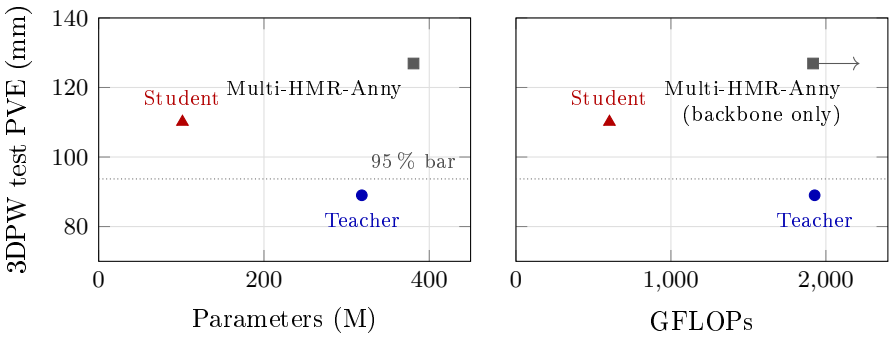


Figure 5.1.: *Model size and compute against 3DPW-test PVE (corrected labels, all 8,190 images; one run per model). Multi-HMR-Anny's GFLOPs count its backbone only (arrow: the true value is higher). Dotted line: the retention threshold against the teacher (93.7 mm).*

Metric	Teacher	Student	Ratio	Bar	Met?
AnnyOne holdout, V1					
PVE (mm)	73.7	119.6	1.62	≤ 77.6	no
PA-PVE (mm)	50.3	80.8	1.61	≤ 52.9	no
MPJPE (mm)	77.9	127.2	1.63	≤ 82.0	no
Recall (%)	87.0	85.4	0.98	≥ 82.7	yes
AnnyOne test-100, V0					
PVE (mm)	77.5	122.7	1.58	≤ 81.6	no
PA-PVE (mm)	54.4	84.2	1.55	≤ 57.3	no
Recall (%)	87	85	0.98	≥ 82.7	yes
3DPW test (v23 labels), V0					
PVE (mm)	89.0	110.1	1.24	≤ 93.7	no
PA-PVE (mm)	57.6	71.8	1.25	≤ 60.6	no
MPJPE (mm)	88.3	112.0	1.27	≤ 92.9	no
Recall (%)	99	99	1.00	≥ 94.1	yes
Cost (L40S, batch 1, one person)					
Params (M)	318.38	101.38	3.14×	fewer	yes
GFLOPs	1927.4	603.3	3.19×	fewer	yes
ms per image	59.7	28.5	2.10×	faster	yes

Table 5.3.: *The Student against its teacher (Final) and the 95 % rule: every error of the Student at most the teacher's divided by 0.95, every recall at least 0.95 of the teacher's, and fewer parameters, fewer FLOPs and a faster forward pass. The rule fails on every error and holds on recall and cost. One run, no seed. Against Multi-HMR-Anny the PVE bars would be 63.5 mm on AnnyOne (most likely in-sample for it; not met) and 133.6 mm on 3DPW (met). Cost: both models timed in the Student's job on one L40S.*

68.8 mm. PA-PVE first removes the best scale, rotation and position. So the Student's lead over the baseline lies in the overall size and orientation of its bodies, and its aligned surface is slightly worse.

This comparison is not fully fair to the baseline. My Student and teacher were trained on converted 3DPW images, and the released baseline was not. The result compares two complete trained systems, and it does not show that my network design is better. On the holdout the bias goes the other way. NAVER's model was trained on the AnnyOne release, and that release includes the 100 images I hold out. Its 60.3 mm is therefore most likely an in-sample number, and mine are not. On 3DPW, which neither NAVER's model nor Base saw before my fine-tune, the teacher is 37.9 mm better (89.0 against 126.9 mm). For this reason it means little that the Student misses the AnnyOne bar (63.5 mm). I did not repeat the run, and I have no labelled real photos on which to test the Student's hands and head.

5.2.2. Model size and speed

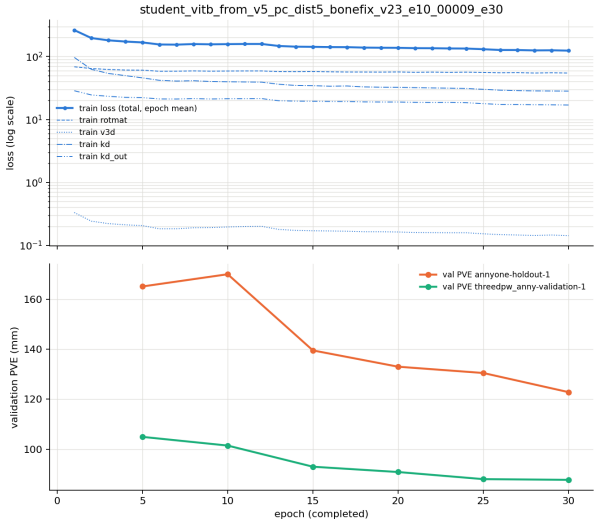
The Student has 101.38 M parameters, 3.14 times fewer than the teacher (318.38 M) and 3.76 times fewer than the baseline (380.94 M; Table 5.2). It needs 3.19 times fewer FLOPs than the teacher. On one L40S, in the same job, a Student forward pass took 28.5 ms and a teacher pass 59.7 ms, so the Student is 2.10 times faster. I compare latencies only within one GPU column. In Figure 5.1 the x-axis is the model size, once in parameters and once in GFLOPs, and the y-axis the 3DPW test PVE. The Student sits to the left of the teacher and above it: smaller, but with a higher error. This benchmark did not time the baseline. The only speed comparison with it on the same GPU is the video in Section 5.5.

5.3. Training for 60 instead of 30 epochs

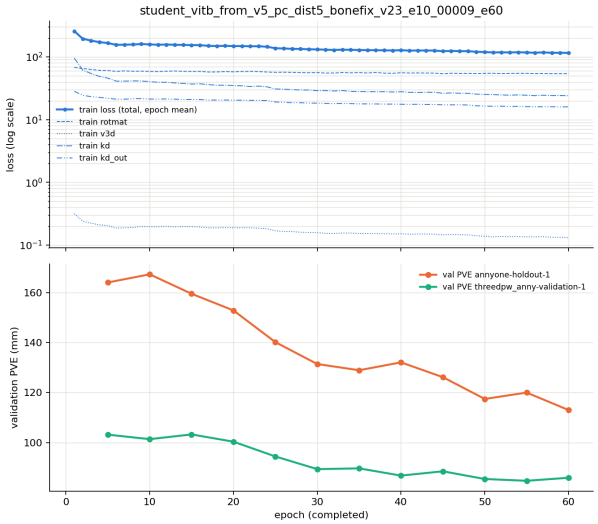
After 30 epochs the Student's holdout error measured during training (setting V0, Figure 5.2) was still falling, from 139.5 mm at epoch 15 to 122.8 mm at epoch 30. Its 3DPW validation error had almost stopped. It went from 88.0 to 87.7 mm in the last five epochs. A simple rule for this is: if the training and the validation curve both still fall, train longer, and if only the training curve falls, stop. To check, I trained the same recipe for 60 epochs.

In Figure 5.2 the x-axis is the epoch. The upper plot of each panel has the training loss on a log scale, and the lower one the PVE in mm, measured every 5 epochs on the holdout and on 300 3DPW validation images. Both runs were trained before I added the option to log a validation loss, so I use this PVE in its place. The holdout PVE rises between epochs 5 and 10, when the copied heads are unfrozen. The large drops in the training loss follow the learning-rate halvings (every 12 and every 24 epochs).

Over the last 20 % of the epochs the training loss and the holdout PVE still fall in both runs, but the 3DPW validation PVE is flat (-0.06 and $+0.06$ % per epoch). The test numbers in Table 5.4 agree. Doubling the epochs lowered



(a) *Student, 30 epochs (reported)*



(b) *Student, 60 epochs*

Figure 5.2.: *Training curves of the Student. Top of each panel: the training loss per epoch, total and its terms (log scale). Bottom: the validation PVE measured during training every 5 epochs on the AnnyOne holdout and on 300 3DPW-validation images. These runs were trained before a validation loss was logged, so the validation PVE stands in for it. Over the last 20 % of the epochs (25–30 and 49–60) the training loss still falls (−0.87 and −0.29 % per epoch) and so does the holdout PVE (−1.21 and −0.38 % per epoch, ending at 122.83 and 113.04 mm), while the 3DPW validation PVE is flat (−0.06 and +0.06 % per epoch, ending at 87.75 and 85.96 mm). The holdout PVE rises between epochs 5 and 10, after the copied heads are unfrozen, and the large drops follow the learning-rate halvings (every 12 and 24 epochs).*

	Teacher	Student 30 epochs	Student 60 epochs	95 % bar
AnnyOne holdout, V1				
PVE (mm)	73.7	119.6	107.7	≤ 77.6
PA-PVE (mm)	50.3	80.8	74.6	
MPJPE (mm)	77.9	127.2	114.5	≤ 82.0
Recall (%)	87.0	85.4	85.0	
3DPW test (v23 labels, 8,190 images), V0				
PVE (mm)	89.0	110.1	110.9	≤ 93.7
PA-PVE (mm)	57.6	71.8	72.5	
MPJPE (mm)	88.3	112.0	112.9	≤ 92.9
Recall (%)	99	99	99	
Cost, batch 1, one person				
ms per image	59.7 / 60.6	28.5	28.9	faster

Table 5.4.: *Training length. The Student of Table 5.3 (30 epochs) and the same recipe trained for 60 epochs, against the teacher and the 95 % bars (teacher error divided by 0.95; the rule is set on PVE and MPJPE). Doubling the epochs lowers the holdout PVE by 11.9 mm but leaves 3DPW where it was (110.1 and 110.9 mm PVE); both runs miss all four bars. Each student job timed itself and the teacher on its own GPU, so the teacher has two times (30-epoch job first). Parameters and FLOPs are the same for both students (101.38 M, 603.3 GFLOPs). One run each, no seed.*

the holdout PVE (setting V1) by 11.9 mm, to 107.7 mm, and the holdout root error from 830.7 to 697.9 mm (teacher: 728.5 mm). On 3DPW the PVE went from 110.1 to 110.9 mm. This change of 0.8 mm is far below the 7.6 mm that two identical pilots differed by on the holdout (Section F.7); the variation on 3DPW itself was not measured. The 60-epoch run still misses all four bars, at the same cost.

Longer training helps on synthetic images and not on real ones. By this rule the 3DPW curve says stop, because its error has not improved since about epoch 25. This means the Student's gap to the teacher on real photos is not a training-length problem. I keep the 30-epoch run as the reported Student. These two runs have no validation loss, and I did not try a third training length.

5.4. Distillation temperature

The temperature T sets how strongly the feature distillation softens the channel values before the Student is compared with the teacher (Section 2.6). The reported Student uses $T = 4$, which I chose without a search, so the question is how much the result depends on it. I set up 5 runs with $T \in \{1, 2, 4, 8, 16\}$ and the reported recipe otherwise. The output distillation has no temperature, so it stays the same. The new runs evaluate after every epoch and log a validation loss. Because the sweep trains $T = 4$ again, it also gives one more measure of run-to-run noise.

	AnnyOne holdout, V1				3DPW test, V0			
T	PVE	PA-PVE	MPJPE	Rec.	PVE	PA-PVE	MPJPE	Rec.
1	116.0	77.8	123.1	85.0	109.9	71.0	111.6	100.0
2	116.7	80.2	124.1	85.0	106.8	71.3	108.0	100.0
4 (sweep)	117.1	81.4	124.2	85.8	110.1	72.0	111.5	100.0
8	117.1	77.8	124.8	82.5	107.1	71.0	108.4	99.0
16	no result: the run stopped after training on a network time-out							
4 (reported)	119.6	80.8	127.2	85.4	110.1	71.8	112.0	99
Teacher	73.7	50.3	77.9	87.0	89.0	57.6	88.3	99
95 % bar	77.6		82.0		93.7		92.9	

Table 5.5.: *Distillation temperature sweep for the Student (errors in mm, recall in %). Only the temperature T of the feature distillation changes: it softens the channel softmax of the projected backbone tokens before the KL term (gradient scaled by T^2); the output distillation is an L1 term without a temperature and stays as it is. Everything else is the reported recipe (30 epochs). The row "4 (reported)" is the Student of Table 5.3; the sweep re-runs $T = 4$ so that its difference to the reported run gives the run-to-run noise: 2.5 mm holdout PVE and 0.0 mm 3DPW PVE. One run per temperature; the $T = 16$ run trained but stopped in a later step on a network time-out.*

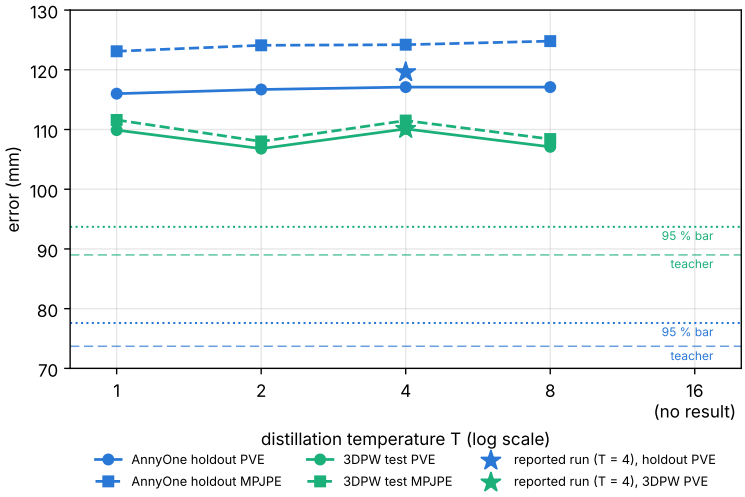


Figure 5.3.: *Student error against the feature-distillation temperature T (logarithmic axis): PVE and MPJPE on the AnnyOne holdout (V1) and on the 3DPW test set; $T = 16$ has no result. Dashed lines: the teacher's PVE (73.7 and 89.0 mm); dotted lines: the 95 % bars (77.6 and 93.7 mm); stars: the reported $T = 4$ run. Numbers in Table 5.5.*

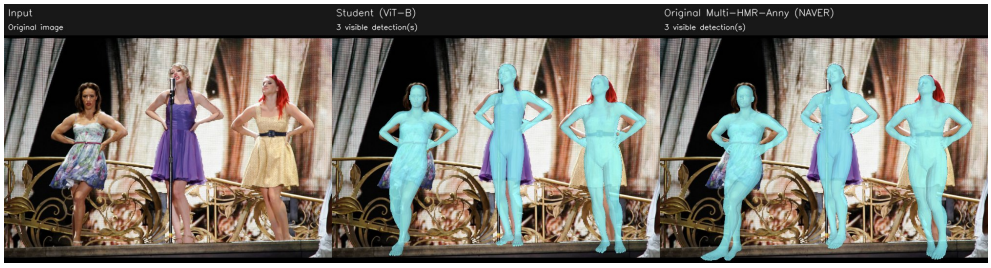


Figure 5.4.: *One real photo of three people on a stage. Left: the input. Middle: the Student (ViT-B). Right: the original Multi-HMR-Anny of NAVER. Both models find all 3 people. No ground truth exists for this photo, so the figure shows only how the meshes sit on the people.*

In Figure 5.3 the x-axis is the temperature on a log-2 scale and the y-axis the error in mm, one line per metric and test set, with the teacher dashed and the bars dotted. The idea is to read off the temperature with the lowest errors and use it.

The lines in Figure 5.3 are almost flat. Over $T = 1$ to 8 the holdout PVE moves by only 1.1 mm (116.0 to 117.1 mm) and the 3DPW PVE by 3.3 mm (106.8 to 110.1 mm). The sweep's own $T = 4$ run differs from the reported one by 2.5 mm on the holdout, so differences of this size are run-to-run noise. Formally $T = 2$ has the lowest combined error, but I would not call it better. No temperature brings the Student close to the 95 % bars (Table 5.5). So the temperature is not what holds the Student back, and I keep $T = 4$. The run with $T = 16$ trained to the end, but a later evaluation step stopped on a network time-out, so it has no numbers.

5.5. Real photos and video

5.5.1. A still photo

Figure 5.4 shows three singers on a stage, with the Student and Multi-HMR-Anny side by side. Both models found all 3 people. Each body sits at the depth its model predicts. There is no tracking, the bodies are not placed from their detection boxes, and the fingers are predicted instead of being set to rest pose. Looking at the overlays, both models put a body with the hands on the hips over each singer, and the baseline's bodies look slightly larger. Since the photo has no ground truth, it only shows that the Student works on a real photo from outside 3DPW, not that it is as accurate as the baseline there.

5.5.2. A walking video

The clip `man_walking.mp4` has 356 frames of one man walking across a square. I ran both models on the same frames with the procedure of Section 4.9 and rendered five panels (Figure 5.5): the input, the raw Student, the Student with temporal filtering, the Student with landmark alignment, and the baseline. The filter blends each frame's prediction with the earlier ones through

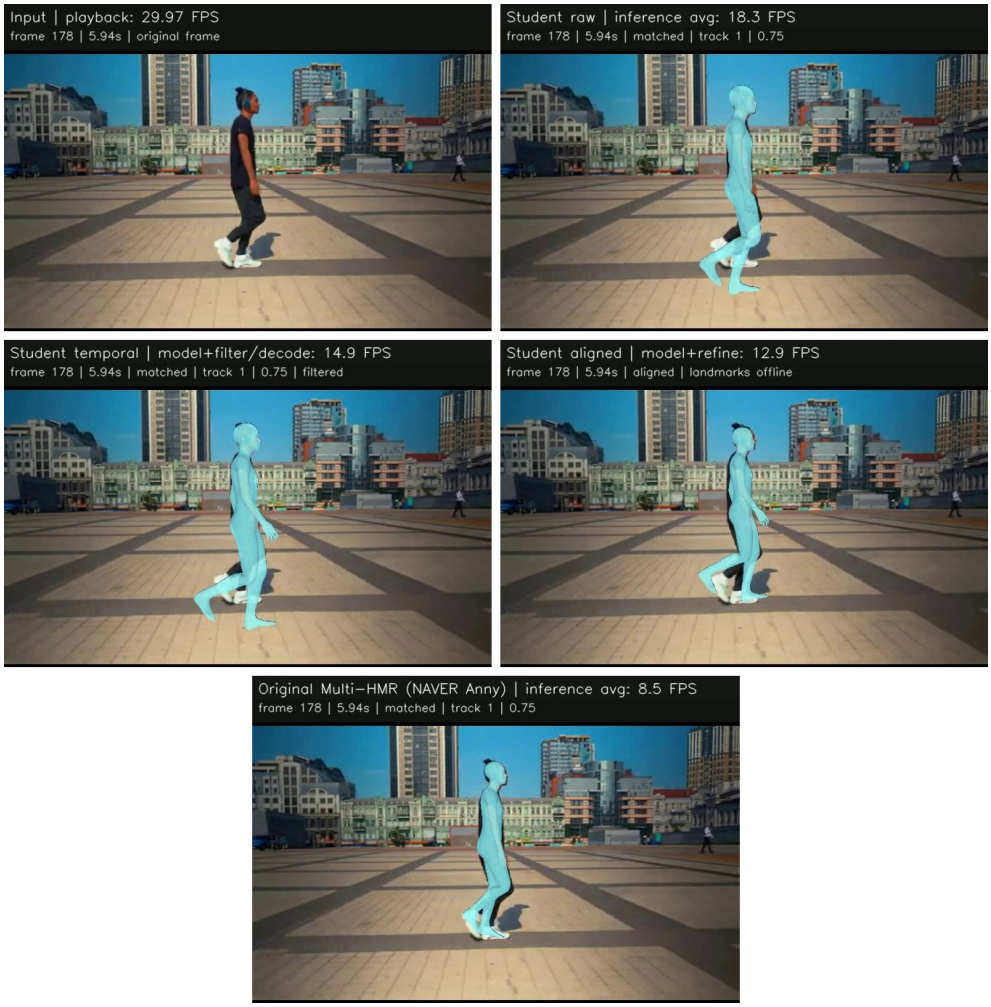


Figure 5.5.: *One frame (178 of 356) of the video comparison. Top left: the input (playback 29.97 FPS). Top right: the raw Student, one prediction per frame (inference 18.3 FPS). Middle left: the Student with temporal smoothing, a causal exponential moving average per track on the bone rotations (blended on the rotation sphere), the shape and the 3D position, with the mesh decoded again through Anny (model and filter 14.9 FPS). Middle right: the Student with landmark alignment: MediaPipe shoulders, hips, knees and ankles, the mesh position and depth fitted so the projected joints match them, then the corrected depth smoothed (model and refinement 12.9 FPS; landmarks extracted offline). Bottom: the original Multi-HMR-Anny of NAVER (inference 8.5 FPS). The FPS labels are burned into the video; their definitions are in Table 5.6. Neither variant retrains the Student. In this frame the raw Student mesh is drawn ahead of the person; the aligned mesh follows the person more closely.*

Panel	FPS	What the FPS includes
Input	29.97	playback rate of the source video
Student, raw	18.3	average model inference: detection, pose and shape, Anny mesh decoding
Student, temporal	14.9	model inference plus the temporal filter and the Anny re-decode
Student, aligned	12.9	model inference, the temporal work if enabled, and the alignment fit; offline landmark extraction excluded
Multi-HMR-Anny (NAVER)	8.5	average model inference, same definition as the raw Student

All FPS are batch-one, sequential timings on one NVIDIA A40 (fp32 with TF32, 672-pixel input) over the same 346 frames after 10 warm-up frames. None of them includes image preprocessing, rendering or video decoding and encoding, so none is end-to-end video throughput.

Table 5.6.: *Video comparison on one walking clip (356 frames). The Student matched a person in 356 of 356 frames, Multi-HMR-Anny in 354 of 356; these are tracker matches, not labelled accuracy. The FPS values come from the run's timing file (job 3657550) and are also printed in the comparison video (Figure 5.5). The raw Student runs about 2.16 times as fast as Multi-HMR-Anny; temporal smoothing and landmark alignment cost part of that gain.*

an exponential moving average per tracked person. The alignment moves the mesh in position and depth until its joints line up with MediaPipe landmarks (Section 4.9.1). Neither retrains the Student. In this run the Student kept a matched person in 356 of 356 frames and the baseline in 354. These counts come from the tracker. They are not a detection recall against labels.

Seen frame by frame, the raw Student mesh sometimes extends beyond the person; in Figure 5.5 it is drawn ahead of the man. Temporal filtering makes its movement steadier. Landmark alignment improves its position and apparent size, so the mesh follows the person more closely, and the depth smoothing reduces sudden changes in size. The fifth panel is the baseline on the same frames, so the two raw models can be compared side by side.

The frame rates in Table 5.6 measure different amounts of work. Only the raw Student and the baseline both time model inference alone: 18.3 against 8.5 FPS on the same GPU and frames, so the Student is about 2.16 times faster. The two networks differ in more than the backbone, though; the baseline's head has 8 cross-attention blocks and mine 2. The filtered and the aligned Student do extra work and run at 14.9 and 12.9 FPS. All four were measured on one NVIDIA A40, so none of them compares with the L40S forward times of Table 5.2. Per frame, the Student's inference took 54.5 ms and the baseline's 117.6 ms; the filter added 12.7 ms and the alignment fit another 10.2 ms.

The clip has no ground truth, so I cannot say that the steadier mesh is also the more accurate one. The landmarks come from another detector and not from a measured body, so agreeing with them is not the same as being right. The motion statistics that the run saves agree with what I see in the video. The mean joint acceleration (relative to the pelvis) falls from 38.0 to 10.5 $\mathrm{m/s^2}$ with the filter, and the mean speed of the head anchor from 1.69 to

0.63 m/s (0.53 m/s after alignment). The predicted gender also varies less over the clip (standard deviation 0.058 against 0.024). The unsmoothed baseline has 29.6 m/s^2. Lower numbers mean less jitter, but a filter can also flatten real motion, so they are not accuracy numbers. I ran only this one clip, with one person and no occlusion.

5.6. Teacher supervision ablations

The rest of the chapter is about the teacher. I did not repeat these comparisons for the Student, so they say nothing about what the same changes would do there. Control and Final share the start point, the schedule, the real-image share and the hand mask. Final adds the head-anchor supervision and the bone correction.

5.6.1. Reconstruction on AnnyOne

Table 5.7a puts Final next to Control, Base and the two pilots. Final has the lowest PA-PVE of every model, 6.1 mm below Control. Its PVE is 3.7 mm below Control's, less than the gap between the identical pilots, and Base keeps the lowest PVE, as it did against every fine-tune. Recall is 2.0 points below Control, at the limit of the variation, and 3.2 below Base. In Figure 5.6 Base's bodies are small and off their people, Final's close to the ground truth. During training Final stayed below Control at all ten epochs on 3DPW validation. On the holdout the two swapped places from epoch to epoch (Figure D.7).

As far as one run can tell, the two changes do not raise the body error, and they give the largest PA-PVE gain in the project. PVE aligns the pelvis, so it cannot see the depth gain, and the neck and fingers are only a few vertices of a whole body (Section 6.2). The recall cost against Base is larger than the variation. A likely reason, which I did not test, is that the detection cell moved from the pelvis to the head, so people with a hidden head are found less often. I did not compare PVE on the same people (the matched sets hold 214, 219 and 222 people), and I did not compute recall only on people with a visible head. There are no bootstrap intervals and no second run of Final.

5.6.2. Real-image supervision

Table 5.7b scores every model with the same script. Final has the lowest error on our 3DPW-Anny labels, 89.0 mm PVE, ahead of Control, Mixed-L (the previous best), Base and Multi-HMR-Anny, and it finds 99 % of the labelled people. In the overlays (Figure D.10, Appendix D) most men get male bodies in Final and female-looking ones in Base, Final's bodies are a little larger, and the seated guitar player is poor in both.

Control, without either change, is already 30.3 mm better than Base. So most of the gain over Base comes from training on 3DPW at all. The two changes add 2.5 mm PVE and 2.1 mm PA-PVE, but these are small single-run differences, and I did not measure the run-to-run variation on 3DPW. Final is 6.5 mm better than Mixed-L, which saw the old labels. Mixed-L also had

	Base	Depth pilot	Neck pilot	Control	Final
PVE (mm)	70.5	74.8	68.8	77.4	73.7
PA-PVE (mm)	53.3	54.0	50.9	56.4	**50.3**
Recall (%)	90.2	87.8	89.4	89.0	87.0
Root error (mm)	1208	544	1429	1402	**728**
Depth slope	0.33	0.38	0.34	0.28	0.42
Depth intercept (m)	2.36	1.59	2.56	2.66	1.75
Size ratio	0.63	0.84	0.58	0.59	0.78
Neck girth pred/GT	0.461	0.457	0.997	0.464	0.996
PVE_{neck} (mm)	27.3	27.7	10.7	27.7	11.0
Global head direction (deg)	12.0	12.7	14.4	13.0	13.1
Fingers (deg)	17.3	17.4	8.3	17.4	8.3
neck02 bone (deg)	175.6	175.5	4.7	175.8	4.8
head bone (deg)	177.9	178.0	11.2	177.7	8.9

(a) *AnnyOne holdout, 100 images, setting V1 (helper bones masked, matched people); root error, depth fit and size ratio with each body at the model's own predicted depth.*

	Base	Mixed-L	Neck pilot	Control	Final	Multi-HMR-Anny
PVE (mm)	121.8	95.5	120.2	91.5	**89.0**	126.9
PA-PVE (mm)	72.0	62.9	68.7	59.7	**57.6**	68.8
MPJPE (mm)[b]	126.9	94.8	124.8	90.9	**88.3**	129.4
Precision (%)	54	73	53	62	64	75
Recall (%)	100	94	100	99	99	97
AnnyOne test-100 PVE[c]	74.3	78.5	72.7	81.2	77.5	(60.3)[a]

(b) *3DPW test, corrected (v23) labels, all 8,190 images, setting V0, same evaluation script for every model. Mixed-L was trained on the labels before their correction and is tested on the corrected ones, like every model here.*

Table 5.7.: *Final against its Control and against Base. One run per model, no seed; two identical runs differed by 7.6mm PVE, so the PVE gap between Control and Final (77.4 vs 73.7) is within run-to-run variation, while PA-PVE, root error, neck and fingers are not. Depth pilot: depth fix, depth weight ×5, 10 epochs; neck pilot: bone correction and a 2-epoch fine-tune. 3DPW numbers are against our converted labels and are not comparable with published 3DPW results.* [a]*Multi-HMR-Anny masks the helper bones by design, so its number is V1 and compares with Final's 73.7 in (a); it was most likely trained on these 100 images.* [b]*Mean over all 163 Anny joints, pelvis-centred.* [c]*The same 100 images as (a) in setting V0 (helper bones unmasked).*

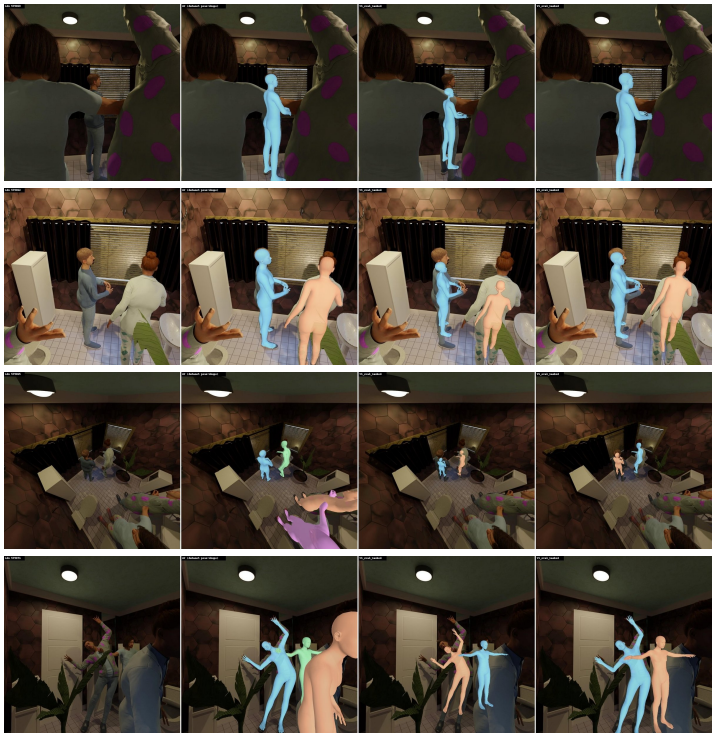


Figure 5.6.: *Four AnnyOne holdout images (index in the first panel), each as image | ground truth | Base | Final. Ground truth drawn with the dataset camera; Base and Final in setting V1 (panels labelled "V1_eval_masked"), each body at its model's own predicted depth. By eye, Base draws its bodies smaller and further back than the ground truth, Final close to its size and place. Mesh colours mark detections, not identities. People without a ground-truth mesh (the foreground arm in 579862) are not counted; people a model misses (the two lying in the foreground of 579865) have no mesh. Synthetic images, one run per model.*

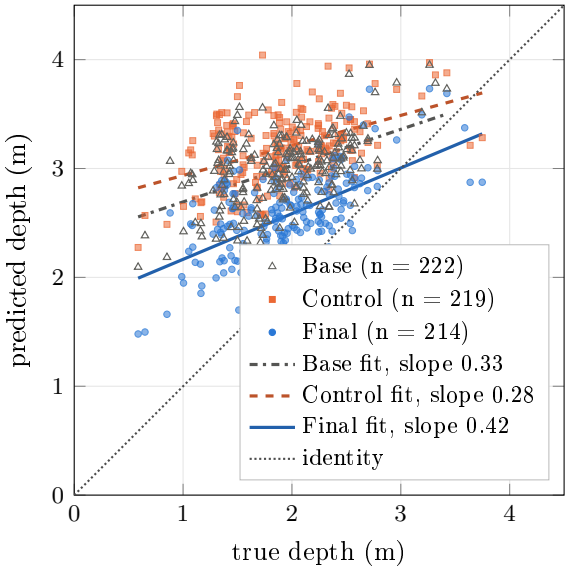


Figure 5.7.: *Predicted against true root depth for every matched person of the AnnyOne holdout (100 images, setting V1, each body at its own predicted depth): Base (222 people), Control (219) and Final (214). Lines: least-squares fits and the identity. Final's fit is* $0.42 \times true + 1.75\,m$*, Control's* $0.28 \times true + 2.66\,m$*, Base's* $0.33 \times true + 2.36\,m$*. One run per model; synthetic images. Statistics in* `data/plot_stats.md`*.*

a different schedule, so I cannot say whether the corrected labels cause part of this. Precision is low for every model because 3DPW labels only one or two actors per sequence: every detected bystander and every ghost body on a half-visible person counts as a false positive. I did not break the result down by sequence, and I looked at only 8 frames by eye.

5.6.3. Camera-space placement

Figure D.1 (Appendix D) shows the depth bug and its fix on one holdout image: Base draws the man far behind his true position and too small, the depth pilot close to his true size. Over all people (Figure 5.7), Control places 99.1 % too far. Final moves the points towards the identity line. The root error falls by 48 % (1,402 to 728 mm), the drawn size rises from 0.59 to 0.78, the depth correlation from 0.46 to 0.56, and the head lands within 7.6 pixels of its true position against 67.1 (Table 5.7a).

The fix survives ten epochs with real images and the bone correction. Every run trained with the head-anchor targets (Table F.3) has a lower root error than every run without them. It moved the offset and the body size, not the slope (0.42). Final's fit crosses the identity at 3.01 m, so near people are still placed too far (94.4 % of all people). The pelvis lands farther from its pixel (41.1 against 26.7 pixels), likely because the head is pinned while the body is still too small. Final's root error is also above the depth pilot's (544 mm), possibly because of the real images or the bone correction. I tested neither

explanation, and I cannot say why the slope does not move. All depth numbers are synthetic, and the single-image figure shows the pilot, not Final.

5.6.4. Bone correction

Figure D.11 (Appendix D) shows the correction on a real frame that no model trained on: Base draws both necks as thin stalks under the head, the corrected model at normal width. Figure F.1 and Table F.4 give the correction without training and after a 2-epoch fine-tune. Final keeps it through ten epochs with real images (Table 5.7a). Its neck02 bone is 4.8 deg off against 175.8 for Control, its neck girth is 0.996 of the true girth against 0.464, and the joint error of the neck region falls from 62.6 to 5.5 deg. The head direction, which got worse with the correction alone (19.1 deg), is back at Control's level.

This means the network learns on top of the correction and does not undo it. The correction is one constant rotation per bone and adds no parameters. I do not know why these bones are flipped in the first place. PVE shows no sign of a cost. Real photos have no neck ground truth. The real frame shows the neck pilot, not Final, and its columns also differ by the fine-tune. But on the holdout the same fine-tune without the correction keeps the girth at 0.461 (Table F.4), so the correction is the likely cause of the wider necks. Flips that cancel each other along a chain (Section F.4) would need a two-sided correction, which I did not implement.

5.6.5. Hand articulation

Predicted fingers are 17.3 deg off in Base and 17.4 deg in Control, worse than fingers left at rest (15.1 deg; Table F.5). Final, with the two flipped hand bones corrected, reaches 8.3 deg (Table 5.7a), and the correction alone gave 8.2. So about half of the finger error was two flipped bones, metacarpal1.L and finger4-1.R, and training keeps the gain. The rest is not explained. Real images add nothing to the fingers, because the hand mask removes the rotation terms of the 38 hand bones for 3DPW people, whose labels hold rest-pose hands.

Figure D.12 (Appendix D) shows predicted fingers on real images, which every earlier demo set to rest pose. In these crops Base's fingers are splayed and stiff, with one finger of the left hand sticking out as the flipped metacarpal1.L would make it. Final's fingers are closed and bent, and they look like hands. In the party photo the hand lies over the armrest, in the guitar frame it is curled as if fretting, and in the bar frame it bends over the edge of the bar. Small hands (Figure D.8) cannot be judged at print size. The real-image evidence is only these three images, judged by eye, and they show the correction and the fine-tune together.

	Base	Final	Final-masked	Final-partial-mask
Recall (%)	47.1	64.7	57.2	63.0
Gender accuracy (%)	46.5	64.2	49.3	47.4
mean gender, men (male = 0)	0.535	0.551	0.717	0.725
mean gender, women (female = 1)	0.551	0.861	0.860	0.888
Child-vs-adult AUC	0.606	0.719	0.708	0.727
Mean age, adult	0.631	0.536	0.638	0.532
teen	0.597	0.524	0.616	0.515
kid	0.613	0.500	0.611	0.506
baby	0.624	0.438	0.524	0.465
Babies with age < 0.45 (%)	2.6	49.3	16.9	34.3
Mean height, adult	0.507	0.500	0.505	0.527
kid	0.451	0.466	0.461	0.497
baby	0.394	0.464	0.478	0.504
Matched children + adults	948	1,305	1,157	1,270
Demo images: standard deviation of the prediction				
Muscle	–	0.04	0.06	n/m
Weight	–	0.04	0.11	n/m
Cup size	–	0.05	0.11	n/m
Firmness	–	0.04	0.11	n/m
Age	–	0.03	0.08	n/m

Table 5.8.: *Body shape on real photos. Upper part: Relative Human validation split (635 images, 2,130 labelled people), detected people only, matched to the labelled boxes at IoU ≥ 0.3; ages and heights are Anny phenotype values in $[0, 1]$, not years; "child-shaped" means a predicted age below 0.45; the AUC ranks kids and babies against adults by predicted age (0.5 = chance). The matched sets differ between models because recall differs. Lower part: spread of the predicted phenotypes over all detected people of the 12 demo images (4 photos and 8 3DPW test frames), not matched to people; not reported for Base, not measured (n/m) for Final-partial-mask. Final-masked masks all eight constant 3DPW phenotypes, Final-partial-mask the seven at 0.5 (age kept supervised; Section 5.7.2). One run per model, no seed. The neck pilot (bone correction and a 2-epoch fine-tune) is the same as Base within run-to-run variation (AUC 0.613).*

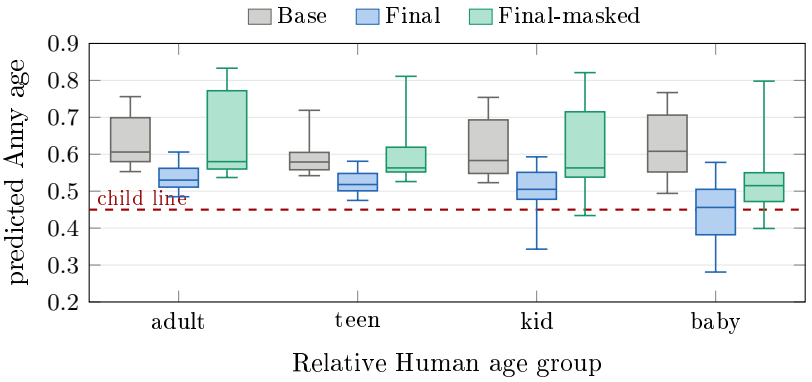


Figure 5.8.: *Predicted Anny age per Relative Human age group for Base, Final and Final-masked, matched (detected) people of the validation split only. Box: quartiles and median; whiskers: 5th and 95th percentile. Dashed: the child line, age 0.45. Ages are Anny phenotype values, not years. One run per model; statistics in* `data/plot_stats.md`.

5.7. Shape of the teacher on real photos

5.7.1. Children and gender

Table 5.8 measures body shape on labelled real photos. Final finds more people than Base and gets the gender right for 64.2 % of them, against 46.5 %. Women are recognised (mean predicted gender 0.861, where 1 is female), but men stay in the middle (0.551). The child-against-adult AUC is the chance that a random child gets a lower predicted age than a random adult, so 0.5 is chance. It rises from 0.606 to 0.719, and 49.3 % of the babies are child-shaped against 2.6 % for Base. In Figure 5.8 the x-axis is the Relative Human age group and the y-axis the predicted Anny age. Base's boxes stand at the same height for every group. Final's step down from adults to babies, and only its baby box reaches below the child line. On the demo photos (Figures D.14 and D.6, Appendix D) most men look male in Final, and the boy keeps an adult body.

Because the AUC only ranks people, a common shift of all ages cannot raise it, so the step is a real separation of children from adults. Part of the lower ages is likely a shift of this kind, caused by the 3DPW age label, which the converter fixes at 0.55 for every real person. The gender gain matches what I would expect from the corrected 3DPW gender labels, because they supervise the shape head on real people for the first time. Children are a measured improvement, not solved, and men are not yet male. I do not give ages in years, because the code uses three age scales that disagree, and I cannot check children on held-out synthetic images, because the holdout has none. Only detected people count, so the matched sets differ between models.

Gender accuracy needs a cut-off (a predicted gender above 0.5 counts as a woman). Table 5.9 adds the gender AUC, which needs none, and I use it as the main gender number. Final reaches 0.780 against 0.562 for Base. The two masked models of Section 5.7.2 reach 0.705 and 0.752, so they still rank

	Base	Final	Final-masked	Final-partial-mask
People with a gender tag	932	1,308	1,163	1,271
Gender AUC	0.562	0.780	0.705	0.752
Accuracy at 0.5 (%)	46.5	64.2	49.3	47.4
Mean prediction, women	0.551	0.861	0.860	0.888
Mean prediction, men	0.535	0.551	0.717	0.725
Best accuracy[a]	0.636	0.723	0.685	0.707
at cut-off	0.62	0.89	0.95	0.93
Spread (sd) of the prediction over all matched people				
Weight	0.101	0.058	0.146	0.120
Muscle	0.084	0.053	0.084	0.055

[a] Cut-off chosen on Relative Human itself, so this row only shows how large the threshold effect is; it is not a usable result. A real cut-off cannot come from the AnnyOne holdout either: its gender labels only span 0.38 to 0.82, with 85 % above 0.5, so it has no fully male body.

Table 5.9.: *Gender on the Relative Human validation split (people matched to a labelled box with a male or female tag; Anny's gender 0 is the male end). The AUC does not depend on a cut-off (0.5 = chance); accuracy uses the cut-off 0.5. The masked models still rank women above men but moved both sexes up, so at 0.5 most men are called women. The partial mask keeps most of the ranking and doubles the weight spread on real photos compared with Final; muscle spread does not come back. One run per model.*

women above men. Their accuracy at 0.5 is near chance only because both sexes moved up, so most men are called women.

A likely reason why men land in the middle is the training data. On the AnnyOne holdout the gender labels span only 0.38 to 0.82, with 85 % above 0.5. If the training images are similar, AnnyOne has no fully male body, and the model saw men labelled 0 only in 3DPW. A cut-off for real photos would have to come from 3DPW validation, which I did not do. The best accuracy in Table 5.9 uses a cut-off chosen on Relative Human itself, so it only shows how large the threshold effect is.

In the rest-pose line-up (Figure D.13, Appendix D) only the body differs. In the pictures, Base gives nearly everyone the same slim, flat body. Final gives the singers a fuller chest and hips, and the man in the bar a broader torso. The tennis men stay slim in all three models, in line with the men's mid-range gender score, and Final-masked looks like Final except for a slimmer man in the bar.

These bodies look much alike, which could mean that Final's shape head cannot vary shape. Table 5.10 checks this on the AnnyOne holdout, where the labels do vary. Final follows weight (Pearson r between prediction and label 0.77), gender (0.73) and cup size (0.69) best. Muscle is the weakest (0.27) and got weaker from Base (0.41) to Final, likely because the converter fixes muscle at 0.5 for every 3DPW person; I did not test this. Every slope of prediction against label is below 1 (Final's largest is 0.61, for weight), so all three models pull their predictions towards the middle.

	Base		Control		Final		Final: spread (sd)	
Phenotype	r	slope	r	slope	r	slope	prediction	label
weight	0.80	0.67	0.80	0.65	0.77	0.61	0.194	0.247
gender	0.77	0.52	0.69	0.54	0.73	0.48	0.087	0.132
cupsize	0.70	0.52	0.68	0.50	0.69	0.47	0.164	0.243
proportions	0.74	0.57	0.66	0.42	0.67	0.50	0.056	0.074
asian	0.65	0.26	0.62	0.26	0.62	0.25	0.036	0.087
height	0.52	0.36	0.55	0.38	0.48	0.37	0.043	0.055
firmness	0.45	0.36	0.45	0.35	0.46	0.34	0.149	0.198
caucasian	0.50	0.25	0.48	0.23	0.46	0.22	0.050	0.104
age	0.44	0.16	0.31	0.14	0.31	0.16	0.036	0.068
african	0.40	0.23	0.35	0.20	0.30	0.17	0.040	0.071
muscle	0.41	0.21	0.35	0.16	0.27	0.13	0.052	0.108
People	222		219		214			

Table 5.10.: *How well each model follows the phenotype labels on the AnnyOne holdout (100 images, setting V1, matched people). r: Pearson correlation between prediction and label; slope: least-squares slope of the prediction on the label (1 would mean no pull towards the middle). The last two columns compare the spread of Final's predictions with the spread of the labels. Every slope is below 1, so all three models pull their predictions towards the middle. Final follows weight, gender and cup size best (r between 0.69 and 0.77), while muscle is the weakest phenotype and got weaker from Base to Final. One run per model.*

This shows that Final can express shape, at least on synthetic images. Base, which never saw 3DPW, already shows part of this narrow spread on real photos. So the gap between synthetic and real images is part of the cause, and not the 3DPW training alone.

5.7.2. Shape-supervision masking

I trained two masked models to test the shape mask of Section 4.4, one run each. Final-masked stops the shape loss on the 8 values the converter fixes for every 3DPW person: age 0.55 and seven values at 0.5. Final-partial-mask stops it only on the seven and keeps age supervised. Gender stays supervised in both. On Relative Human (Table 5.11), Final-masked loses most of Final's gain (gender 49.3 %, babies child-shaped 16.9 %), but on the demo images it brings back variety: the spread of weight, cup size and firmness rises from 0.04–0.05 to 0.11. Final-partial-mask keeps the children (AUC 0.727, 34.3 % of babies child-shaped) and doubles the weight spread on Relative Human (0.058 to 0.120; Table 5.9). At the 0.5 cut-off it loses gender as the full mask does, with both sexes on the female side (men 0.725, women 0.888).

Both masked models keep body errors close to Final's. The mask therefore changes only what the shape head predicts (the 3DPW mesh targets still hold the neutral body). Without the mask, Final learned the seven constants and predicts nearly the same muscle, weight, cup size and firmness for everyone

	Final	Final-masked	Final-partial-mask	Rule
Masked 3DPW phenotypes	none	8 (age + 7)	7 (age kept)	
AnnyOne holdout, V1				
PVE (mm)	73.7	71.7	73.0	
PA-PVE (mm)	50.3	50.7	51.2	
Recall (%)	87.0	86.2	87.4	
Root error (mm)	728	682	680	
3DPW test, v23 labels, V0				
PVE (mm)	89.0	91.1	89.9	
PA-PVE (mm)	57.6	59.0	58.9	
Relative Human validation				
Recall (%)	64.7	57.2	63.0	
Gender accuracy (%)	64.2	49.3	47.4	≥ 60
Child-vs-adult AUC	0.719	0.708	0.727	≥ 0.7
Babies with age < 0.45 (%)	49.3	16.9	34.3	
Demo images, standard deviation of the prediction				
Muscle	0.04	0.06	n/m	> 0.08
Weight	0.04	0.11	n/m	> 0.08

Table 5.11.: *The three shape-mask arms. Final supervises all converted 3DPW phenotypes; Final-masked masks the 8 constants of the converted labels (age 0.55 and seven values at 0.5); Final-partial-mask masks only the seven at 0.5 and keeps age supervised. Gender stays supervised in all three. AnnyOne holdout: 100 images, setting V1; 3DPW test: all 8,190 images, corrected labels, setting V0; Relative Human: validation split, matched people (full columns in Table 5.8); demo: 12 images, not matched to people; n/m: not measured. Rule: the thresholds, fixed before the first masked run, that a masked arm had to meet to replace Final. One run per arm, no seed; two identical runs differed by 7.6 mm PVE.*

on real photos. With age supervised the children's gain stays, so the age label carries it. Gender accuracy falls although its label stays supervised. One reading I did not test is that Final's gender gain depends on the seven constants on the 3DPW men as much as on the gender label.

Before the first masked run I fixed a rule: a masked model would replace Final if gender accuracy stayed at 60 % or more, the child-against-adult AUC at 0.7 or more, and the spread of muscle and weight rose above 0.08. Both masked runs fail on gender and pass on the AUC. So Final stays the thesis model and the Student's teacher. Final-partial-mask keeps the children and more shape variety, but it loses gender at the 0.5 cut-off. I cannot say whether the varied values are right, because no real data set I have gives muscle or weight. Final-partial-mask had no demo run, so I do not know whether it brings the variety back on the demo images. And I do not know why gender falls.

6. Discussion

Here I use the results to answer the three questions of Section 1.2.

6.1. Answers to the research questions

Real-image supervision. With the converted 3DPW labels I can train and test Anny models on real images. I checked them independently against 3DPW's own body joints, and the median distance is 14.1 mm. But 8 phenotypes get the same value for every person, and many hand and head rotations stay at rest (Section 3.6).

Control already reaches 91.5 mm PVE on converted 3DPW without the anchor and bone corrections, against 121.8 mm for Base. Final, with the corrections, reaches 89.0 mm. This means that most of the gain on real images comes from adding 3DPW to the training mix, and not from the corrections or from a change in the network.

On the synthetic holdout Final follows weight, gender and cup size best (Pearson r 0.69 to 0.77, Table 5.10). Every slope is below 1, so its shape head varies the body but pulls the predictions towards the middle. Muscle is the weakest phenotype, likely because the converter fixes it for every 3DPW person, which I did not test.

On real photos I use the gender AUC, because it does not depend on a cut-off (Table 5.9). Men still get a gender value in the middle. The likely reason is the training data. The AnnyOne holdout's gender labels span only 0.38 to 0.82, and if the training images are similar, the model saw men labelled 0 only in 3DPW (Section 5.7.1). Final-partial-mask keeps the children but loses the gender split at the cut-off, so I kept Final as the teacher (Section 5.7.2).

Compact reconstruction. The Student has 3.14 times fewer parameters and 3.19 times fewer FLOPs than its teacher, and it ran 2.10 times faster in the paired L40S benchmark. Its PVE rose from 73.7 to 119.6 mm on the AnnyOne holdout and from 89.0 to 110.1 mm on converted 3DPW. So the Student is smaller and faster, but it is also less accurate.

Longer training did not close the gap on real images. After 60 epochs the holdout PVE fell from 119.6 to 107.7 mm, but on 3DPW it went from 110.1 to 110.9 mm, a change within the run-to-run variation (Section 5.3). I have not tested why. Possible reasons are the smaller backbone, the share of 3DPW in each batch and the teacher's own 3DPW error. The temperature is not the reason either: between $T = 1$ and 8 the errors changed by less than the run-to-run noise (Section 5.4).

Against Multi-HMR-Anny the Student is 3.76 times smaller and has a lower 3DPW PVE (110.1 against 126.9 mm), but a slightly higher PA-PVE (71.8 against 68.8 mm; Section 5.2). Only my models were trained on converted 3DPW. To separate my network and the distillation from the training data, I would need a baseline fine-tuned on the same data.

Video. On the walking clip the Student kept a tracked body in 356 of 356 frames. The baseline did so in 354. These are tracker matches, not recall against labels (Section 5.5). By eye, temporal filtering made the motion steadier, and landmark alignment put the mesh on the person and reduced the jumps in size. The clip has no ground truth and the landmarks come from a 2D detector, so this is no evidence of better 3D accuracy (Section 4.9.1). On one NVIDIA A40 the raw Student ran at 18.3 FPS and the baseline at 8.5 FPS on the same frames. The filtered and aligned frame rates in Table 5.6 also include the time of those steps.

6.2. Interpreting geometry and placement

PVE lines up the two pelvises before it measures vertex distances. So a model can have a low PVE and still put the person at the wrong depth. Control and Final show this: the root error fell from 1,402 to 728 mm, but the PVE only went from 77.4 to 73.7 mm. That is why I report the placement error with the PVE.

From Control to Final the PVE over the neck skin fell from 27.7 to 11.0 mm on the holdout (Table F.4). But the neck has only 171 of the body's vertices, so it moved the whole-body PVE by about 0.2 mm (my estimate from the vertex counts, not a measurement). For this reason I also judged the bone correction by bone angles and neck geometry (Appendix F).

I trained the same short pilot twice, and the two runs differed by 7.6 mm PVE. In other words, a few millimetres between two single runs can be training noise alone (Section F.7). Two runs give no general threshold, so I treat small differences with care.

6.3. Limitations

Children are a measured improvement, not solved. On Relative Human the teacher separates them from adults better than Base, but some still get adult bodies. The holdout has no children, and the 3DPW labels fix age and body build for everyone. The holdout gender labels have no fully male body, so a gender cut-off set there would mean nothing. All of these shape results were measured on the teacher. I did not run the shape tests on the Student.

When a person's true depth grows by one metre, the teacher's predicted depth grows by only about 0.42 m, and a large placement error remains (Section 5.6.3). Consistent anchors give the model a cleaner target, but one image still cannot tell a small person close by from a large person further away. One

clip of one walking person also says little about occlusion, people cut off by the frame or crowds.

My 3DPW errors are measured against fitted Anny labels, with MPJPE over 163 Anny joints, so they cannot be compared with published 3DPW results. The label discrepancy is not an offset that can be subtracted from them (Section 3.6). My training never used the holdout, but for Multi-HMR-Anny it is most likely in-sample, that is, part of its training data (Section 5.2).

Each model comes from one training run without a fixed seed, and I did not switch off the Student's loss terms one at a time. The movement statistics of the walking clip mix the person's own movement with camera motion and tracking errors. Latency depends on the hardware, precision, input resolution and number of people. The paired benchmark and the per-video timing were measured in different ways, so I report them separately.

The repository records the settings and the source of each number, but the cluster scripts contain paths that only work on my cluster. Before releasing weights or converted labels I have to check the terms of use of the datasets, the body model and the checkpoints, which I have not done.

6.4. What I report and why

A model can find a person and still give them the wrong body, and an accurate body can still shake over a video. Speed is a separate question again. The tracking counts, the metrics on labelled images, the video and the frame rates each answer one of these questions. For this reason I report all of them, and not just one error value or one good-looking example.

7. Conclusion and future work

7.1. Conclusion

Multi-HMR reconstructs several complete human bodies from one image. My project builds on this framework with the Anny body model, converted 3DPW labels and a smaller distilled student. I also added temporal filtering and landmark alignment to improve the video outputs.

To train on real photos, I converted 3DPW's SMPL labels into Anny labels. The converter matches the joints of the two body models, adds virtual torso joints and keeps one median shape per person and sequence. In an independent check the converted joints lie a median 14.1 mm from 3DPW's, and some phenotypes are the same for every person (Section 3.6).

The ViT-B student trains on AnnyOne and converted 3DPW. It reuses the prediction heads of its ViT-L teacher, Final, through a learned projector, and learns to imitate the teacher's features and outputs. Both models use consistent head anchors and one fixed output rotation correction, which I analyse in the appendices. At inference only the student and the Anny decoding layer run.

The student is cheaper: 101.38 M parameters and 603.3 GFLOPs against the teacher's 318.38 M and 1927.4 GFLOPs, and 28.5 against 59.7 ms per forward pass on the L40S GPU. The teacher is more accurate. On the AnnyOne holdout and the converted 3DPW test set its PVE was 73.7 and 89.0 mm, against 119.6 and 110.1 mm for the student. So the student failed the 95 % rule I set beforehand (Section 5.2.1). Its 3DPW PVE is lower than the 126.9 mm of the released Multi-HMR-Anny, but only my models were trained on converted 3DPW.

I also trained the student for 60 epochs. The holdout PVE fell to 107.7 mm, but the 3DPW PVE moved only from 110.1 to 110.9 mm, far below the run-to-run noise on the holdout (Section 5.3). So longer training alone does not close the gap on real images, and I keep the 30-epoch student. A sweep of the distillation temperature from 1 to 8 changed the errors by at most 3.3 mm, inside the run-to-run noise, so the temperature is not the limit either (Section 5.4).

On a walking clip of 356 frames, the student kept a tracked person in 356 frames and the baseline in 354 (Section 5.5). By eye, the filter made the motion steadier, and the alignment improved the mesh's position and size, so it followed the person more closely. The network still works frame by frame, and neither step is a learned temporal model (Section 4.9.1). The clip has no ground truth, so this shows steadier output, not better 3D accuracy.

7.2. Future work

Temporal evaluation and modelling. I would run the filter and the alignment on the 3DPW test sequences, whose converted labels exist at 10 frames per second, and measure PVE per frame against the raw student, so that steadiness and accuracy are measured together. Then I would test clips with occlusion, a moving camera or several people, and report each frame rate with its GPU, precision and input resolution. A learned temporal model (Section 2.7) is a separate project.

Accuracy at reduced cost. After the temperature sweep, I would try other weights for the distillation losses, other ways of transferring the teacher's features and more real images per batch. A fair comparison also needs Multi-HMR-Anny fine-tuned on the same 3DPW data.

Real-image shape. To replace the constant phenotypes, I would use age-group labels, body measurements or more Anny fits on real photos. Tests with children should keep two questions apart: can Anny make child proportions, and does the network recognise a child in a photo?

Reliability and reproducibility. Each model comes from one run. I would repeat the runs, with uncertainty estimates and a larger holdout set, to see which differences are real. A portable configuration with release checks would make the system easier to verify. I have not run any of these.

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A. Model names, runs and reproduction

Every number in this thesis can be traced in two steps. Each headline number is a macro in one of the files `numbers_*.tex` of the thesis source, and the comment next to it names the job and the line of its log, for example `3611336:191`. A job is one piece of work that Slurm, the scheduler of the cluster, ran for me, and the job id is the number Slurm gave it. Table A.2 maps each job to the model, the log and the checkpoint. Table A.1 maps every model name to the run name it had during the project. Job ids and old run names appear only in this appendix.

A.1. Model names

"L" is my ViT-L line, every model that keeps the ViT-L backbone. L-stage 1 to 4 are the steps of the transfer schedule that turned NAVER's released model into mine. Base is the end of that schedule and the start of every later fine-tune. Final and Control are the two runs of the last teacher experiment (exp7). Both are the same 10-epoch fine-tune of Base, one with the two corrections and one without them. The number after "Student-S/" is the old version number of a ViT-S student, and Student is the ViT-B model distilled from Final.

"Student, 60 epochs" is the Student's recipe trained twice as long, for the experiment of Section 5.3. It is not the reported model. The sweep of Section 5.4 trains the same recipe at 5 distillation temperatures.

Table A.1.: *Model names: the thesis name of every model, the run or checkpoint name it had during the project ("ep." = the epoch kept), the weights it started from, the backbone, the training data and the schedule, with the setting that is new in the row. Backbone: L, B, S = DINOv2 ViT-L, ViT-B, ViT-S with 14-pixel patches; "all" = every block trained, "last 4" = the last 4 of 24 blocks, "+p" = a linear projector to the heads' width 1024. Data: A = AnnyOne; A+P = 70 % AnnyOne and 30 % 3DPW train, with the old labels (before their correction, Chapter 3) or the corrected v23 labels. Schedule: epochs × iterations per epoch × batch, then the learning rate, halved every 30 epochs unless noted (planned epochs where the run stopped early). Checkpoints lie in* `logs/anny_model/<run>/checkpoints/` *with copies in* `models/multiHMR/`*; Table A.2 gives the path of every model with a number in the thesis, and Appendix C the full flags of Final, Control and the Student. "Not recorded": no script, log or document in the repository states it.*

Thesis name	Old run / checkpoint name	Starting weights	Backb.	Data	Schedule; what is new
Released starting point and baseline					
Multi-HMR-Anny	`multiHMR_672_L_anny` (NAVER's release)	trained by NAVER	L, own decoder	A, BEDLAM, COCO, MPII	not recorded; evaluated with its own model class
Our ViT-L line ("L"), the transfer schedule					
L-stage 1	`anny_s1_pretrained_bb_v2`	Multi-HMR-Anny backbone, fresh heads	L, all	A	150 × 1,000 × 4, 10^{-4}, from epoch 177 at $3 \cdot 10^{-5}$; side branch, no checkpoint kept
L-stage 2	`anny_s2_frozen_bb_v2`, ep. 299 ("v2")	Multi-HMR-Anny backbone, fresh heads	L, frozen	A	150 × 1,000 × 4, 10^{-4}, from epoch 283 at $3 \cdot 10^{-5}$
L-stage 3	`anny_s2_partialft_v3`, ep. 54 ("v3")	L-stage 2, epoch 299	L, last 4	A	300 × 1,000 × 4, 10^{-5}; helper bones held at rest
L-stage 4	`anny_s2_boosted_v4`, ep. 299 ("v4")	L-stage 3, epoch 54	L, last 4	A	300 × 1,000 × 4, 10^{-5}; neck and hand bones weighted ×4
Base	`anny_s2_shape_v5`, ep. 99 ("v5")	L-stage 4, epoch 299	L, last 4	A	100 × 1,000 × 4, 10^{-5}; shape weight 20
Mixed-L	`anny_teacher_v9_mixed`, ep. 62 and 149 ("v9")	Base	L, last 4	A+P, old	150 × 1,000 × 4, 10^{-5}, halved every 60
Fresh-heads-L	`anny_v10_scratch_mixed`, ep. 290 ("v10", not from scratch)	Multi-HMR-Anny backbone, fresh heads	L, last 4	A+P, old	300 × 1,000 × 4, 10^{-5}; 3DPW from the first step, no neck or hand weight
The last experiment: two arms and two shape-mask variants					
Control	exp7 `v5_root_dist1_nofix_v23_e10`, ep. 9; job 3611339	Base	L, last 4	A+P, v23	10 × 1,000 × 4, 10^{-5}; 3DPW hands masked, pelvis targets
Final	exp7 `v5_pc_dist5_bonefix_v23_e10`, ep. 9; job 3611336	Base and its bone correction	L, last 4	A+P, v23	as Control, with head targets, depth and translation weights 5, bone correction

Thesis name	Old run / checkpoint name	Starting weights	Backb.	Data	Schedule; what is new
Final-masked	exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask`, ep. 9; job 3612325	as Final	L, last 4	A+P, v23	as Final, 8 converter constants masked for 3DPW people
Final-partial-mask	exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask_noage`, ep. 9; job 3614042	as Final	L, last 4	A+P, v23	as Final, the 7 constants other than age masked; not adopted
ViT-S students ("Student-S"), history; teacher Base unless noted					
Student-S/1	distillation v1, `anny_distill_vits`	ImageNet DINOv2-S, random heads	S, all	A	$300 \times 1{,}000 \times 4$, 10^{-4}; feature distillation
Student-S/2	v2, `anny_distill_vits_v2_a`	Student-S/1, epoch 199	S, all	A	$200 \times 1{,}000 \times 4$, $2 \cdot 10^{-5}$, halved every 60
Student-S/3	v3, `anny_distill_vits_v3`	ImageNet DINOv2-S; 54 of 67 head tensors from Base	S, all	A	$150 \times 1{,}000 \times 4$, $5 \cdot 10^{-5}$, halved every 60; output distillation
Student-S/4	v4, `anny_distill_vits_v4`	NAVER's `multiHMR_672_S` backbone; heads as /3	S, all	A	$150 \times 1{,}000 \times 8$, $7 \cdot 10^{-5}$, halved every 60
Student-S/5	v5, `anny_distill_vits_v5`	as Student-S/4	S, all	A	as /4; heads frozen for 5 epochs
Student-S/6	v6, `anny_distill_vits_v6`, ep. 137	NAVER's ViT-S backbone; all 67 head tensors from Base	S+p	A	as /4; stopped at epoch 138 by the time limit
Student-S/6b	v6b, `anny_distill_vits_v6b`, ep. 140	Student-S/6, epoch 137	S+p	A	$150 \times 1{,}000 \times 8$, $2 \cdot 10^{-5}$, halved every 60; teacher in bfloat16
Student-S/6c	v6c, `anny_distill_vits_v6c`, ep. 149	Student-S/6b, epoch 140	S+p	A	$150 \times 1{,}000 \times 16$, $3 \cdot 10^{-5}$, halved every 60
Student-S/mixed	v7, `anny_distill_vits_v7_mixed`, ep. 130	Student-S/6b, epoch 140	S+p	A+P, old	$150 \times 1{,}000 \times 8$, $2 \cdot 10^{-5}$, halved every 60
Student-S/final	v8, `anny_distill_vits_v8_final`, ep. 52	Student-S/6c, epoch 149; teacher Mixed-L	S+p	A+P, old	$150 \times 1{,}000 \times 16$, $3 \cdot 10^{-5}$, halved every 60
ViT-B student (requirement 1b)					
Student	exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30`; job 3614191 (done 19 Oct)	NAVER's `multiHMR_672_B` backbone; every head tensor from Final; teacher Final	B+p, all	A+P, v23	$30 \times 1{,}000 \times 8$, $3 \cdot 10^{-5}$, halved every 12; Final's targets and masks; holdout 119.6, 3DPW 110.1 mm, 95 % rule not met
Student, 60 epochs	exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e60`, ep. 59; job 3623342	as Student	B+p, all	A+P, v23	as Student, but 60 epochs, halved every 24; holdout 107.7, 3DPW 110.9 mm, 95 % rule not met; training-length experiment, not the reported model
Distillation-temperature sweep (submitted 6 Nov, results pending)					

Thesis name	Old run / checkpoint name	Starting weights	Backb.	Data	Schedule; what is new
Student, $T = t$	exp8 (submitted by exp9) `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30_kdT<t>_sweep`; jobs **[TODO: ids pending]**	as Student	B+p, all	A+P, v23	as Student, with feature-distillation temperature $t \in \{1, 2, 4, 8, 16\}$; evaluation after every epoch and validation loss on (both only measure)

A.2. Environment

All training, evaluation and demo jobs ran on the DFKI Pegasus Slurm cluster, from the repository clone in `/netscratch/najib/multi-hmr`. The repository is public on GitHub (`saadnajib/Multi-Person-Whole-Body-Human-Mesh-Recovery-in-a-Single-Shot-with-Annyone-dataset-and-3dpw-dataset`). Checkpoints, logs and datasets are not in it. Each job script prints the short git hash of the clone into the run's `settings.txt`, so a run can be matched to a commit.

Every job runs inside the container image `ubuntu+20.04_v4.sqsh` (in `/netscratch/najib/ContainerImages/`), started by `srun` with the option `--container-image` and the mounts `/fscratch`, `/netscratch`, `/home/najib` and `/dev/shm`. Inside it, `slurm/env.sh` is sourced first. It activates the conda environment `multihmr` (Python 3.9, PyTorch 2.0.1 with CUDA 11.7, xformers 0.0.20, torchvision 0.15.2) and moves the Hugging Face, Torch and Anny caches to `/netscratch/najib`. It also sets one thread per data-loader worker and strips the `X | None` type hints from the downloaded DINOv2 hub code, because Python 3.9 rejects them. `slurm/egl.sh` then installs the EGL libraries for the headless renderer. The video launcher also tests a real render and stops before loading a checkpoint if that test fails.

The landmark detector of the video pipeline, MediaPipe 0.10.14, runs in its own Python 3.10 environment, with the packages pinned in `requirements-video-landmarks.txt`. There, `scripts/video/extract_pose_landmarks.py` reads the video once on the CPU and writes a JSON file before the GPU job starts. `docs/video_inference.md` creates this environment at `/netscratch/najib/venvs/video-landmarks`.

That PyTorch build has GPU kernels up to compute capability 8.6. For this reason the training and evaluation scripts ask for A100, A40, L40S and RTX A6000 nodes and refuse an H100 or H200 (compute capability 9.0). The V100 nodes `gifu` and `glendale` are excluded: every converter task sent there failed within a minute with an argument-parser error whose cause I do not know. Only the 3DPW converter ran on H100 nodes, through a small virtual environment with PyTorch 2.1.2 for CUDA 12.1 on top of `multihmr` (`PY_VENV=/netscratch/najib/venv_cu121`). Steps that need only NumPy (building the 3DPW label files, the re-fit check) run on the login node. The model-cost script needs a GPU node, because `model.py` imports the renderer and the login node has no EGL.

A.3. Runs, jobs, logs and checkpoints

Table A.2 lists every job behind a number, a table or a figure of the thesis. All of them have finished except the 5 sweep jobs, which I submitted on 6 November. The table still lacks the job ids of the sweep, of the five-panel video and of the stage photo.

Table A.2.: *The Slurm jobs behind every number, table and figure of the thesis, with the log and the checkpoint or output. Model names as in Table A.1. Paths are relative to the project folder on the cluster; the log of job j is* `logs/<logname>_j.out` *(array jobs:* `_j_<task>.out`*; the video and image jobs:* `logs/<logname>-j.out`*, with a hyphen). E7 stands for* `results/experiments/exp7_final_model/`*, M for* `models/multiHMR/`*.*

Model or item: what the job did	Job(s)	Log name; checkpoint or output
Models		
Base: holdout trace (V1 PVE, placement)	3570222; re-trace 3571596	`diag_holdout`; `logs/anny_model/anny_s2_shape_v5/checkpoints/00099.pt`
Mixed-L: holdout trace	3571963	`diag_holdout`; `M/anny_teacher_v9_mixed_00062.pt`
Fresh-heads-L: holdout trace	3571794	`diag_holdout`; `logs/anny_model/anny_v10_scratch_mixed/checkpoints/00290.pt`
Final: training, holdout trace, 3DPW test	3611336 (smoke test 3611333)	`exp7_final_model`; `E7/v5_pc_dist5_bonefix_v23_e10/train/checkpoints/00009.pt`, copy `M/exp7_v5_final_00009.pt`
Control: the same	3611339	`exp7_final_model`; `E7/v5_root_dist1_nofix_v23_e10/`
Final-masked: the same	3612325	`exp7_final_model`; `E7/v5_pc_dist5_bonefix_v23_e10_shpmask/`
Final-partial-mask: the same (done 19 Oct)	3614042	`exp7_final_model` (holdout table lines 191–215); `E7/v5_pc_dist5_bonefix_v23_e10_shpmask_noage/`
Student: distillation from Final, 30 epochs, own evaluation (done 19 Oct)	3614191 (smoke test 3613926)	`exp8_distill_vitb`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt`; `final_table.json` in the same run folder
Multi-HMR-Anny: holdout and 3DPW test with my code	3614192 (3613928 crashed: FFN width)	`eval-test3dpw` (task 0); `M/multiHMR_672_L_anny.pt`
Model cost: parameters, GFLOPs, latency of Multi-HMR-Anny and Final (A100)	3614947	`model_cost`; `results/diagnostics/model_cost_naver_vs_final.json`
Training length and distillation temperature (added 6 Nov)		
Student, 60 epochs: the same recipe trained twice as long, own evaluation	3623342	`exp8_distill_vitb`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e60/train/checkpoints/00059.pt`; `final_table.json` in the same run folder
Loss curves of the 30- and 60-epoch runs, from their tensorboard files (Figure 5.2)	none (login node)	`scripts/analysis/plot_loss_curves.py`; `results/experiments/exp9_kd_temperature/loss_curves/<run>_loss_curves.png`, `.pdf`, `.csv`; thesis copies `figures/img/F23_student_loss_curves_e30.png`, `_e60.png`

Model or item: what the job did	Job(s)	Log name; checkpoint or output
Temperature sweep, 5 runs (submitted 6 Nov)	3717507, 3717530, 3717538, 3717581, 3717598 ($T = 1, 2, 4, 8, 16$)	`exp8_distill_vitb` (one job per temperature, submitted by `slurm/experiments/exp9_kd_temperature.sh`; record `logs/exp9_kd_temperature_<date>.txt`); collector `scripts/analysis/kd_temperature_sweep.py` writes `results/experiments/exp9_kd_temperature/kd_temperature_sweep.csv`, `.png`, `.pdf`
Evaluations and demos		
3DPW test (v23): Base, Mixed-L, neck pilot (Final and Control in their own jobs)	3612073, 3612074, 3612075	`eval-test3dpw` (task 0), lines 117–125
3DPW test (v23): Student-S/6c, Student-S/final, also AnnyOne test-100	3615020_2, 3615020_4	`eval-test3dpw`, line 119 (_2) and 118 (_4)
Relative Human: Base, neck pilot, Final, Final-masked, Final-partial-mask	3611285_0, _1, 3612131, 3613616, 3616821	`eval_rh`; `results/diagnostics/relative_human_val_<tag>/`
Demos: neck pilot, Final, Final-masked, each next to Base	3581312, 3612078, 3613617	`demo-compare`; `demo_montage_photos/`, `demo_montage_3dpwtest/`
Demos: predicted fingers (Base, Final); rest pose (Base, Final, Final-masked)	3615018, 3615019	`demo-compare`; scripts `slurm/demo/run_demo_compare_hands.sh`, `_rest.sh` (copies made on the cluster)
Video, raw Student and Multi-HMR-Anny on `man_walking.mp4` (frame and match counts)	3642415	`video-compare` (log `logs/video-compare-3642415.out`); `results/video_compare/video_3642415/comparison.mp4`
Video, five panels with temporal filter and landmark alignment (Figure 5.5)	3657550	`video-compare`; `slurm/demo/run_video_compare.sh` with `TEMPORAL=ema`, `ALIGNMENT=landmarks` and `LANDMARKS=<json>`; landmarks from `scripts/video/extract_pose_landmarks.py` (separate MediaPipe 0.10.14 environment); `results/video_compare/video_3657550/`: `comparison.mp4`, `timing.csv`, `student_temporal/`, `student_aligned/alignment.json`; thesis copy `figures/img/F25_video_panels.jpg` (frame 178 of the video in the final presentation)
Stage photo, Student and Multi-HMR-Anny (Figure 5.4)	3657637	`image-compare`; `slurm/demo/run_image_compare.sh`, which runs `demo_image_compare.py`; `results/image_compare/images_3657637/comparison/`; thesis copy `figures/img/F26_image_compare.jpg` (cut from the final presentation)
3DPW labels		
Label check before and after the re-fit	3575369, 3611284 (v23)	`check_3dpw_conversion`; `results/diagnostics/check_3dpw_conversion_v23/`
Re-fit with converter v2.3, 87 person-sequences	3590385, 3602813	`smpl2anny-all`; `threedpw_anny_v23/labels/`; `logs/check_refit_all_v23.log`
Label rebuilds from the old fits	3579266, 3581320	`3dpw-labels`; `threedpw_anny_full/labels_rebuilt/`
Pilots and diagnostics (outputs in `results/experiments/` and `results/diagnostics/`)		
Phenotypes; youngest 50 training people	3569948; 3569950	`diag_pheno`; `diag_young`
Real-photo ages with the crop camera: Base, sampler pilot, its control	3569949, 3572419, 3572420	`diag_cropK`

Model or item: what the job did	Job(s)	Log name; checkpoint or output
Depth pilots, 2 epochs (×1, control, ×5, ×20) and 10 epochs (×1, control, ×5)	3571598, 3571599, 3572418, 3572483; 3571933, 3571934, 3572482	`exp1_depth_anchor`
Sampler pilot and control; batch ages only	3571955, 3571956; 3571601	`exp2_children`
Head-loss pilot; two identical controls (noise)	3571957; 3571958, 3571960	`exp3_neck_global_rot`; `exp4_fingers` (3571960)
Finger pilot (control 3571960: crop trace, hand size)	3571959	`exp4_fingers`
Geodesic pilot and its L1 control	3573520, 3573521	`exp5_neck_fix`
Per-bone dumps of eight checkpoints	3573518, 3573519	`diag_neck_bones`; `results/diagnostics/neck_bones/`
Bone correction: Base (with the neck pilot), Fresh-heads-L	3579014, 3579015	`exp6_bone_fix`; `results/experiments/exp6_bone_fix/anny_s2_shape_v5_00099_n200_s0/bone_fix.json`; neck pilot `M/exp6_v5_bonefix_00001.pt`

The scripts behind these jobs and the exact commands that produced the thesis numbers are listed in the file `docs/thesis/REPRODUCE.md` of the repository. Each command runs from the login node in the repository clone. The flags each training run passes are in Appendix C.

A.4. What cannot be reproduced exactly

The training code sets no random seed. The `--seed` of the holdout trace only picks which training images the bone correction is estimated on. The Student script passes its seed to that trace and nowhere else (`exp8_distill_vitb.sh`, lines 257 and 1142). The sweep runs through the same script, so this also holds for the 60-epoch run and for every temperature. The order in which training images are drawn (with replacement) therefore differs between two runs of the same command. So does the number of optimiser steps skipped for non-finite gradients (16 in Final, 10 in Control, out of 10,000).

Two runs of one 2-epoch configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table F.7). A repeat of Final or Control should land within such a margin, not on the same numbers. Final and Control also ran on different GPU types (Final on an A100, Control on an A40, both with TF32). This changes the speed, and it can change the last digits of the arithmetic. Evaluating a fixed checkpoint involves no sampling. I did not test whether it repeats exactly across GPU types.

Apart from the temperature, the sweep runs evaluate after every epoch instead of every fifth and also compute the training loss on the validation images (`--val_loss 1`). Both additions only read values out of the model; neither changes a weight. Because of this, the sweep's own $T = 4$ run repeats the reported recipe and will give a second estimate of run-to-run variation.

There are three more things I cannot repeat. The history scripts of the L line and of Student-S were edited after their runs, so their recipes in Table A.1

are as recorded there, not re-run. `train.py` keeps only the last two or three checkpoints of a run, so intermediate epochs are gone. The checkpoints, the AnnyOne and 3DPW data and the Relative Human annotations are not in the repository, and 3DPW and Relative Human come with their own licences.

A.5. Evidence that the video comparison ran

For the video comparison of job 3642415 I have the log `video-compare-3642415.out` and the matching error log. It used the 30-epoch Student checkpoint `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt` with the correction selected by that run's `student_fix_arg.txt`. The baseline was `models/multiHMR/multiHMR_672_L_anny.pt` and the input was `man_walking.mp4`. Both checkpoints loaded with 100 % state coverage, meaning every saved weight found its place in the model. The renderer reported an NVIDIA L40S and passed its test mesh.

The final log lines report 356 processed frames for each model, 356 student matches and 354 baseline matches. The output is `results/video_compare/video_3642415/comparison.mp4`. The job ran before the temporal filter and the later timing code existed. So its log does not give the inference speed of the two models measured on the same frames, and it has no number for how much the pose or the body shape changes from frame to frame.

The five-panel video of Section 5.5 came from a later run of the same launcher with `TEMPORAL=ema` and `ALIGNMENT=landmarks`. It adds the filtered and the aligned Student panels (Section 4.9.1) and a frame-rate label on every panel. This run is job 3657550 (`results/video_compare/video_3657550/`). It ran on one NVIDIA A40 in fp32 with TF32 at 672 pixels, timed 346 frames after 10 warm-up frames, and used the filter constants of Section 4.9.1. Its own summaries give the same matched-frame counts as job 3642415: 356 of 356 for the Student and 354 for the baseline.

The frame-rate label on each panel counts a different amount of work (Table 5.6; definitions in `docs/video_inference.md`, lines 387–440). `timing.csv` has one row per model panel, and the input panel's playback rate is a column of those rows. The raw Student and the baseline show `inference_fps`, model inference alone. The filtered panel shows `processing_fps` from `student_temporal/summary.json`, which also counts the filter and the Anny re-decode. The aligned panel shows the same quantity from `student_aligned/summary.json`, which counts the filter work as well as the alignment fit, but not the offline landmark extraction.

The video implementation and the EGL correction are in commits `1d5621b` and `7b0b5e1`, and the revised FPS implementation is `a0fa59f`. The video comparison is launched through `slurm/demo/run_video_compare.sh`, and `docs/video_inference.md` describes how to run it, where each timing starts and stops, and which files it writes. The stage photo of Section 5.5.1 was made by `demo_image_compare.py`, normally launched through `slurm/demo/run_image_compare.sh`, whose defaults are the same Student,

correction and baseline (`docs/image_comparison.md`). It is job 3657637, with its output in `results/image_compare/images_3657637/`.

B. SMPL–Anny joint correspondence

The converter has to know which Anny bone stands for each SMPL joint. The two skeletons are the ones in Table 2.1: SMPL has 24 joints and Anny has 163 bones, of which the root and 18 rotate in the fit. Some joints of the two rigs sit at different places, so names alone are not enough. The converter compares the two rest skeletons (the default standing pose) before it fits the posed frames.

For each hip, Anny has two possible bones, the upper-leg bone and the pelvis bone. I take the one whose head (the point where the bone starts) lies about as far from the pelvis as SMPL's hip joint; for the v23 fits this is the upper-leg bone. For the spine, I order the Anny spine bones by their height, scaled so that the hip midpoint is 0 and the neck is 1. I then keep the ordered triple of bones whose heights are together closest to the heights of SMPL's three spine joints. For the limbs and the head, the converter takes the Anny bone of the same body part from a fixed list of names: the shoulder, for example, goes to the head of the upper-arm bone. Each run saves the bones it picked together with the fits.

For the torso joints, the converter fits a virtual joint (Equation (3.2)): a point that sits at a fixed offset from the Anny bone head. The converter measures this offset once, in the rest pose. Let $\mathbf{j}_k^S$ be the position of SMPL joint k, $\mathbf{o}_A$ and $\mathbf{o}_S$ the hip midpoints of the Anny and the SMPL rest skeleton, s the ratio of Anny's to SMPL's hip-midpoint-to-neck length, $\mathbf{Q}_{S\to A}$ the matrix that lines up the axes of the two rest skeletons, and $\mathbf{h}_b^0$ and $\mathbf{Q}_b^0$ the rest head and orientation of the bone b assigned to joint k:

$$\mathbf{j}_k^{S\to A} = \mathbf{o}_A + s\mathbf{Q}_{S\to A}(\mathbf{j}_k^S - \mathbf{o}_S), \qquad \boldsymbol{\delta}_k = (\mathbf{Q}_b^0)^\top(\mathbf{j}_k^{S\to A} - \mathbf{h}_b^0). \tag{B.1}$$

In words, the first formula moves the SMPL joint into Anny's rest skeleton. It shifts the joint from SMPL's hip midpoint to Anny's, scales it by s and turns it with $\mathbf{Q}_{S\to A}$. The second formula then measures the offset from the Anny bone head, in that bone's own frame. The offset stays fixed and turns with the bone, so it is not an extra joint fitted in each frame.

Each fit stores its own correspondence in three arrays: `anny_bone_names`, `joint_offsets_local` and `offset_joints`. To reproduce a particular fit, use these arrays and not this table. The code is in `scripts/data/smpl_to_anny.py`. Appendix G shows how geometric matching and virtual joints changed the fit error, and why matching by name failed in my first converter.

SMPL target	Anny correspondence and fitting point
Pelvis	root bone with a measured torso offset
Left/right hip	upper-leg or pelvis bone, chosen by its distance from the pelvis (the upper-leg bone in v23), with an offset
Spine1, spine2, spine3	spine bones matched by normalised rest height, in order, with offsets
Neck	neck candidate with a measured rest-frame offset
Left/right collar	clavicle bone with an offset
Shoulder, elbow, wrist	heads of the matching upper-arm, lower-arm and wrist bones
Knee, ankle, foot	heads of the matching lower-leg, foot and toe bones
Head and hand targets	heads of the matching bones; their pose follows from the parent bones that the fit is allowed to rotate

Table B.1.: *Correspondence used by the converter for the v23 labels. The exact bone names and offsets are saved with each fit and can differ between Anny rig versions. The fit rotates the root and 18 body bones; the others, including most fingers and the finer head and neck bones, keep their rest rotation.*

C. Flags and weights

This appendix lists the command-line flags I passed to the training script, so that each run can be repeated. Table C.1 covers Final and Control. The Student of Section 4.6 and the 60-epoch run of Section 5.3 share Table C.2. The temperature sweep of Section 5.4 has its own, Table C.3.

I list a flag when its value differs from the default in the code, or when it keeps the default but matters for the result. Every other flag keeps its default. The loss weights are in Table 4.1 of Chapter 4. The commands that start Final, Control and the Student are in the file `docs/thesis/REPRODUCE.md` of the repository (Appendix A). I started the 60-epoch run with `EPOCHS=60` in front of the same Student command, and the sweep with `slurm/experiments/exp9_kd_temperature.sh`.

Table C.1.: *Training flags of Final and Control. "Value" is Final's. Where Control differs, its value follows in brackets. Paths are relative to the project folder on the cluster. Both runs start from Base and train for 10 epochs of 1,000 iterations at batch 4, that is 40,000 images, at a constant learning rate.*

Flag	Default	Value (Control)	Meaning
Start and network			
`--pretrained`	none	Base checkpoint	weights the run starts from, heads included
`--pretrained_remap`	1	0	no renaming of weight names; Base was saved by this code
`--backbone`, `--img_size`	ViT-S, 336	`dinov2_vitl14`, 672	DINOv2 ViT-L, 48 × 48 tokens of 14 pixels
`--freeze_backbone`, `--unfreeze_last_n_blocks`	0, 0	1, 4	backbone frozen except its last 4 of 24 blocks and final norm
`--bone_fix`	off	Base's `bone_fix.json` (`none`)	constant output rotation on neck02, head, metacarpal1.L and finger4-1.R, stored in the checkpoint (Section 4.7.2)
Targets and losses			
`--gt_center_joint`	root	`person_center` (`root`)	the joint the depth, translation, cell and offset targets describe: the head, which the network anchors (Control: the pelvis)
`--alpha_dist`, `--alpha_transl`	1, 1	5, 5 (1, 1)	weights of the depth and translation terms
`--alpha_shape`, `--alpha_j2d`, `--alpha_v2d`	1, 1, 1	20, 0.1, 0.1	shape and 2D loss weights
`--boost_neck_weight`, `--boost_hand_weight`	1, 1	4, 4	extra weight on neck, head and hand bones, renormalised to mean 1
`--start_2d_epoch`	10	0	2D terms from the first epoch
`--mask_helper_joints_train`	0	1	hold the 88 helper bones at identity, no rotation targets
`--use_anny_shape`	0	1	phenotype targets from the dataset

Flag	Default	Value (Control)	Meaning
`--mask_3dpw_hands`	0	1	no rotation targets for the 38 hand bones of 3DPW people
`--mask_3dpw_shape`	0	not passed	shape mask for 3DPW people; 1 only in Final-masked and Final-partial-mask
Data			
`--train_data`, `--val_anny_n`	BEDLAM, 0	AnnyOne, 100	AnnyOne for training, its last 100 images as the holdout
`--exclude_indices_file`	none	the 55 manifest images	never sample the diagnostic-manifest images
`--mix_3dpw_labels`	none	`threedpw_anny_v23/labels`	corrected 3DPW labels mixed in, 30 % of a batch on average (default ratio 0.3)
`--val_3dpw_n`	0	300	evaluate on 300 evenly spaced 3DPW validation images
Optimisation, bookkeeping, job			
`--batch_size`, `--learning_rate`	2, $5 \cdot 10^{-6}$	4, 10^{-5}	images per step; Adam learning rate
`--n_iters_per_epoch`, `--max_iter`	100, 10,000	1,000, 10,000	10 epochs of 1,000 steps
`--amp`	1	0	no mixed precision (it produced NaNs with the Anny layer); TF32 stays on
`--num_workers`, `--nb_max_ckpt`, `--log_freq`	0, 10, 100	12, 2, 50	bookkeeping; output in `results/experiments/exp7_final_model/<run>/train`
`TIME`, `EVAL_TEST` (script)		5 h, 1	Slurm limit on A100, L40S, RTX A6000; after training the holdout trace and the 3DPW test (v23)

Table C.2.: *Training flags of the Student where they differ from Final's. Every other flag is as in Table C.1. This includes the targets, the masks, the loss weights and the 3DPW mixing, so the Student also learns the corrected depth. There is no shape mask. The Student trains for 30 epochs of 1,000 iterations at batch 8, that is 240,000 images, with a 20-hour Slurm limit. The script passes the colour-jitter flags (`-brightness 0.2` and so on), but the code never reads them, so there was no augmentation. Of the flags below, the 60-epoch run of Section 5.3 differs only in the two values in brackets. Its job settings also differ: it has its own run name, ending in `_e60`, and the script computes its time limit from the epoch count instead of using 20 hours. That run's settings file is not in the repository. I took the two values from the script's rule (the learning rate halves every 2/5 of the epochs) and from the recorded launch command, which set only the epoch count. The loss curves of Figure 5.2 drop at the expected epochs.*

Flag	Default	Value (60 epochs)	Meaning
`--backbone`	ViT-S	`dinov2_vitb14`	DINOv2 ViT-B
`--pretrained`	none	`models/multiHMR/multiHMR_672_B.pt`	NAVER's ViT-B as the backbone start
`--pretrained_remap`, `--load_only_backbone`, `--freeze_backbone`	1, 0, 0	1, 1, 0	rename NAVER's weight names; load the backbone only and train all of it
`--head_dim`	0	1024	heads at the teacher's width, fed by a 768 to 1024 projector

Flag	Default	Value (60 epochs)	Meaning
`--distill_teacher_ckpt`, `--teacher_bf16`	none, 0	Final, 1	frozen teacher, its backbone in bfloat16; turns the run into distillation
`--init_heads_from_teacher`, `--freeze_heads_epochs`	0, 0	1, 5	copy every matching head tensor from the teacher; keep them frozen for 5 epochs
`--lambda_kd`, `--lambda_kd_out`	1.0, 0	1.0, 1.0	feature distillation (KL, temperature 4) and output distillation (L1)
`--bone_fix`	off	the teacher's	Final's correction in training; the Student's own, estimated on 200 training images, for evaluation
`--batch_size`, `--learning_rate`	2, $5 \cdot 10^{-6}$	8, $3 \cdot 10^{-5}$	images per step; Adam learning rate
`--max_iter`, `--lr_decay_every`	10,000, 30	30,000, 12 (60,000, 24)	30 epochs; learning rate halved after epochs 12 and 24 (60-epoch run: after 24 and 48)
`--eval_freq`, `--nb_max_ckpt`	1, 10	5, 3	evaluate every 5 epochs, keep 3 checkpoints

Table C.3.: *Training flags of the distillation temperature sweep that differ from the Student's (Table C.2). Each temperature T is one full Student run. I repeat $T = 4$ to measure how much two runs with the same training settings differ; its result is pending (Section 5.4). The 30- and 60-epoch runs were trained before* `--val_loss` *existed, so they have no validation loss.*

Flag	Default	Value	Meaning
`--kd_temperature`	4	1, 2, 4, 8, 16	softens the channel softmax of the feature distillation; the output L1 has no temperature
`--eval_freq`	1	1	evaluate after every epoch (Student: every 5); this changes nothing in training
`--val_loss`	0	1	a new flag that computes the training loss on the validation images in evaluation mode, without gradients, so it does not change the training either
`RUN` (script)		`..._kdT<t>_sweep`	own output folder per temperature
`TIME` (script)		16 h	Slurm limit per temperature (Student: 20 h); each temperature is one job

D. Extra figures

The chapters refer to the figures below but do not show them, so that the main text stays short. This appendix has both depth examples and two more bone-correction pairs on the holdout. It also has a second neck pair on 3DPW, the other demo frames and photos, and the in-training curves of Final and Control. The last two figures show all twelve demo images with predicted fingers and nine more rest-pose line-ups.

I judged every image panel by eye, so none of them is a measurement. Where a body is placed from the detector box, a wrong depth cannot be seen in the picture.

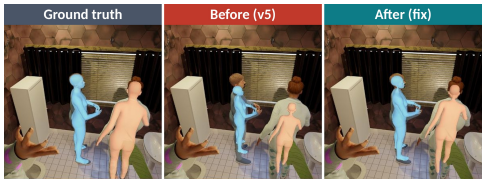


Figure D.1.: *Holdout image 579862 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). The predicted body goes from 0.76 to 0.95 of its true height. Synthetic image. This is one of the good cases, not a typical one. Image 579860 is on the right, in Figure D.2.*



Figure D.2.: *Holdout image 579860 rendered at each model's own predicted depth: ground truth, Base, and the same depth pilot as on the left. In the Base panel the ground-truth head shows above a shrunken body that is placed too far away. Synthetic image. This is one case only, and not a typical one.*

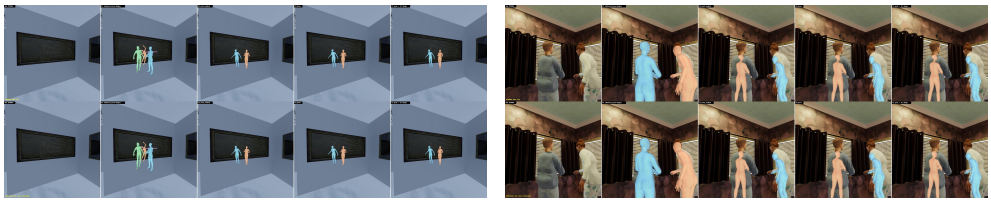


Figure D.3.: *Holdout images 579864 (left) and 579866 (right). In each image the top row is Base as trained and the bottom row is the same Base checkpoint with the bone correction applied to its output, with no further training (Section F.4). Synthetic images at the model's own predicted depth, so the depth bug still draws the bodies too small. I judged the necks by eye; there is no metric. These go with Figure F.1.*

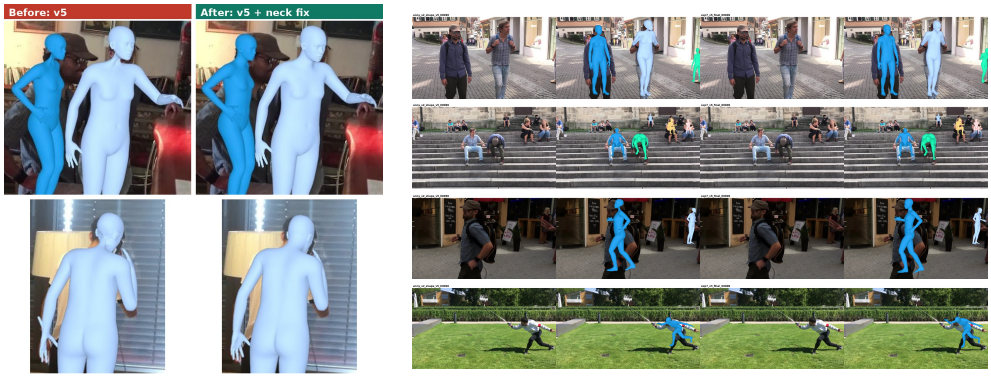


Figure D.4.: *Left: 3DPW test frames* `downtown_bar_00` *701 (top) and* `office_phoneCall_00` *440 (bottom), Base next to the neck pilot (the bone correction and a 2-epoch fine-tune); necks ringed. Bodies placed from the detector box, fingers at rest pose. Judged by eye, no metric. The two columns differ by the correction and also by the fine-tune.*

Figure D.5.: *Right: the other four 3DPW test frames of the demo, from top to bottom* `downtown_walking`*, stairs,* `downtown_runForBus` *and fencing. Each shows the photo and Base, then the photo and Final (labels in the image). Placed from the detector box, fingers at rest pose. Judged by eye, no metric. The first four frames are in Figure D.10.*



Figure D.6.: *The four real photos of the demo (garden and tennis in the top row, stage and party below). None of them was used in training. Each shows the photo and Base, then the photo and Final. Placed from the detector box, fingers at rest pose. Judged by eye, no metric. The tennis photo is also in Figure D.14.*

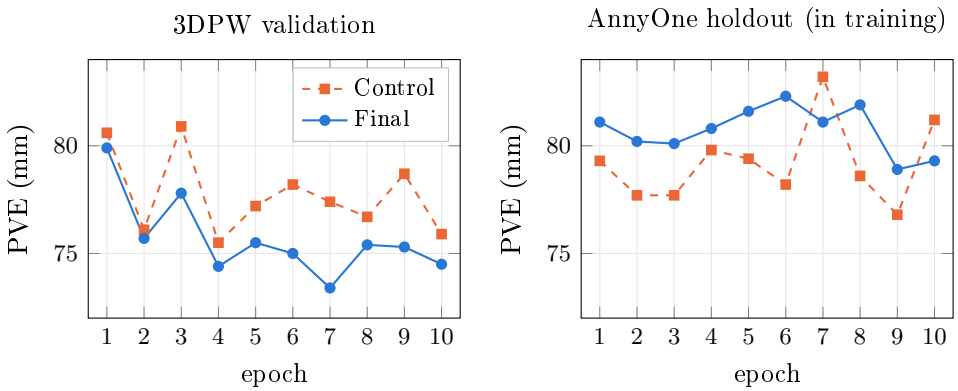


Figure D.7.: *PVE of Final and Control after each training epoch (one run each, no seed). The x-axis is the epoch and the y-axis the PVE; lower is better. Left: 300 images of the corrected 3DPW validation labels. Right: the AnnyOne holdout as evaluated during training, in setting V0 (helper bones unmasked), so the last-epoch values are higher than the V1 numbers of Table 5.7a (73.7 mm for Final and 77.4 mm for Control). Values from the training logs; read in Section 5.6.1.*



Figure D.8.: *All twelve demo images with the predicted fingers (not set to rest pose), each as photo and Base, then photo and Final (labels in the image). Row by row: the garden and tennis photos; the stage photo and fencing; stairs and* `downtown_walking`*;* `downtown_runForBus`*, the party photo and guitar;* `downtown_bar`*,* `office_phoneCall` *and the weekly market. Demo setting, placed from the detector box. Base still has the flipped base bone of the left index finger (metacarpal1.L) and of the right ring finger; Final has the correction. In most images the hands are only a few pixels wide and cannot be judged, so Figure D.12 zooms in on the three where they can. Judged by eye, no metric.*



Figure D.9.: *The other nine demo images in rest pose, each as photo and Base, photo and Final, photo and Final-masked (labels in the image). From top to bottom: the garden photo, the 3DPW test frames fencing, stairs,* `downtown_walking` *and* `downtown_runForBus`*; then the party photo and guitar,* `office_phoneCall` *and the weekly market. Every detected person stands in rest pose, so only the body shape differs between the models. Demo setting, placed from the detector box. At this size only large differences in build can be seen. Judged by eye, no metric. The other three images are in Figure D.13.*



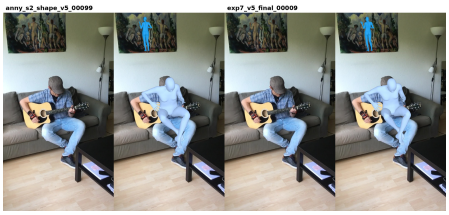
(a) `downtown_bar`



(b) *weekly market*



(c) `office_phoneCall`



(d) *guitar*

Figure D.10.: *Four 3DPW test frames, each shown as photo and Base (left half), then photo and Final (right half; the labels in the image name the two checkpoints). Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The other four demo frames are in Figure D.5 (Appendix D).*

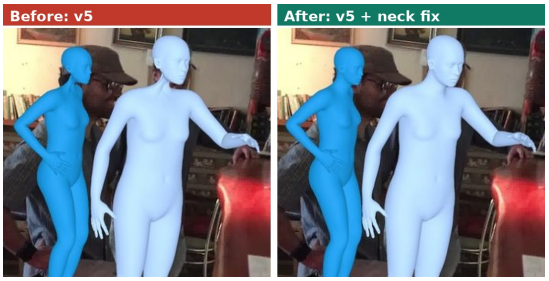
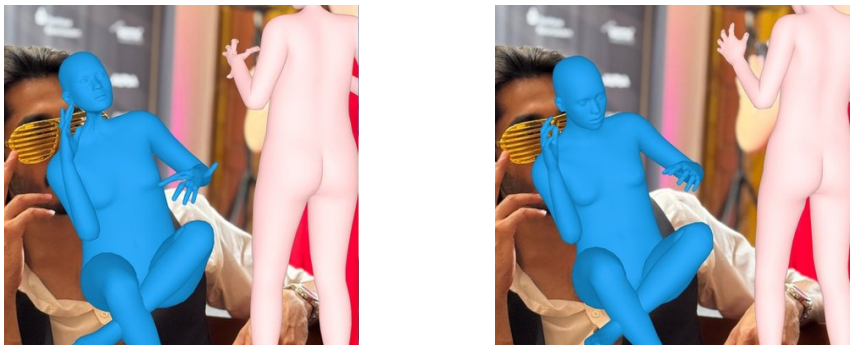


Figure D.11.: *Real 3DPW test frame* `downtown_bar_00` *701: Base (left) and the neck pilot, fine-tuned 2 epochs with the correction (right); necks ringed. Bodies placed from the detector box; judged by eye. The columns differ by the correction and the fine-tune. A second frame is in Figure D.4.*

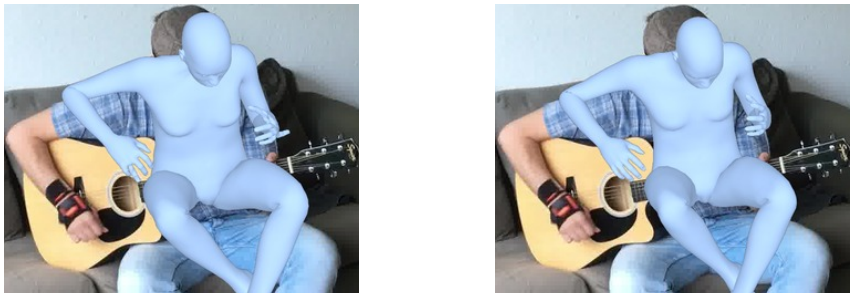
D.1. Figures of the teacher ablations

These figures belong to the teacher ablations in Section 5.6 and to the shape analysis in Section 5.7.1. I moved them here to keep Chapter 5 short; the text there refers to each of them.

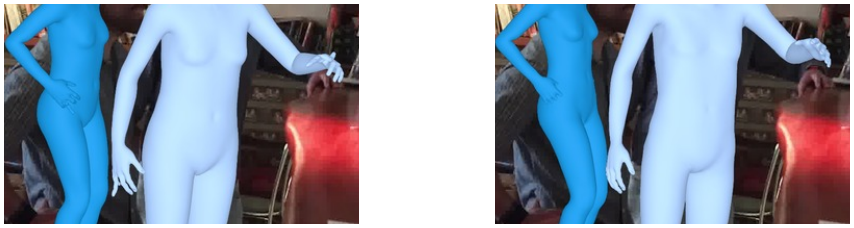
Base, predicted fingers | Final, predicted fingers



(a) party photo: the hand at the face and the hand over the armrest

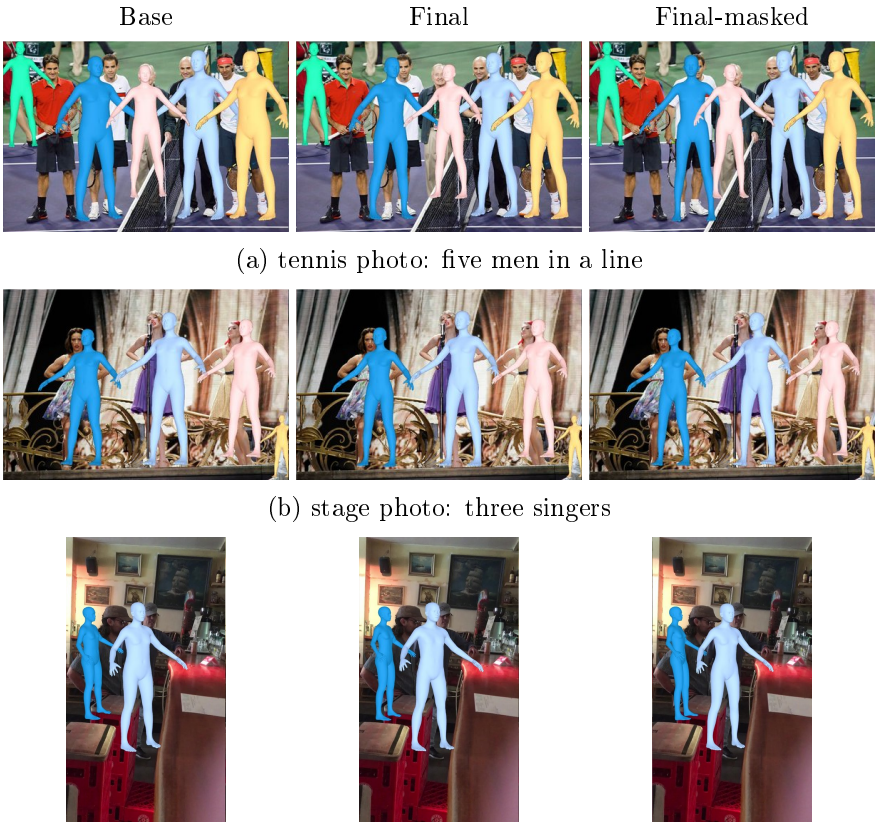


(b) 3DPW test frame, guitar: strumming hand (left) and fretting hand (right)



(c) 3DPW test frame `downtown_bar`: the hand on the bar and the hanging hand

Figure D.12.: *Predicted fingers (not set to rest pose) on three demo images, zoomed on the hands; Base left, Final right. Bodies placed from the detector box, so meshes sit beside their people. Base still carries the two flipped hand bones; Final has the correction. Neither model puts the fingers on the real ones. Three images, judged by eye; the columns differ by the correction and the fine-tune. All twelve montages are in Figure D.8.*



(a) tennis photo: five men in a line

(b) stage photo: three singers

(c) 3DPW test frame `downtown_bar`: two men

Figure D.13.: *Rest-pose line-up on three demo images: every detected person drawn in rest pose over the photo, so only body shape differs; Base, Final and Final-masked from left to right. Bodies placed from the detector box; judged by eye, one run per model. The other nine images are in Figure D.9.*

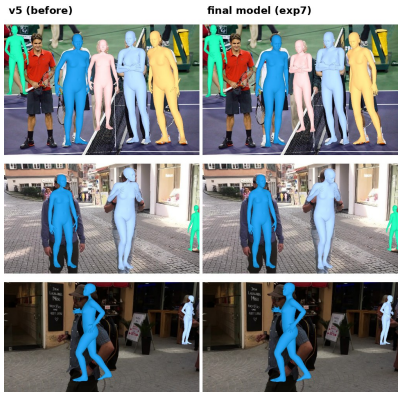


Figure D.14.: *Three real photos never used in training (a tennis photo and the 3DPW test frames* `downtown_walking_00` *and* `downtown_runForBus_00`*): Base left, Final right, bodies placed from the detector box. Most men get male bodies, women stay female, the boy stays an adult body. Judged by eye; no metric exists for these images.*

E. Pilot results in full

Before the final runs I tested six changes in pilots. A pilot is a short run, usually a 2-epoch fine-tune of Base, that tests one change. I compare each pilot with a control, which is the same training without the change, and Table E.1 names the control of each pilot. Appendix F explains what each result means. The second section is older history, the ViT-S students I trained first and the stages that produced Base.

E.1. The six pilots

Every pilot fine-tuned Base on AnnyOne only, at learning rate 10^{-5} and batch 4, for 2 epochs (8,000 images) unless noted. I evaluated all of them on the 100-image AnnyOne holdout in setting V1. Table E.1 gives the success rule that I wrote into each script before the run. The full depth runs are in Table F.3.

The table uses three terms that have not come up before. The geodesic angle is the angle of the single rotation that turns the predicted bone into the true one. The loss I normally train with does not use this angle. It compares the two rotation matrices element by element with L1. A flagged bone is one that the per-bone dump of Base in Appendix F marks as flipped or as turned by a constant offset. Age bias is the mean of the predicted minus the true age phenotype, so it says whether the predicted age is too high or too low on average.

Table E.1.: *The six pilot experiments. V1, 100-image AnnyOne holdout, matched people; fine-tunes of Base, one run each, without a seed. The rule is the success rule written into each script before the run. Head direction is the global head orientation error.*

Pilot	What changed	Success rule	Result	Control
Depth pilots (2 and 10 epochs)	training targets at the head joint instead of the pelvis; depth and translation weights ×1, ×5, ×20	slope toward 1 and intercept toward 0 m, with PVE ≤ 71.5 mm and recall ≥ 89.2 %	rule not met (PVE or recall missed in every run, controls too); root error 1.21 → 0.54 m, size 0.63 → 0.84, slope 0.33 → 0.38 (×5, 10 epochs)	same schedule, targets at the pelvis, weight ×1 (one 2-epoch, one 10-epoch)
Sampler pilot	age-balanced sampling: children (age < 0.3) per batch 2.1 → 9.7–11.1 %	10 manifest children: PVE < 47.8 mm and |age bias| < 0.078; holdout PVE ≤ 71.5, recall ≥ 89.2	inconclusive: children 48.5 vs. 53.1 mm, age bias 0.078 vs. 0.085; baby age floor 0.100 → 0.102	same run with the sampler off (age ×3 and height ×2 shape weights in both)

Pilot	What changed	Success rule	Result	Control
Head-loss pilot	L1 loss on the camera-frame orientation of head, neck03 and both wrists, weight 5	head direction $<$ 12.0 deg; PVE $\leq$ 71.5 mm, recall $\geq$ 89.2 %	head direction 11.3 deg, within the controls' 11.7–13.6; safety limit missed (71.8 mm, 89.0 %)	two identical runs at weight 0
Finger pilot	region-normalised rotation loss (body, hands, head-neck averaged separately); a test-time person crop	fingers below the rest-pose floor 15.1 deg without a PVE cost	no effect: fingers 17.5 vs. 17.4 deg; crop 17.2 deg at body PVE 124.8 mm	same run with the loss off (one of the two identical runs above)
Geodesic pilot	geodesic angle instead of L1 on the rotation matrices	flagged bones down by $\geq$ 3 deg more than the control, or neck and head under 15 deg	no movement: neck02 175.6 $\rightarrow$ 175.0, head 177.9 $\rightarrow$ 177.1 deg	same run with the L1 loss
Bone-correction pilot	constant output rotation on neck02, head, metacarpal1.L, finger4-1.R; no training, then a 2-epoch fine-tune	bones $<$ 15 deg, girth $\geq$ 0.9; no training: head direction $\leq$ Base + 3 deg; fine-tuned: guards against the control	no training: fails (head direction 19.1 vs. limit 15.0 deg); fine-tuned: passes (neck02 4.7 deg, girth 0.997, fingers 8.3 deg)	the geodesic pilot's L1 control

Pilot noise. Each pilot is one run without a seed, so first I need to know how much two identical runs differ. Table F.7 (Section F.7) gives that. I trained one configuration twice with the same settings, and the two runs differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points. Restarting a fine-tune of Base cost about 2.4 mm on its own. In other words, a difference of less than about 10 mm PVE, 3 deg or 2 recall points between a pilot and its control can be training noise alone.

In the pilots, the root error, the bone angles, the girth and the finger error changed by much more than this noise. The PVE and the recall of every pilot changed by less. For this reason I compared each pilot only with its own control. I chose the two changes that went into Final because of what the diagnostics showed, and not because of a PVE gain.

E.2. The ViT-S students and the stages that produced Base

I trained Base in stages (the names are in Table A.1, the numbers in Table E.2a). L-stage 1 trained new Anny heads together with every block of NAVER's ViT-L backbone. I did not continue it, and no checkpoint of it is

Run	Change against the previous run	PVE	PA-PVE	MPJPE	Rec.
L-stage 1	fresh heads, all blocks trained (side branch)	–	–	–	–
L-stage 2	fresh heads, backbone frozen	145	–	–	–
L-stage 3	+ last 4 of 24 blocks, helper mask	73.6	–	–	–
L-stage 4	+ neck and hand loss ×4	73.6	56.5	73.2	–
Base	+ shape loss weight 20	74.3	57.5	74.4	90

(a) *The transfer schedule that produced Base (Table 4.3); recall (Rec.) in %. L-stages 2–4: evaluation during training; Base: the standalone script.*

			AnnyOne holdout, V0					3DPW test, v23[c]	
Model	Teacher	Params	PVE	PA-PVE	MPJPE	Prec.	Rec.	PVE	PA-PVE
Base (ViT-L)	–	318.4	74.3	57.5	74.4	81	90	121.8	72.0
Mixed-L	–	318.4	78.5	57.4	78.6	83	89	95.5	62.9
Student-S/2	Base	32.8	193.4	128.2	210.7	–	86	–	–
Student-S/6b[a]	Base	37	126.9	92.1	132.0	77	88	–	–
Student-S/6c	Base	37	**114.5**	**84.6**	**118.4**	81	89	142.6	91.0
Student-S/mixed[b]	Base	37	120.6	88.2	125.2	78	87	–	–
Student-S/final[b]	Mixed-L	37	120.6	86.9	125.0	76	87	121.6	83.7

(b) *The ViT-S students against their ViT-L teachers; parameters in millions, precision (Prec.) and recall (Rec.) in %. Student-S/2: feature distillation only; Student-S/6b: heads fed through a projector at the teacher's width; Student-S/6c: Student-S/6b at batch 16; Student-S/mixed and Student-S/final: continued with 3DPW images. [a]Evaluation during training. [b]30 % 3DPW images per batch, old labels. [c]Corrected labels, the script of Table 5.7b; Final: 89.0 / 57.6 mm.*

Table E.2.: *History, before the depth fix, the bone correction and the corrected 3DPW labels. AnnyOne holdout (100 images), setting V0, errors in mm; one run per model, no seed. "–": not measured with the same script. The students' old-label 3DPW numbers are retired.*

kept. L-stage 2 started again from NAVER's backbone, kept it frozen and trained only new Anny heads. From there each stage started from the one before. L-stage 3 unfroze the last 4 of the 24 blocks and added the helper mask. L-stage 4 weighted the loss on the neck and hand bones four times. Finally, Base added the shape loss weight of 20.

Before the ViT-B Student I tried smaller students to meet the size requirement. I trained a series of ViT-S students, Student-S/1 to Student-S/final, with Base as their teacher. ViT-S is the vision transformer one size below the Student's ViT-B. All runs in Table E.2, the stages and the students, came before the depth fix, the bone correction and the corrected 3DPW labels. So the table shows the numbers I measured at the time, in setting V0 (the students are in Table E.2b). The best student, Student-S/6c, had 1.54 times the PVE of its teacher.

Four things I learned from these students went into the Student's training. With feature distillation alone, nothing trains the heads, and Student-S/2 has by far the largest error in Table E.2b. For this reason the Student starts with a copy of the teacher's heads. The copied heads only worked when the projector fed them their input. With a new, randomly initialised input layer in front of them they did not. Heads at the teacher's width also did better than narrower ones. So the Student keeps every head of Final at its full width. The starting point mattered most. A backbone that NAVER had already trained for mesh recovery gave the largest gain, which is why the Student starts from NAVER's ViT-B. Last, the students were held back more by their data than by their size, so the Student trains on AnnyOne and on the 3DPW training images with the corrected labels.

Two settings made the ViT-S training steps faster. I ran the matrix products in TF32, a reduced-precision mode of the A100, and I kept the teacher's backbone in bfloat16, a 16-bit number format. Together they cut a ViT-S student's training step from 1.24 to 0.47 s on an A100, and the evaluation numbers did not change. The Student's run keeps both settings: TF32 as in Final (Table C.1) and the teacher in bfloat16 (Table C.2). I took these two step times from an earlier version of this report; the training log behind them was not kept.

F. Supporting diagnostic experiments

When I rendered the predictions of Base, three things were visibly wrong. The necks were thin and twisted, the fingers were bent into claws, and children received adult bodies. Before I changed any training setting, I tested each idea I had about the causes on its own, and several of these ideas turned out to be wrong. Chapter 4 only defines the two corrections that came out of this work; the evidence for them is in this appendix. Unless I say otherwise, every number below comes from Base on the 100-image AnnyOne holdout in setting V1. Each variant was trained once, without a fixed seed, so I compare every difference with the run-to-run variation measured in Section F.7.

F.1. The three visible failures

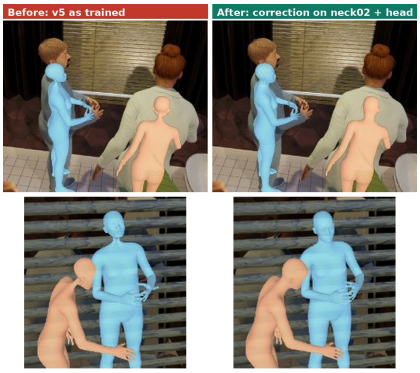


Figure F.1.: *Holdout images 579862 (top) and 579863 (bottom). Left: Base as trained. Right: the same checkpoint with the output correction of Section F.4, without training. Synthetic images, bodies at the model's own predicted depth, so the depth bug of Section F.3 still draws them too small. Necks judged by eye.*

In Figure F.1 (left) Base draws each neck as a thin stalk pinched under the head. The fingers curled into claws, so every real-photo demo set them to rest pose. A standing boy in a demo photo received an adult body (Figure D.14).

In my first draft I had written down a cause for each failure without testing it. For the necks I blamed the skinning of the stacked neck bones, which I called "a property of the rig" (the rig is the skeleton that moves the mesh). For the fingers I blamed their small size in the image. For the children I blamed an age the network had never learned, and as evidence I pointed to five small

people in the demo photos who were predicted at adult ages (0.57–0.67). The neck cause turned out to be wrong, and the small size of the fingers explains only a small part of their error. The child evidence was wrong too. Four of the five small people were adults who crouched, sat or were cut off by the frame, and their predicted height phenotype was 0.47–0.56. Only the boy was a child. He received age 0.60 and height 0.48, which is an adult body. The number that measures the thin necks is the neck girth in Table F.4, the predicted neck circumference divided by the true one.

F.2. Six structural checks

check	sample	result	verdict
1 labels decode onto the people (no network)	manifest, 55 people	84.5 % of joints inside the image, 84.9 % inside the box (the rest is truncation); dataset camera right on 100 % of images	labels fine
2 body model, age sweep	body model	skeletal height 0.53 m at age 0, 1.55 m at age 0.8; head/height 0.200 to 0.106	child shape exists
3 single-joint moves	body model	all 16 pass (96–100 % of moved vertices in the intended part)	rig fine
4 gradients	4 holdout images	zero gradient only for the 88 helpers; all 75 active bones and 11 shape outputs get gradient (598 at the root to 4.7 at the fingers); two of three camera outputs get none	path fine
5 prediction vs. truth, shape swap	holdout, manifest	neck length 99.3 % of truth; true shape lowers the 10 manifest children's PVE 47.8 → 41.6 mm	shape head is the gap
6 capped overfit	40 manifest training images	training PVE −36 % (80.7 → 51.8 mm), holdout +45 %	learns; two warnings

Table F.1.: *Six checks on Base before any training change. The manifest holds 40 training images (10 adults, 10 children, 10 with visible hands, 10 hard neck poses) and 15 holdout images without children; it is excluded from every later fine-tune. Check 6 trains a throw-away copy for 400 steps and reports V0. Synthetic images.*

Before I blamed the network, I checked the parts around it: the labels, the body model, its rig and the path of the gradient from the loss back to the weights. Table F.1 lists the six checks. Check 1, for example, decodes the labels and draws them over the image without any network, so a wrong label or a wrong camera would show up there (Figure F.2). The rig, the label decoding and the gradient path were all fine, so the cause had to lie in what the training targets and the losses asked the network to do.

I also looked at how the total loss was split at the end of Base's training. The shape term made up 25.7 % of it, the translation term 1.9 % and the distance term 0.6 %. This matters for placement. The 3D terms move both pelvises to the same point before they compare the bodies, so they cannot tell whether a



(a) *holdout 579888, a close-up with hands* **(b)** *training image 462901, low camera*

Figure F.2.: *Check 1 on two manifest images: image | ground-truth mesh | joints and box, decoded through the training body model and drawn with the dataset camera; no network. By eye the mesh lies on the person, down to the fingers. In (a) the frame cuts head and legs: truncation, not a decoding error (the black corner is in the dataset image). Synthetic; two of four rendered cases.*

person stands at the right distance. Only the distance and translation terms can, and before the depth fix the two together were only 2.5 % of the loss. Two details in the log are easy to misread. The log calls the detection term `bce`, but it is the focal loss of CornerNet [LD18] and not a binary cross-entropy. And the neck and hand boost (Table 4.1) only moves rotation-loss weight from some bones to others; it adds no loss on top.

The overfit run of check 6 gave two warnings. The rotation loss barely moved while the images were memorised (44.6 → 40.8), and the translation and 2D terms rose (1.67 → 1.89, 58 → 75, first against last five epochs), possibly a first sign of the depth bug. That evidence is weak: the values are single-batch prints, and the half-run averages fall (2.10 → 1.96, 69 → 63).

Two faults were present that these checks could not find. The first was a depth target built at a different joint than the one the mesh is placed by. The checks compare pelvis-centred quantities, so a placement error does not show in them. The second was a bone turned the wrong way by the same amount in every person. The checks averaged bone errors by group, so one bad bone disappeared in the mean of its group.

F.3. The hidden depth bug

quantity	truth	Base
pelvis depth, mean (sd)	1.92 m (0.49)	3.00 m (0.32)
people placed farther than truth	–	100 %
corr. of true depth with depth error	–	−0.76
skeletal height (3D)	1.409 m	1.405 m ($r = 0.89$)
head / pelvis 2D error, median	0	46 / 14 px
corr. of PVE with depth error	–	−0.03

Table F.2.: *Where Base places people. V1, 100-image AnnyOne holdout, all 222 matched people; slope, intercept and drawn size are in Table F.3. The bodies have the right metric size but stand at the wrong distance, and PVE does not see this. One run, synthetic images.*

run (depth loss)	root (m)	size	slope	icpt. (m)	head px	pelvis px	PVE	recall
Base	1.21	0.63	0.33	2.36	64.5	23.4	70.5	90.2
2-epoch fine-tunes of Base, AnnyOne only								
control ×1	1.11	0.66	0.33	2.28	58.1	23.3	72.3	89.8
fix ×1	0.69	0.78	0.30	1.93	9.6	37.3	77.3	87.8
fix ×5	0.66	0.80	0.44	1.62	8.5	38.1	76.8	87.8
fix ×20	0.53	0.87	0.43	1.44	8.8	31.4	75.7	87.4
10-epoch fine-tunes of Base, AnnyOne only								
control ×1	1.44	0.59	0.34	2.59	64.0	24.3	72.2	89.4
fix ×1	0.73	0.78	0.30	1.98	7.6	38.5	75.5	87.0
fix ×5	0.54	0.84	0.38	1.59	7.2	33.1	74.8	87.8
10-epoch fine-tunes of Base with 30% 3DPW (v23 labels)								
Control ×1	1.40	0.59	0.28	2.66	67.1	26.7	77.4	89.0
Final ×5	0.73	0.78	0.42	1.75	7.6	41.1	73.7	87.0
other checkpoints trained with pelvis targets								
Mixed-L	1.39	0.59	0.31	2.60	–	25.1	75.0	88.6
Fresh-heads-L	1.49	0.57	0.28	2.71	–	24.0	74.1	89.0

Table F.3.: *Placement before and after building the training targets at the head joint. V1, 100-image AnnyOne holdout, matched people; metrics as in Table 4.4, head and pelvis px = 2D error, PVE in mm, recall in %. Control rows build the targets at the pelvis, fix rows at the head; ×1/5/20: distance and translation weight. Final also has the bone correction. One run per row, no seed; each fix row is read against the control of the same length (none was run at ×5 or ×20).*

Base places every person too far from the camera (Table F.2, Figure 5.7), and the people nearest to the camera are pushed back the most. The predicted skeleton has the right height, so a body that is placed too far away is drawn at 0.63 of its true size in the image. PVE does not notice any of this, because it moves the predicted pelvis onto the true pelvis before it measures.

I found the cause in the code. The mesh is placed in the image by its head joint, but the anchor targets had been built at the pelvis (Section 4.7.1). So the network was asked to put the head where the pelvis is. A debug print on one batch showed anchor offsets of -2 to -19 cells of the token grid, where the legal range is ± 0.5 of a cell. I did not test on its own why this pushes people away, but the likely reason is the following. When the mesh hangs from a head that is in the wrong place, the 2D terms can only reduce the mismatch by drawing the body smaller in the image, and a smaller body means a body farther away. The direct depth terms did not hold against this, and the mismatch is largest for people near the camera.

The fix is to build the anchor targets at the head, the joint the mesh is placed by. Table F.3 compares runs with this fix against control runs that have the same training schedule but keep the pelvis targets. All five fix runs reach a root error (the 3D distance between predicted and true pelvis) of 0.53–0.73 m, while both controls stay at 1.11–1.44 m. The drawn size rises from 0.59–0.66 to 0.78–0.87, and the head lands within 7.2–9.6 pixels of its true position instead of 58.1–64.0. Every fix run lies below every control, far outside the run-to-run

variation. Two other checkpoints trained with pelvis targets carry the bug too. It stayed hidden for so long because PVE aligns the pelvis first and the real-photo demos place each body from its detection box.

The fix did not give correct depth. It moved the intercept and the body size, but barely the slope of predicted against true depth (0.30 with the switch alone, 0.38–0.44 with a stronger depth loss, against 0.33 for Base). Predictions are still pulled toward 2.5–2.9 m, for a reason I have not found. The pelvis 2D error rose to 31.4–41.1 pixels. The likely reason is that the head is now pinned in place while the body is still drawn too small, which I did not check. Every fix run is also worse than its control, by +2.6 to +4.9 mm PVE and −1.6 to −2.4 recall points. Each of these gaps on its own is within the run-to-run variation, but all of them go in the same direction, so the fix most likely costs some accuracy. I had written a success rule before the runs (PVE $\leq$ 71.5 mm, recall $\geq$ 89.2 %). By that rule every run failed, and so did the controls.

F.4. A per-bone dump of the neck

	Base	corr., no train.	corr. + 2-ep. FT	2-ep. control	Control 10 ep.	Final 10 ep.
neck02 (deg)	175.6	4.5	4.7	175.7	175.8	4.8
head bone (deg)	177.9	12.4	11.2	177.9	177.7	8.9
metacarpal1.L (deg)	180.0	0.0	0.0	–	180.0	0.0
finger4-1.R (deg)	n.d.	10.2	10.3	–	180.0	9.6
neck girth, pred./true	0.461	1.004	0.997	0.461	0.464	0.996
PVE_{neck} (mm)	27.3	10.8	10.7	27.9	27.7	11.0
body PVE (mm)	70.5	71.3	68.8	76.9	77.4	73.7
PA-PVE (mm)	53.3	52.2	50.9	55.2	56.4	50.3
recall (%)	90.2	90.2	89.4	91.1	89.0	87.0
head direction (deg)	12.0	19.1	14.4	13.8	13.0	13.1
fingers (deg)	17.3	8.2	8.3	17.4	17.4	8.3

Table F.4.: *The half-turned bones and their correction. V1, 100-image AnnyOne holdout, matched people. Bone errors: local medians (about 180 deg = half a turn). PVE_{neck} over the 171 neck-skin vertices. Corr.: the bone correction (four constants from 200 training images); FT: fine-tune. 2-epoch columns: AnnyOne-only fine-tunes of Base; Control and Final: 10 epochs with 30% 3DPW. n.d.: not dumped. One run per column, no seed; PVE gaps below about 10 mm are within run-to-run variation (Table F.7).*

My earlier neck numbers were group means (62.7 and 89.5 deg) that hid which bone was wrong, so I dumped the error of every bone of the neck chain (Table F.4, first column). neck02 is 175.6 deg wrong, a half-turn about its own axis (twist 175.7, swing 3.3 deg), and the head bone 177.9 deg. neck03 is right relative to its parent (5.7 deg) but 173.3 deg off in the camera frame. The two half-turns cancel, so the head points the right way (12.0 deg), but the skin between them is twisted thin, which is what the low girth shows.

The same flip is present in all eight checkpoints I dumped from the ViT-L line that leads to Base and continues from it: L-stages 2 to 4, Base, Mixed-L and three short fine-tunes of Base (Table 4.3). In all of them neck02 is off by 175.0–176.0 deg and the girth is 0.46–0.47. The flip is already there in L-stage 2, before the helper mask (helper bones held at rest), the neck boost and the shape weight of 20 were added, so none of these caused it. One of the three fine-tunes is the 10-epoch depth fix with $\times 5$ weight from Table F.3, and it keeps the flip too, so the depth bug is not the cause of the neck either. Fresh-heads-L is a separate ViT-L run that started from the Multi-HMR-Anny backbone with new heads and saw 3DPW from the first step, without the neck and hand weight. Its neck is normal (girth 1.00), but other bones are flipped instead: two bones in the left arm whose errors cancel each other, and three metacarpals.

So the explanation in my draft was wrong. The rig itself is fine, because it passes all 16 single-joint moves, and the rest pose, in which every rotation is the identity, cannot show a flip at all. Changing the loss did not remove the flip either. The neck boost ($\times 2.41$ for the boosted bones after renormalisation) left it unchanged. A geodesic loss, which penalises the angle between the predicted and the true rotation, moved neck02 only from 175.6 to 175.0 deg. The likely reason is that neck02 and the head have to turn back together, which I did not test. For this reason I corrected the output instead, with one constant rotation per flipped bone (Section 4.7.2): 178.8, 179.2, 179.8 and 179.7 deg for neck02, head, metacarpal1.L and finger4-1.R.

Without training, the correction restores neck02, girth and neck-skin error. But the head direction worsens to 19.1 deg, above the 15-degree limit I had set beforehand, likely because the head's error differs per person. A 2-epoch fine-tune brings it to 14.4 deg (control 13.8). This is why I fine-tuned the teacher with the correction already in place. A constant added after training removes the half-turn that every person shares, but it cannot follow an error that differs from person to person. The differences in body PVE lie within the run-to-run variation, and Figure D.3 shows two more holdout pairs. This section leaves two questions open. I do not know why the bones flipped in the first place. The network outputs each rotation as six numbers (the 6D form), and their singular start value would explain the tilted metacarpal1.L, but not the twist of neck02. I also cannot measure the necks on real photos, because they have no ground truth, so Figure D.11 can only be judged by eye.

F.5. The fingers

In Table F.5 Base's predicted fingers are 17.3 deg off and fingers at rest 15.1 deg, so the prediction was worse than none. My first explanation was that the hands are too small in the image for the network to see the fingers. They are small: at the median a hand spans 2.25 tokens, the small square patches of the image that the encoder works on, and for 41 % of hands fewer than two. But when I cropped the image around each person at test time, so that the hands became larger, the finger error stayed at 17.2 deg while the body PVE

(a) finger rotation error on the holdout setting	fingers (deg)	reference (deg)
Base, predicted fingers	17.3	–
Base, fingers set to rest pose	15.1	–
Base, test-time person crop (PVE 124.8 mm)	17.2	17.3 (paired)
region-normalised rotation loss, 2 ep.	17.5	17.4 (control)
Base + bone correction, no training	8.2	17.3 (Base)
Base + bone correction, 2-ep. fine-tune	8.3	17.4 (control)
Final (10 ep., 30% 3DPW)	8.3	17.4 (Control)
Fresh-heads-L, three metacarpals corrected	17.8	32.0 (uncorr.)

(b) hand size in tokens (14-px patches of the 672-px input)	median length	under 2 tokens
full image, all hands	2.25	41%
person crop at test time	4.2	0%
children's hands, full image	0.7	80%

Table F.5.: *Fingers. (a) Finger rotation error, V1, 100-image AnnyOne holdout, matched people, against the matching run without the change; rest pose keeps the flipped metacarpal1.L. (b) Hand length in input tokens. One run per row, no seed.*

nearly doubled. A region-normalised rotation loss did not help either. It only moves the hands' share of the rotation loss from about 70 to 33 %, so the hands are weighted differently but not more strongly. Then the per-bone search on training images found that metacarpal1.L and finger4-1.R are half-turned, like neck02. Correcting these two bones roughly halves the finger error, both with and without training.

So about half of the finger error came from two flipped bones, and the small size of the hands explains much less. There are two things I did not do. I never tried a separate hand-crop branch in the network, and with the correction in place I did not measure again the error of fingers held at rest pose, the floor I compared with above. Figure D.12 shows the fingers on real photos.

F.6. Children and body shape

Table F.6(a) is my draft's shape-weight test on 10 holdout images: raising the shape weight from 1 to 20 moved the age bias from +0.199 to −0.002 and the height bias from −0.154 to −0.026. Part (b) shows that this holds only near the mode, the most common value. In both tails, the rare low and high values, the predictions move toward the training median. The holdout has no children, so the young tail is measured on training images.

On Relative Human (Table 5.8) Base reads children as short adults. Kids get a mean age of 0.61, babies 0.62 and adults 0.63 (age AUC 0.61, where 0.5 is chance). None of 89 kids and 1 of 38 babies fall below the child threshold; the smallness goes into height instead. Gender carries no signal (accuracy 46.5 %). On my demo photos I took the medians over all 299 predicted people, one shape line per detection in the demo log (about 30 distinct people, nearly all

(a) predicted minus true phenotype, 10 holdout images (draft measurement)

phenotype	true mean	true std	weight 1	weight 20 (Base)	reading
age	0.553	0.036	+0.199	−0.002	near the mode
height	0.579	0.026	−0.154	−0.026	near the mode
proportions	0.763	0.070	−0.017	−0.001	near the mode
firmness	0.757	0.039	−0.167	−0.190	unchanged
gender	0.694	–	−0.042	−0.063	≈
weight	0.382	–	+0.056	+0.064	≈

(b) Base on the full holdout and in the young tail

holdout, all 222 people (nobody younger than 0.513)	age bias −0.015 (abs. 0.037); gender −0.008 (0.068)
holdout, true age 0.6–0.8 ($n = 53$, mean 0.67)	predicted age 0.57
50 youngest training people (in-sample, true age 0.015)	predicted age 0.10; height −0.25
training medians (AnnyOne)	gender 0.588, age 0.601, height 0.500

Table F.6.: *Shape head, phenotype units* $[0, 1]$. *(a) The stage before Base (shape weight 1) against Base (weight 20) on the same 10 holdout images. (b) Base, V1, on the full holdout and the 50 youngest training people (in-sample). In both tails predictions move toward the training median.*

adults). These medians (gender 0.59, age 0.61, height 0.50) equal the training medians.

The children and the female-looking men are the same failure. When the image gives the shape head only weak cues, it falls back to the training median. I see two possible reasons, and my evidence cannot tell which one matters more. The first is the gap between synthetic and real images. The second is the loss itself. The shape outputs go through a sigmoid and are trained with an L1 loss, and an L1 loss is smallest at the median. The sigmoid also flattens at its ends: its gradient is 0.25 at an output of 0.5 but only 0.015 at an output of 0.015, so an output near the edge of the range barely moves during training. On top of that, only 2.53 % of the training people are below age 0.3. The result looks like a short adult because Anny's child shape appears only below an age of about 0.4–0.5. For example, the shoulders are 2.24 head widths wide at age 0.1 and 3.00 at 0.6. So a low height at age 0.6 gives a short adult (at least 1.22 m), and proportions barely matter (leg over stature −0.016).

Chapter 5 gives two more views of the shape head: how well each predicted phenotype agrees with the truth on the holdout (Table 5.10), and how well the models rank gender on real photos (Table 5.9). For Base, Control and Final the slope of predicted against true value is below 1 for every phenotype, which is the same pull toward the middle as above. Final follows weight (Pearson $r = 0.77$) far better than muscle ($r = 0.27$). Two things I cannot give here. I cannot say which of the two reasons above dominates, and I cannot give ages in years, because the code uses three age scales that disagree with each other.

F.7. Pilot noise and restart cost

quantity	measured	one run resolves
two identical controls: PVE	76.3 vs. 68.7 mm (7.6 mm)	about 10 mm
head direction	13.6 vs. 11.7 deg (1.9 deg)	about 3 deg
recall	0.8 points apart	about 2 points
six arms, Base recipe	PVE 68.7–76.3 mm, recall 88.6–90.2 %, head 11.3–13.7 deg	
restart cost	72.9 vs. 70.5 mm for Base (2.4 mm)	read against own control
root error	no measured floor	fix 0.53–0.73, control 1.11–1.44 m

Table F.7.: *Run-to-run variation of the 2-epoch pilots (fine-tunes of Base at learning rate 10^{-5}, no seed). V1, 100-image AnnyOne holdout, matched people. Restart cost: what two more epochs of fine-tuning Base cost with nothing changed.*

Two short pilots that I trained with identical settings differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table F.7). The code sets no seed. Fine-tuning Base for two more epochs with nothing changed also made it worse by about 2.4 mm; I call this the restart cost. For this reason I compare each pilot with its own control and never with Base directly.

Two runs are not enough for a general threshold, and I computed no bootstrap intervals. So I use these differences as a rough yardstick and not as a significance test. Every change this appendix builds on is far larger than the yardstick: the root error of the depth fix, neck02 from 175.6 to 4.5 deg, the girth from 0.46 to 1.004, and the fingers from 17.3 to 8.3 deg. A PVE change of a few millimetres, on the other hand, is smaller than the yardstick. For this reason I cannot say whether the depth fix and the neck pilot cost PVE or not.

F.8. What did not work

Six training changes did not fix any of the three failures (Appendix E, Table E.1). An age-balanced sampler gave no clear answer on the 10 children it could be tested on, who come from the training images. A global head orientation loss stayed within the spread of two identical controls. The region-normalised hand loss, the geodesic rotation loss and the neck and hand boost did not remove the flipped bones, as the sections above show. A $\times 20$ depth loss gave no gain over $\times 5$. None of the six touched the faults I found later, a wrong anchor target and flipped bones, which the loss could not see. The two changes that did work, the head-anchor targets and the bone correction, came out of the checks in this appendix.

G. Supporting conversion experiments

Chapter 3 describes the final converter, version 2.3. This appendix shows how it got there. No earlier version produced the labels used for the main results.

G.1. Development of the converter

The first converter (v1) matched SMPL joints to Anny bones by name. Its mean joint error on `downtown_bar_00` was 34.5 mm. On four test scenes the error was 30 to 37 mm on every frame, and all four actors came out with the same body. The error did not change with the pose, and the body did not change with the person. This told me that the shape parameters were absorbing a fixed mismatch between the two skeletons instead of fitting each actor.

So I put both models in their rest poses and compared them. This showed two wrong choices that the bone names alone could not reveal. Anny's rig numbers the spine from the top down and SMPL from the bottom up, so SMPL's lowest spine joint, `spine1`, had been mapped to a bone 25 cm above it. SMPL's hips had been mapped to `pelvis.L/R`, 4 cm from the hip joint, instead of `upperleg01.L/R`. Version 2 picks the spine bones by height along the body axis and the hip bones by distance from the pelvis (Appendix B). Limbs and head fell to 1 to 10 mm, but the torso stayed at 43 to 73 mm (Table G.1).

The torso stayed high because the two rigs put their torso joints at different points of the body. SMPL's pelvis joint sits 9 cm above its hips. Anny's root is at hip height and 7 cm in front. For nine torso joints I therefore measure once, in the rest pose, the offset $\boldsymbol{\delta}_k$ from the Anny bone head b to the SMPL joint k, written in the bone's own coordinate frame. The fit then moves the *virtual joint* $\mathbf{p}_b = \mathbf{h}_b + \mathbf{Q}_b\boldsymbol{\delta}_k$ onto the SMPL joint, with $\mathbf{h}_b$ and $\mathbf{Q}_b$ the posed bone head and orientation (as in Equation 3.2). So the fitted point keeps a fixed offset from its bone and turns with it. The offsets are 11.6 cm (pelvis), 4.3 cm (hips), 3 to 7 cm (spine) and 5 cm (collars). With them (v2.1) the torso fell to 13 to 22 mm, and four varied test scenes to 15.8 to 20.1 mm (one occluded frame 43.5 mm).

Age was still wrong. In every version up to v2.1 it came out at exactly 0.67 for every actor, because joints cannot tell one adult age from another. Version 2.2 fixes age at 0.55. On all 87 person-sequences it fitted 22,915 frames with a median fit error of 18.2 mm. Every number the converter printed for these fits looked fine. But the fits still had two of the three defects that I describe

in the next section.

converter	mean (mm)	max	worst joints (mm)
v1: correspondence by name, neutral SMPL targets	34.3	37	all joints, common floor
v2: correspondence by geometry, 3DPW joints, gender from metadata (reversed until v2.3)	24.6	30	spine1 73, collars 55–59, spine3 58, pelvis 53
v2.1: + virtual torso joints, bone mask, warm start	15.8	18.8	shoulder 36, wrists 25–27

Table G.1.: *Mean joint error over all 24 joints on* `outdoors_fencing_01` *(38 frames, including full lunges) for three converter versions. These are converter fit errors against SMPL joint targets (v1: the joints of the neutral SMPL model; v2 and v2.1: 3DPW's own joints), not network predictions. The geometric correspondence is worth about 10 mm, the virtual joints a further 9 mm.*

G.2. Three label defects and the v2.3 re-fit

Figure 3.1 in the main text shows one checked frame. Here I give the history behind that check.

The converter grades its own fit, so it cannot see errors in what it is asked to fit, or in what happens to the fits afterwards. For this reason I wrote an independent check that reads the finished label files. On sampled images from every sequence it compares the labels with 3DPW's joints, surface and 2D keypoints, and it confirms that the data loader passes them on unchanged. It also checks that the gender is the right way round, that every valid actor has a label, and that the head turn is kept. On the first label set ("old labels", v2.2 fits, about 2,000 images) its grade was FAIL, with three defects (Table 3.3).

Defect 1: gender reversed. The converter wrote 1.0 for men and 0.0 for women. That is MakeHuman's convention and the reverse of Anny's, so every 3DPW man was labelled with a female body. When I refitted with the flipped value, the joint error fell for 72 of 73 men (median 19.2 to 15.3 mm). For women the joint error rose (13.7 to 16.3 mm), but the body surface came closer to 3DPW's surface for 27 of 31.

Defect 2: missing actor. In 407 of 960 sampled two-person images, one valid actor had no label. The converter fits every third valid frame of each person. It counted these frames from that person's own first valid frame, so each person had their own frame grid. After a gap in which one actor was not valid, the two grids no longer lined up, and the two actors were fitted on different frames. The detection loss then treats a visible, real person as background.

Defect 3: missing labels. The training split had 4,451 label images where the fits implied 6,437 (31% missing). The image files of 8 sequences were incomplete on the cluster (`courtyard_captureSelfies_00`: 480 of 750 files), and the label builder skipped every frame that had no image file, without counting how many it had skipped. A rebuild after re-extracting the images recovered every label (Table 3.2, "rebuilt"). It still had defects 1 and 2, so I never trained on it.

To fix defects 1 and 2, version 2.3 adds two switches: `-gender_convention anny` writes the gender the right way round, and `-frame_grid shared` puts every person of a sequence on one frame grid. Their defaults reproduce v2.2 exactly. I re-fitted all 87 person-sequences with both switches on, and the independent check passed on all 60 sequences (Table 3.3, middle block). 9 single-person outdoor sequences became 1 to 3 mm worse. The likely reason is a different warm start (each frame's fit starts from the previous one), which I did not test.

I call the labels built from the v2.3 fits the v23 labels: 16,694 images and 22,749 people (Table 3.2). That is fewer images than the rebuilt set but nearly the same number of people (22,753 in the rebuilt set), because both actors now share each image. I dropped no person for a high fit error or a small bounding box. I did drop 163 frames, all from `outdoors_crosscountry_00`, because the person cannot be visible in them. With 3DPW's own camera, the ground-truth joints of this sequence sit at a depth of -0.3 m, behind the camera. The 3D fit is fine, but image labels for this sequence cannot exist with this camera, and no label set ever contained it. Figure G.1 shows three more v23 frames.

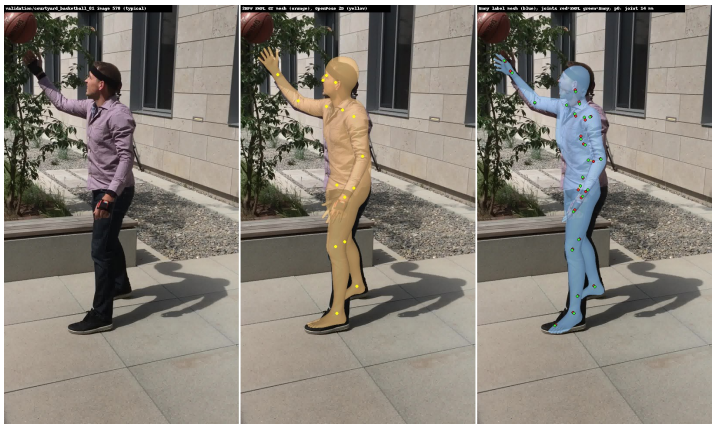
I ran the same check on the v23 labels (3,314 people in 2,360 images, 59 sequences). The grade moved from FAIL to WARN (Table 3.3). The median body-joint discrepancy of the check fell from 18.7 to 14.1 mm. Flipping the gender now makes the fit worse, so the gender is the right way round, and no sampled two-person image misses an actor. The head turn relative to the chest is now kept. In the old test labels it was mostly lost (6.3° against 20.0° in SMPL). Two rows of Table 3.3 stay at WARN: the limb-length ratios, and the people with a joint more than 100 mm off, mostly at the hip. The likely reason is that Anny and SMPL differ in limb proportions and in where they put the hip joint. Figure G.1(b) shows one of the three people that the check renders as its worst cases, chosen by the largest discrepancy at a single joint; the value printed in the panel is that person's mean joint discrepancy.



(a) `courtyard_arguing_00`, *image 24 (training split), two people*



(b) `outdoors_climbing_02`, *image 633 (training split), one of the check's worst cases*



(c) `courtyard_basketball_01`, *image 570 (validation split), a typical frame*

Figure G.1.: *Three more converted labels from the v23 set, judged by eye. These are labels, not network predictions. Each row shows the image, 3DPW's SMPL label (orange) and the converted Anny label (blue; red dots SMPL joints, green dots Anny joints), with each person's joint discrepancy printed above the right panel. In (a) both actors have a label; in the old labels one actor had no label in 407 of 960 sampled two-person images (defect 2). In (b) the Anny body looks thinner than the climber.*

Bibliography

[Bar+24] Fabien Baradel, Matthieu Armando, Salma Galaaoui, Romain Brégier, Philippe Weinzaepfel, Grégory Rogez, and Thomas Lucas. "Multi-HMR: Multi-Person Whole-Body Human Mesh Recovery in a Single Shot". In: *European Conference on Computer Vision (ECCV)*. 2024. URL: https://github.com/naver/multi-hmr.

[Bla+23] Michael J. Black, Priyanka Patel, Joachim Tesch, and Jinlong Yang. "BEDLAM: A Synthetic Dataset of Bodies Exhibiting Detailed Lifelike Animated Motion". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023.

[Bog+16] Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, Javier Romero, and Michael J. Black. "Keep it SMPL: Automatic Estimation of 3D Human Pose and Shape from a Single Image". In: *European Conference on Computer Vision (ECCV)*. 2016, pp. 561–578.

[Bra+26] Laura Bravo-Sánchez, Matthieu Armando, Romain Brégier, Grégory Rogez, Serena Yeung-Levy, and Fabien Baradel. "All-Age Human Mesh Recovery". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Findings track*. Code released as Anny-Fit. 2026, pp. 3677–3687. arXiv: 2605.04728. URL: https://github.com/naver/anny-fit.

[Bré+26] Romain Brégier, Guénolé Fiche, Laura Bravo-Sánchez, Thomas Lucas, Matthieu Armando, Philippe Weinzaepfel, Grégory Rogez, and Fabien Baradel. "Human Mesh Modeling for Anny Body". In: *Computer Vision – ECCV 2026*. Cham: Springer Nature Switzerland, 2026, pp. 170–188. ISBN: 978-3-032-37314-4. arXiv: 2511.03589. URL: https://github.com/naver/anny.

[Cho+21] Hongsuk Choi, Gyeongsik Moon, Ju Yong Chang, and Kyoung Mu Lee. "Beyond Static Features for Temporally Consistent 3D Human Pose and Shape from a Video". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021, pp. 1964–1973.

[Cho+22] Vasileios Choutas, Lea Müller, Chun-Hao P. Huang, Siyu Tang, Dimitrios Tzionas, and Michael J. Black. "Accurate 3D Body Shape Regression using Metric and Semantic Attributes". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022, pp. 2718–2728.

[Dos+21] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, et al. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale". In: *International Conference on Learning Representations (ICLR)*. 2021.

[Goe+23] Shubham Goel, Georgios Pavlakos, Jathushan Rajasegaran, Angjoo Kanazawa, and Jitendra Malik. "Humans in 4D: Reconstructing and Tracking Humans with Transformers". In: *International Conference on Computer Vision (ICCV)*. 2023.

[Hes+18] Nikolas Hesse, Sergi Pujades, Javier Romero, Michael J. Black, et al. "Learning an Infant Body Model from RGB-D Data for Accurate Full Body Motion Analysis". In: *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2018, pp. 792–800.

[HVD15] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. "Distilling the Knowledge in a Neural Network". In: *arXiv preprint arXiv:1503.02531* (2015).

[KAB20] Muhammed Kocabas, Nikos Athanasiou, and Michael J. Black. "VIBE: Video Inference for Human Body Pose and Shape Estimation". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2020, pp. 5253–5263.

[Kan+18] Angjoo Kanazawa, Michael J. Black, David W. Jacobs, and Jitendra Malik. "End-to-end Recovery of Human Shape and Pose". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2018.

[Kan+19] Angjoo Kanazawa, Jason Y. Zhang, Panna Felsen, and Jitendra Malik. "Learning 3D Human Dynamics from Video". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 5614–5623.

[Kol+19] Nikos Kolotouros, Georgios Pavlakos, Michael J. Black, and Kostas Daniilidis. "Learning to Reconstruct 3D Human Pose and Shape via Model-fitting in the Loop". In: *International Conference on Computer Vision (ICCV)*. 2019, pp. 2252–2261.

[LD18] Hei Law and Jia Deng. "CornerNet: Detecting Objects as Paired Keypoints". In: *European Conference on Computer Vision (ECCV)*. 2018, pp. 734–750.

[Lop+15] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. "SMPL: A Skinned Multi-Person Linear Model". In: *ACM Transactions on Graphics (TOG)* 34.6 (2015), pp. 1–16.

[Mak] MakeHuman Community. *MakeHuman: open-source tool for creating 3D human characters*. `https://static.makehumancommunity.org/`. Accessed 18 Oct 2026.

[Mar+18] Timo von Marcard, Roberto Henschel, Michael J. Black, Bodo Rosenhahn, and Gerard Pons-Moll. "Recovering Accurate 3D Human Pose in The Wild Using IMUs and a Moving Camera". In: *European Conference on Computer Vision (ECCV)*. 2018.

[NAV26a] NAVER LABS Europe. *Anny: a free-to-use human body model for all ages, version 0.6.1*. `https://github.com/naver/anny`. Code under Apache License 2.0; MakeHuman-derived assets under CC0 1.0; accessed 18 Oct 2026. 2026.

[NAV26b] NAVER LABS Europe. *Multi-HMR: official code and pre-trained checkpoints, including `multiHMR_672_L_anny`*. `https://github.com/naver/multi-hmr`. README model table and news entry of 17 Feb 2026 (Anny checkpoint); accessed 18 Oct 2026. 2026.

[ODM+23] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. "DINOv2: Learning Robust Visual Features without Supervision". In: *arXiv preprint arXiv:2304.07193* (2023).

[Pat+21] Priyanka Patel, Chun-Hao P. Huang, Joachim Tesch, David T. Hoffmann, Shashank Tripathi, and Michael J. Black. "AGORA: Avatars in Geography Optimized for Regression Analysis". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021.

[Pav+19] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. "Expressive Body Capture: 3D Hands, Face, and Body from a Single Image". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 10975–10985.

[Raj+22] Jathushan Rajasegaran, Georgios Pavlakos, Angjoo Kanazawa, and Jitendra Malik. "Tracking People by Predicting 3D Appearance, Location and Pose". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022, pp. 2740–2749.

[Rom+15] Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. "FitNets: Hints for Thin Deep Nets". In: *International Conference on Learning Representations (ICLR)*. 2015.

[RTB17] Javier Romero, Dimitrios Tzionas, and Michael J. Black. "Embodied Hands: Modeling and Capturing Hands and Bodies Together". In: *ACM Transactions on Graphics (TOG)* 36.6 (2017), pp. 1–17.

[She+23] Xiaolong Shen, Zongxin Yang, Xiaohan Wang, Jianxin Ma, Chang Zhou, and Yi Yang. "Global-to-Local Modeling for Video-based 3D Human Pose and Shape Estimation". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023, pp. 8887–8896.

[Shi+24] Soyong Shin, Juyong Kim, Eni Halilaj, and Michael J. Black. "WHAM: Reconstructing World-grounded Humans with Accurate 3D Motion". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.

[Sun+21] Yu Sun, Qian Bao, Wu Liu, Yili Fu, Michael J. Black, and Tao Mei. "Monocular, One-stage, Regression of Multiple 3D People". In: *International Conference on Computer Vision (ICCV)*. 2021.

[Sun+22] Yu Sun, Wu Liu, Qian Bao, Yili Fu, Tao Mei, and Michael J. Black. "Putting People in their Place: Monocular Regression of 3D People in Depth". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022.

[Wei+22] Wen-Li Wei, Jen-Chun Lin, Tyng-Luh Liu, and Hong-Yuan Mark Liao. "Capturing Humans in Motion: Temporal-Attentive 3D Human Pose and Shape Estimation from Monocular Video". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022, pp. 13211–13220.

[Zen+22] Ailing Zeng, Lei Yang, Xuan Ju, Jiefeng Li, Jianyi Wang, and Qiang Xu. "SmoothNet: A Plug-and-Play Network for Refining Human Poses in Videos". In: *European Conference on Computer Vision (ECCV)*. 2022.

[Zho+19] Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. "On the Continuity of Rotation Representations in Neural Networks". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019.